BIO-SBI

Simulation experiments for planning organoid lineage experiments

Sequential edition: the experiments in the order of the argument

Two-type branching models (LGR5$^+$ = A, KRT20$^+$ = B),
neural posterior estimation, identifiability and robustness

Status of results	10 November 2026 (all planned experiments complete; exact BH posteriors preliminary)
Repository	`BIO-SBI`, branch `main`, results up to commit `c2b639c`
Companion	`report/report.pdf` (same content, ordered by experiment ID)
Data	synthetic (all experiments) and measured CRC0322 organoid counts (Part V)
Audience	group planning the wet-lab organoid experiments

How to read this report. The executive summary (p. 1) is self-contained and follows the same line of argument as the body. Section 1 shows how the experiments depend on each other and maps every experiment ID to its section here and in the ID-ordered report. Section 2 defines the model and every diagnostic. Each experiment section opens with a blue *Why this next* box (the earlier result that raised its question), followed by *Question*, *Setup*, *Result*, *Comment*, and ends with a green *Leads to* box. Every experiment of the original plan is complete; dashed orange boxes in the roadmap mark follow-ups (Part VI). Parameters: A→AA etc. are outcome probabilities per cell event; τ_A, τ_B are mean cell lifetimes (Bellman–Harris only).

Executive summary

Scope. All simulation sweeps of E0–E4 are finished (1126 runs, incl. the E4 extensions to 1024–4096 organoids per endpoint, 5k-vs-20k simulation checks at the real-like BH design, at 512 and at 4096 per endpoint, and the full transition model at 4096, §8), all on synthetic data from one reference parameter vector $\theta^\star$ (Table 2), 5 seeds per setting. Two later experiments test the role of the prior with truths swept or drawn from the prior: the E2c rerun (90 runs, §32) and X5 (355 runs; X5a §6, X5c §7, X5d §14, X5b §15); X2 (40 runs, §3) separates dataset from training noise. New in this version: X3 (GW fitted to BH data, 25 runs, §19), the follow-ups of E1b (all seeded organoids recorded, 90 runs, §24) and E2 (asymmetric mis-typing, 80 runs, §29), the BH timing P0 (Appendix A.1) and the first fit of *measured* data, CRC0322 organoids (12 models × 5 data variants, Part V). Also new: X1 (four inference methods on five decisive settings, 94 runs, §4) and X4 (combined realistic errors at 50k simulations, 20 runs, §34). With them every experiment of the original plan is complete. New since 8 November: the **exact likelihood** (35 GW runs on the same datasets as NPE, §9; BH Fisher information and preliminary, not converged BH posteriors, §13.1) and a speed benchmark (Appendix B); they **correct the B-row conclusion** of Part I from structural to practical non-identifiability. Galton–Watson (GW, time in generations; mapping to days nominal) and Bellman–Harris (BH, time in hours; real-like design = days 6 and 10 = 144/240 h with 165/101 organoids). Every error is compared with the well-specified control of the same sweep.

Headline — the real CRC0322 organoids (Part V). The model class used in every synthetic experiment (shared outcome probabilities, one A founder, B cells not dividing, constant lifetimes) **cannot describe the measured organoids**: posterior support mass 0.01–0.04 and 44–54% of the summary features outside the 95% predictive interval; it predicts a strongly positive A–B correlation across organoids where the data show a negative one (z = −21 and −60 on days 6 and 10). Of twelve models fitted to five data variants (3D mean, the three 3D samples, 2D sections), only the combination of **organoid-level fate heterogeneity** (each organoid draws its own outcome probabilities around the shared ones) and **density-dependent growth** (lifetimes lengthen as the organoid grows) fits both the 2D and the 3D data: support 0.73–0.94 over variants and two training seeds, PPC 0–10% (3D) and 10–15% (2D). *Identifiable*: A self-renewal A→AA ≈ 0.6 (0.62 ± 0.06 in 3D, 0.64 ± 0.08 in 2D); A division time early in growth ≈ 16 h in the 3D counts (16.9 ± 1.7 h; 28 h in 2D, a scale difference); carrying capacity $K \approx 50$ cells (54 ± 13 in 3D; 18 in 2D); strong organoid-to-organoid heterogeneity, $\kappa \approx 3$–4. *Not identifiable at this design*: whether KRT20$^+$ cells divide or switch back to LGR5$^+$, and the founder mix — as E0 and E1a predicted from synthetic data (practically: the data are too weak at 266 organoids, not the model degenerate, §9).

Key findings, in the order of the argument

Reading key (X2, §3). One trained network is reliable: training noise is ≈ 20% of the posterior sd, and the seed-to-seed spread of the experiments below is about half training, half dataset noise. Posterior sds are the uncertainty to quote; seed sds are a stability check.

Methods check (X1, §4). At realistic sizes the inference method is not the problem (at large designs it can be: §9): amortised NPE, sequential NPE, NLE and NRE agree on the identifiability findings — B→A 0.19–0.22 with B divisions free, 0.71–0.72 (5/5) with them fixed; BH ridge A→AA 0.30–0.34 (truth 0.55) — and on the flagged/silent pattern: too-short fixed lifetimes are silent for every method (PPC ≤ 3%), a lifetime prior 50% too long is flagged by every method that finishes (PPC NPE 35%, NLE 76%, NRE 14%). NPE is the best calibrated (SBC error 0.025–0.039 vs 0.045–0.110) and the fastest (1.4–3.2 min per run; NRE 1.4–2×, SNPE 4–9×, NLE 9–31× longer). **SNPE hangs when the posterior leaves the prior** (loud setting 0/5 runs finish, silent 4/5): it has no warning of its own. NLE and NRE are less calibrated at 5k simulations.

I. Identifiability and the prior (Part I).

1. **B's row is prior-dominated if all transitions are free** (E0): B→A = 0.20 vs truth 0.80 at 5k–50k simulations, every design (uniform Dirichlet puts 0.03% mass above 0.8; a flat prior gives 0.53). X5 shows this is not a tail effect: the estimate barely follows the truth (0.04 → 0.22 for truths 0.1 → 0.9) and 1024 organoids do not fix it.
2. **Constraints, not priors or realistic numbers of organoids** (X5a, X5c): with B divisions free no prior (uniform, Jeffreys, flat) and no sample size up to 1024 organoids recovers B→A; with them fixed every truth 0.1–0.9 is recovered under every prior. A weak biological prior (A mostly divides, B rarely) is the best soft option (B-division mass 0.54 → 0.11, lowest error, calibrated) but recovers B→A only partly (0.59 at truth 0.9; constraint 0.79).
3. **NPE at 4096 organoids per endpoint fails** (§8): with B divisions free, even 4096 organoids per endpoint on three endpoints and a 20k-simulation network give B→A 0.17–0.34 (0/20 runs covered) and put about half of B's probability (0.49–0.57, the prior mean 0.5) on divisions that do not exist, with a clean PPC (≤ 0.6%); in GW also A's row is biased (A persist 0.16 vs 0.03). We first read this as structural non-identifiability.

4. **Corrected by the exact likelihood: practical, not structural** (§9). On the same datasets the exact GW posterior (no simulations, no summaries) recovers **B→A = 0.78 ± 0.02** (B-division mass 0.04, A's row MAE 0.011) at 4096 × 3: the failure there was NPE's. At the E0 and real-like designs the exact posterior is as prior-dominated as NPE's (B→A 0.25–0.27; Cramér–Rao sd 0.27–0.28, more than the prior sd): there the data are too weak. Early endpoints help partly ({2,3} × 133: 0.56 ± 0.15). With B divisions fixed exact and NPE agree (0.77 vs 0.71–0.75). Whether B divides must still be fixed or measured — because feasible designs cannot decide it, not because the model cannot.

II. Design and amount of data (Part II).

5. **Endpoints matter more than organoids, in both simulators** (design comparison, E4): days/generations {4,6,10} beat every other set at every size; one late endpoint plateaus; a late extra endpoint ({6,10,14}) adds little. GW: contraction 0.80 → 0.95 → 0.96, MAE 0.052 → 0.023 → 0.019 → 0.020 (16 → 512 → 2048 → 4096 organoids/endpoint); with {4,6,10} every probability except A→AB (error 0.02–0.03, intermittently covered) is recovered by ≈128, and at 5k simulations multi-endpoint designs plateau at ≈1024 per endpoint (MAE ≈ 0.02; flat to 4096). A single endpoint never catches up: at 4096 organoids MAE 0.046, worse than three endpoints at 32 (0.042). BH: 0.50 → 0.79 → 0.80 → 0.82, MAE 0.114 → 0.060 → 0.058 → 0.057. **The plateau is set by the network, not the data**: at 4096 organoids per endpoint, 20k instead of 5k training simulations on the same datasets lower GW MAE 0.020 → 0.017 and SBC error 0.064 → 0.044, and raise BH run coverage 0.76 → 0.93 (§12.1).
6. **The well-specified GW model is recovered**: at the real-like design MAE 0.033 over the 9 outcome probabilities, contraction 0.89, A→AA 0.52 (truth 0.55); across all E4 designs A→AA 0.47–0.55 — provided B's divisions are fixed to 0 (E0).

III. Bellman–Harris timing (Part III).

7. **BH baseline bias = lifetime/self-renewal ridge.** Well-specified BH gives A→AA ≈ 0.32 and τ_A ≈ 16 h (truth 0.55, 20 h), 0/5 covered, at every design. Fixing lifetimes to the truth gives 0.51 (5/5), MAE 0.109 → 0.059 (§13). With the standard 5k-simulation NPE the posterior stays on the ridge up to 4096 clones per endpoint (up to 12 288 organoids, E4): A→AA 0.29–0.41, τ_A 15.7–17.1 h, 0/5 covered in every design, τ_A contraction 0.99 — confidently on the wrong point. But with 20k simulations at 4096 clones (days 4/6/10) it moves most of the way to the truth: A→AA 0.40 → 0.50 (3/5 covered), τ_A 17.0 → 18.9 h (3/5), lifetime error 0.27 → 0.15 (§12.1). At realistic sizes 20k simulations do *not* move it: real-like design A→AA 0.32 → 0.33, τ_A 15.9 → 15.4 h (0/5); with 512 per endpoint on days 4/6/10 it starts (A→AA 0.38 → 0.44, τ_A 2/5 covered). The ridge resolves gradually with data if the network is trained enough; at realistic sizes lifetime information remains necessary. The exact BH likelihood confirms a *practical* ridge: nonsingular Fisher information, Cramér–Rao sd 0.23 for A→AA at the real-like design, $\approx 6 \times 10^3$ clones for sd 0.05 (§13.1). Preliminary exact posteriors (not converged) sit closer to the truth along the ridge than NPE on the same datasets (B divisions fixed: A→AA 0.46–0.49 vs 0.33–0.39, τ_A 17.7–18.3 vs 15.7–16.6 h): part of NPE's bias at this design may be the network and summaries, as in GW; their widths are not reliable.
8. **Averaged over the prior, both simulators are calibrated** (X5d): 90% coverage, SBC ≈ 0.03; contraction 0.68 GW vs 0.51 BH — single-truth failures are about truths and priors, not broken inference. A→AA is learned well on average: the BH ridge bias is specific to this tail truth.
9. **Lifetime information is the practical lever at realistic sizes** (X5b, E3a): a right literature lifetime prior (±25%) moves A→AA 0.32 → 0.46 (MAE 0.109 → 0.068); a lifetime prior 50% too long is detected (support ≈ 0, PPC 0.27), and so is a fixed global lifetime longer than A's; a slightly too short one biases silently. Lifetime CV 0.18–0.5 is harmless, near-exponential cycles are not (E3b); quiescence is not identifiable (E3c).
10. **GW as an approximation of BH (X3, §19)**: on BH data (τ_A 20 h, τ_B 30 h) a GW model with a generation length τ at or below A's lifetime fits without any warning (support 0.62–0.84, PPC 0). With τ = 20 h it recovers A's row better than the true BH model (A→AA 0.52, 5/5 covered; MAE 0.086 vs 0.116): fixing τ fixes the time scale and removes the ridge. The price is a silent bias in B's row: B's slower cycle is read as persistence (B persist 0.40 vs 0.10, B→A 0.48 vs 0.80, 0/5). τ longer than A's lifetime fails loudly (τ 24 h: support 0.05, PPC 0.21; 30 h: no posterior samples). GW is usable only with τ set from measured division times of the fastest type.

IV. Misspecified biology and assay (Part IV).

11. **Founder composition**: founder type not learnable in BH (contraction ≤ 0.12), partly in GW (0.68 for truth 0.5); inferring it removes no shift. Founder size: stays at prior in BH; in GW it tracks the truth (0.71 for $p_1 = 0.6$) and inferring it removes the misfit and halves the shifts.
12. **Observed-organoid fraction** (detected/seeded): 0.94–0.95 at $\theta^\star$; dominated by A death (0.3 ⇒ 0.54–0.58); GW exact and BH simulated agree to 0.02 on average; A and B death trade off (one constraint, not

two).

13. **Recording extinct organoids (E1b follow-up, §24)**: with every seeded organoid recorded, death rates are sharper in BH (A death sd 0.015–0.033 vs 0.045–0.057 detected-only) but overall accuracy does not improve, and in GW it worsens (MAE 0.063 vs 0.032 at A death 0.03). Survivor-only summaries plus the extinct fraction recover part of the loss (GW 0.057, 0.068 at A death 0.03, 0.10) but do not beat detected-only data at equal design size.
14. **Size of the tested errors (vs paired control; BH | GW):**
    - founders 50% B: A→AB -0.11, τ_B $+4$ h | shifts ≤ 0.07 (silent in both);
    - 40% multi-cell clones: τ_A -1.8 h, B→A -0.07 | A→AA $+0.13$, B→A -0.25 (0/5 covered, flagged);
    - B death fixed at 0.4 (true 0.1): B→A -0.19 | -0.23 to -0.26; A death fixed at 0.2 (true 0.03): A persist -0.10, τ_A -2 to -3 h | no fit; deaths fixed at 0: ≤ 0.07 in both (silent);
    - mis-typing ignored, $e = 0.2$: A→AB $+0.09$ (0.7 sd) | $+0.15$ (2.1 sd; already 1.4 sd at $e = 0.1$); modelled: information loss (A→AB contraction 0.30 → -0.22 | 0.85 → 0.37);
    - BH only: lifetime CV 0.18–0.5 harmless, exponential A→AA $+0.16$, B→A -0.18; quiescence (A persist) not identifiable.
15. **Mis-typing direction matters (E2 follow-up, §29)**: stem cells read as differentiated (A→B label, e_A) are the harmful direction. In GW, ignoring $e_A = 0.10$–0.15 doubles the error (MAE 0.040 → 0.066–0.086, coverage 0.64–0.76) and lowers B→A to 0.63–0.66, partly flagged (support 0.42–0.50); B read as A at 0.10 is almost harmless (MAE 0.040, coverage 0.84). Modelling the known confusion removes the bias. In BH the effect is small next to the ridge error (MAE ≈ 0.12 in every setting).
16. **Hidden third type (GW, E2c): resolved by the rerun with prior-drawn truths** (§32). The original failure of even the true model (support ≈ 0) was the tail truth (prior probability 10^{-6}–10^{-10}), not the method. In the prior bulk, ignoring C is flagged when C is read as B (9/10 runs at $q = 0$, 7/10 at $q = 0.5$) but *not* when C reads as A (2/10 at $q = 1$, while A/B MAE 0.18 and coverage 0.42 at contraction 0.86: silent confident bias). Modelling C (free or trajectory-constrained) is calibrated (PPC 0, coverage 0.92–0.93) but not identifiable at this design: contraction 0.16–0.20, A/B MAE 0.055–0.059 = prior-mean baseline 0.057; the wrong trajectory constraint is not detected.
17. **A free time scale hides errors** (Figs. 35, 36, §33). In BH the PPC stays ≤ 0.033 for every error except both deaths fixed wrong (0.096) and a global lifetime longer than A's (0.35–0.51). GW, without lifetimes, flags more: A death fixed too high (support ≈ 0), B death fixed at 4× truth (support 0.13–0.33), 40% multi-cell founders (PPC 0.12, support 0.16). Still silent in GW: wrong founder type, B death at 2× truth (B→A 0/5 covered), ignored mis-typing up to 2 posterior sd.
18. **Error budget of the baseline model (X4, §34)**: with 10% B founders, 5% symmetric mis-typing and lifetime CV 0.5 acting together at the real-like design (50k simulations), the baseline BH model has MAE 0.106 vs 0.083 for the well-specified control: ≈ 0.08 from the ridge (A→AA 0.32, τ_A 15.4 h, 0/5) plus ≈ 0.02 from the assay errors, **none of it flagged** (PPC 0, support 0.90). Modelling the errors makes it worse (MAE 0.147): the founder row is not learnable (founder A 0.54 vs 0.9, at its prior). A hidden type C on top is only weakly flagged (PPC 2.1%, support 0.80) and is absorbed into B's row (τ_B 52 h vs 30 h, B→A 0.06). Calibration is good in every arm (SBC 0.011–0.016). This is the budget of the baseline class, which the real data reject (Part V).

V. Real data (Part V).

19. **The synthetic baseline model is rejected, in the way the synthetic experiments made recognisable**: support mass ≈ 0 and PPC 44–54% flag it, as they flagged the large errors of Part IV; the decisive misfit is the sign of the A–B correlation across organoids. B dividing and an inferred founder type remove most of that misfit but not the size misfit; fate heterogeneity brings the 3D support to 0.37–0.44; density dependence plus whole-row heterogeneity fits (support 0.73–0.94, PPC 0–15%). Multi-cell seeding does not help.
20. **What the final model can and cannot say** follows the synthetic lessons: A self-renewal (≈ 0.6), A's early division time, K and κ are recovered in a synthetic check of the same network; B's row (E0) and the founder mix (E1a) are not, and their posteriors must not be read as findings.

Timing (P0, Appendix A.1; speed, Appendix B). On an idle machine a BH run at the real-like design costs the same as a GW run (≈ 45 s at `budget=dev`); earlier minute-long run times were contention. Exact inference: GW 3–24 ms per likelihood evaluation, 10–66 min per posterior as run (NUTS); BH 1.3–2.7 s per evaluation on 20 threads, $\approx 10^5$ evaluations (≈ 3.5–4 h on 16 threads, SMC + MCMC) per reliable posterior (AMIS, 1–2 h, gave unreliable weights); NPE 0.5–17 min of training, then ≈ 0.03 s per dataset. Exact inference is for validating NPE on a few datasets, not for SBC.

Recommendations for the wet-lab design

- **Endpoints:** three, including an early one — e.g. days 4, 6, 10 (best in both simulators). A single day-10 endpoint is the weakest design, and no number of organoids compensates (GW: 4096 per endpoint at one endpoint is worse than 32 per endpoint at three).
- **Organoids:** ≈ 128 per endpoint. In GW this recovers every outcome probability with {4,6,10} (A→BB from 32, B persist from 64, B→A from 128; A→AB within 0.03 but only intermittently covered); gains stop at ≈ 1024 per endpoint. In BH it gives contraction 0.71; A→BB and B death need ≥ 128, A→AB ≥ 256; self-renewal is not recovered at realistic sizes without lifetime information, whatever the training budget (real-like design with 20k simulations: A→AA 0.33; 512 per endpoint: 0.44; only 4096 with 20k comes close: 0.50, §12.1).
- **Measure independently:**
  (a) *A (LGR5$^+$) division time* by live imaging — the single most valuable measurement (it is what separates BH, MAE 0.109, from the GW target, 0.033); per type, not pooled. It is also what a GW approximation needs (X3: τ from the fastest type), and what would test the real-data estimate (16.9 h early in growth, lengthening with organoid size);
  (b) *whether B (KRT20$^+$) cells divide* — by live imaging, lineage tracing or a proliferation marker on KRT20$^+$ cells. At feasible designs the counts cannot decide it: the exact likelihood is as prior-dominated as NPE at 96–266 organoids, and late endpoints need thousands of organoids per endpoint (§9); so B→A is recovered only once B divisions are fixed;
  (c) *seeding*: number of organoids seeded per well (observed fraction, sensitive to A death; recorded as extinct organoids it sharpens the death rates in BH, §24) and whether wells contain multi-cell seeds ($\gtrsim 15\%$ multi-cell clones bias self-renewal and B→A in GW; then fix or infer the founder-size distribution);
  (d) *mis-typing rate* (2D vs 3D comparison), above all how often LGR5$^+$ cells are called KRT20$^+$ (the harmful direction, §29), and pass it to the model as an asymmetric confusion: ignored, it is tolerable only up to $e \approx 0.05$ once posteriors are as narrow as in GW;
  (e) *cell-cycle variability*: only whether CV is < 0.5 or near-exponential;
  (f) *founder type* and *quiescence* (e.g. Ki67) if they matter biologically — the counts cannot provide them;
  (g) *apoptosis* only if death rates are to be fixed; otherwise keep them free (a value too high biases B→A; in BH silently).
- **Biological constraints help more than data at feasible sizes**: B divisions = 0 (if KRT20$^+$ cells do not divide) is what makes B→A recoverable; neither a different prior nor 1024 organoids replaces it (X5a, X5c). The data could decide it in principle (exact likelihood at 4096 per endpoint on three endpoints: B→A 0.78 $\pm$ 0.02), but not at realistic sizes, and amortised NPE did not even at 4096 (§8, §9); with B divisions free, NPE's B-row error also leaks into A's row. Early endpoints (generations 2–3) carry the most information per organoid but do not decide it at 266 organoids (B→A 0.56 $\pm$ 0.15). In BH the B row is weaker still: B's lifetime must be known as well (§13.1). If it is not known that B cells do not divide, measure it (item (b) above). A soft biological prior ("B rarely divides") helps but does not replace it; with the constraint keep B's prior near-uniform (an informative symmetric one shrinks B→A toward 1/3).
- **Simulation budget:** train every BH fit and every large design ($\gtrsim 1024$ per endpoint) on ≥ 20k simulations with an epoch cap of ≥ 500 (the 20k runs at 4096 stopped at the 200-epoch cap) and check SBC. At the real-like design 20k does not fix the ridge but lowers the probability error (MAE 0.109 → 0.079); 5k results at large sizes are lower bounds on attainable accuracy (§12.1).
- **Literature division times** can enter as an informative lifetime prior ($\pm 25\%$ gets most of the benefit of fixing them, X5b); check PPC and support mass for conflict (a prior that is too long is flagged; too-short lifetimes bias silently, E3a).
- **Real-data fits must model organoid heterogeneity and density dependence** (Part V): the homogeneous constant-lifetime model of the synthetic experiments is rejected by the CRC0322 data. Report only the identifiable quantities (A self-renewal, A's early division time, K, κ) and treat B's row and the founder mix as unresolved until measured (B divisions, founder type at seeding).
- **Hidden third type**: endpoint counts of two labels cannot identify a three-type model (E2c rerun). If C may exist, find out how it is labelled: read as B, ignoring it is usually flagged; read as A, it biases the A/B rows silently. Modelling it needs a C-specific marker, trajectory data that justify constraints, or much richer designs — a free C model returns the prior.
- **Inference for real-data fits** (X1, X4): keep amortised NPE as the default (fast, best calibrated, built-in support-mass check); confirm key real-data conclusions with NRE (a different approximation at about twice the cost); use SNPE only after the model has passed NPE's support and PPC checks — it hangs when the posterior leaves the prior. Quote the unflagged error of the model class (X4: ≈ 0.1 MAE for the baseline) next to the posterior sd, and do not free parameters the data cannot determine (founder mix) to "model" an error. Where an exact likelihood exists, check NPE against it on a few datasets (§9): NPE failed silently

at 4096 organoids per endpoint where exact inference succeeded. Exact GW costs minutes per posterior, exact BH ≈ 3.5–4 h on 16 threads, so use them for validation, not for SBC (§B).

Methodological lessons

- **Prior tails**: a truth with $< 2\%$ prior mass is pulled to the prior only where the data are weak or confounded along a ridge (A→AA = 0.55 in BH); where the model is identifiable the data win at every truth position (X5a, B divisions fixed: B→A 0.1–0.9 recovered under every prior). Where it is confounded at the available data size (B divisions free; practically, not structurally, §9) the prior decides at every position, and neither a flat prior nor up to 1024 organoids fixes it. With 10^{-6} or less (original E2c) amortised NPE extrapolates outside the prior support, while the same model is calibrated for truths in the bulk. SBC and prior-averaged coverage (X5d) look fine in all these cases: report prior quantiles of the truth, check prior sensitivity, and compare hand-picked truths with prior-drawn ones.
- **Simulation budget vs data size — confirmed**: at 1024–4096 organoids per endpoint the SBC error of the 5k-simulation amortised NPE doubles (GW 0.03 → 0.06; BH up to 0.039) and accuracy is flat from 2048 to 4096. The 20k check at 4096 (same datasets) separates the two: GW MAE 0.020 → 0.017, SBC 0.064 → 0.044; BH coverage 0.76 → 0.93, A→AA 0.40 → 0.50. The plateau was the network's, and at very large designs an apparently data-limited ridge is partly a training limit (E4, §12.1); training hit the 200-epoch cap there. At realistic sizes the same check (epoch cap 500, not binding) shows the reverse: the ridge stays (A→AA 0.32 → 0.33) while other parameters improve. Test the budget before attributing a bias to the data.
- **Training noise is small; quote the posterior sd** (X2, §3): at the real-like design the sd of posterior means over 10 training seeds on one dataset is ≈ 20% of the posterior sd (median 0.24 GW, 0.18 BH; its variance ≈ 4% of the posterior variance). Dataset and training contribute about equally to the seed-to-seed spread (training share 0.53 GW, 0.70 BH), and posteriors are wider than the spread of posterior means across datasets (GW B→A: sd 0.123 vs 0.043). So the posterior sd, not the seed sd, is the uncertainty to quote; differences $\gtrsim 0.3$ posterior sd between arms are not training artefacts; narrow posteriors are the exception (GW A→AA: training noise 0.65 posterior sd).
- **Ridges and free time scales**: endpoint counts fix the growth rate; lifetime, self-renewal and quiescence trade off along it (posterior corr. +0.7, −0.6). The prior picks the point, and the free lifetime also absorbs (hides) errors that GW detects (E1a number, E1c).
- **Compare with paired controls**, not with the truth (BH baseline is biased); 5 seeds resolve shifts of ≈ 0.02–0.03 (probabilities) and 0.5–1.5 h.
- **Diagnostics**: SBC checks the inference model only; PPC and support mass are relative checks and vary between well-specified GW runs (support 0.07–0.89, PPC up to 0.10); support mass ≈ 0 = the model cannot explain the data (numbers from such runs are not estimates). A clean PPC does not imply a correct posterior: with B divisions free at 4096 organoids per endpoint NPE's PPC is $\leq 0.6\%$ while B→A is 0.17–0.34, a point the exact likelihood rejects by 116–178 nats (§8, §9).
- **On measured data the checks select models; a synthetic check selects the estimates** (Part V): support mass and PPC separated twelve candidate models clearly (support 0.00–0.06 vs 0.73–0.94). But the final model's B row is as confident-looking as its A row; only a synthetic recovery check of the same trained network shows which parameters are identifiable. Run both for every real-data fit.
- **Method-independence (X1) holds at realistic sizes**: the weak B row, the BH ridge and silent mis-specifications are found by every method (NPE, SNPE, NLE, NRE and, for GW, the exact likelihood); a different method does not rescue weak data.
- **NPE can fail silently at large data — validate it against exact inference** (§9): at 4096 organoids per endpoint the 5k/20k amortised NPE passed SBC (0.05–0.07) and the PPC, yet put B→A at 0.27–0.34 where the exact posterior is 0.78 ± 0.02; we had read this as structural non-identifiability. A negative NPE result at a large design is not evidence about the data. Where the likelihood is tractable (GW; BH with integer lifetime shape) compare NPE with exact inference on a few datasets (cost: §B); otherwise check the Fisher information / Cramér–Rao bound or a profile likelihood before calling a parameter non-identifiable. SNPE is not a diagnostic: when the posterior leaves the prior it does not warn, it hangs.

Done since the previous version (8 November): the exact likelihood for GW and BH and its comparison with NPE on identical datasets (§9, §13.1), which corrects the B-row conclusion of Part I, and the speed benchmark (Appendix B). **Earlier:** the real CRC0322 data (Part V); X1 methods comparison (§4); X4 combined errors (§34); X3 GW fitted to BH data (§19); P0 BH timing (Appendix A.1); asymmetric mis-typing (§29); all seeded organoids as data, with survivor summaries (§24). **Every experiment of the original plan is now done. Follow-ups** (Part VI): mis-typing in the real-data model; alternative forms of heterogeneity and density dependence; survivor summaries (extinct fraction) in real fits once seeded organoids are counted; the synthetic conclusions (design, error budget) re-checked under the final model class; converged exact BH

posteriors (the present ones are preliminary) and NPE on the early designs; the open biology questions Q1–Q5. The results point to *measurements* rather than more simulations: A's division time, whether B cells divide, the LGR5→KRT20 mis-typing rate and the number of seeded organoids.

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1 Roadmap: how the experiments follow from each other

The experiments were planned as one-error-at-a-time sweeps (`TODO.md`), but they were not independent: several of them were launched because an earlier result raised a question. This report follows that chain. The argument runs in six parts.

Two methodological checks come first, before Part I: X2 (§3) measures how much of the seed-to-seed spread is training noise, which is how every later comparison is read; X1 (§4) repeats the decisive settings of Parts I, III and IV with four inference methods and shows that, at realistic sizes, the findings belong to the data, not to NPE (the exact likelihood, §9, later shows where NPE itself fails).

I. **Identifiability and the role of the prior** (Part I). The first fits (E0) could not recover B's switching back to A. Before anything else could be measured we had to know whether this was the prior or the model, which X5a (truth position, prior shape), X5c (sample size), the full model at 4096 organoids per endpoint with a 20k network (§8) and finally the exact likelihood (§9) settle: it is the switching/division confounding, *practical* (feasible designs are too weak; at 4096 per endpoint the exact likelihood recovers B→A where NPE did not), and fixing B divisions to 0 is the baseline for everything after.

II. **Design and amount of data** (Part II). With an identifiable model, the design comparison and E4 ask which endpoints and how many organoids. In GW everything is recovered; in BH self-renewal is not recovered with the standard 5k-simulation network at any size up to 4096 organoids per endpoint — a 20k-simulation network moves it gradually with size (not at the real-like design, a little at 512, most of the way at 4096).

III. **Bellman–Harris timing** (Part III). That failure is the lifetime/self-renewal ridge, practical rather than structural by the exact BH Fisher information (§13.1). X5d and X5b ask whether it is the truth's position in the prior and whether lifetime information removes it; E3a–E3c test the timing assumptions such information would rest on; X3 asks whether GW, whose fixed generation acts as a fixed lifetime, can stand in for BH.

IV. **Misspecified biology and assay** (Part IV). Against well-understood GW and BH baselines, each real-data error is introduced alone: founders (E1a), death (E1b, E1c), mis-typing (E2a/b), a hidden third type (E2c and its rerun, which reuses the tail-truth lesson of Parts I and III). The GW/BH contrast keeps returning: a free time scale absorbs errors. Two follow-ups sit next to their parent experiments: all seeded organoids as data (after E1b) and asymmetric mis-typing (after E2). A synthesis section collects detectability, and X4 combines the realistic errors into the error budget of the baseline model.

V. **Real data** (Part V). The measured CRC0322 organoids, read with all of the above: the synthetic baseline is rejected by the checks of Part IV, a ladder of models finds that organoid-level fate heterogeneity and density-dependent growth are needed, and the final model's estimates are sorted into identifiable and not along the lines of E0 and E1a.

VI. **Planned follow-ups** (Part VI): every experiment of the original plan is done; what remains are real-data follow-ups, the open biology questions Q1–Q5 and measurements. Engineering (device benchmark, P0 timing, speed of exact inference vs NPE) is in Appendix A.

Figure 1 draws the dependencies; Table 1 lists every experiment with the result that motivated it, what it led to, and its section in this report and in the ID-ordered report (`report/report.pdf`).

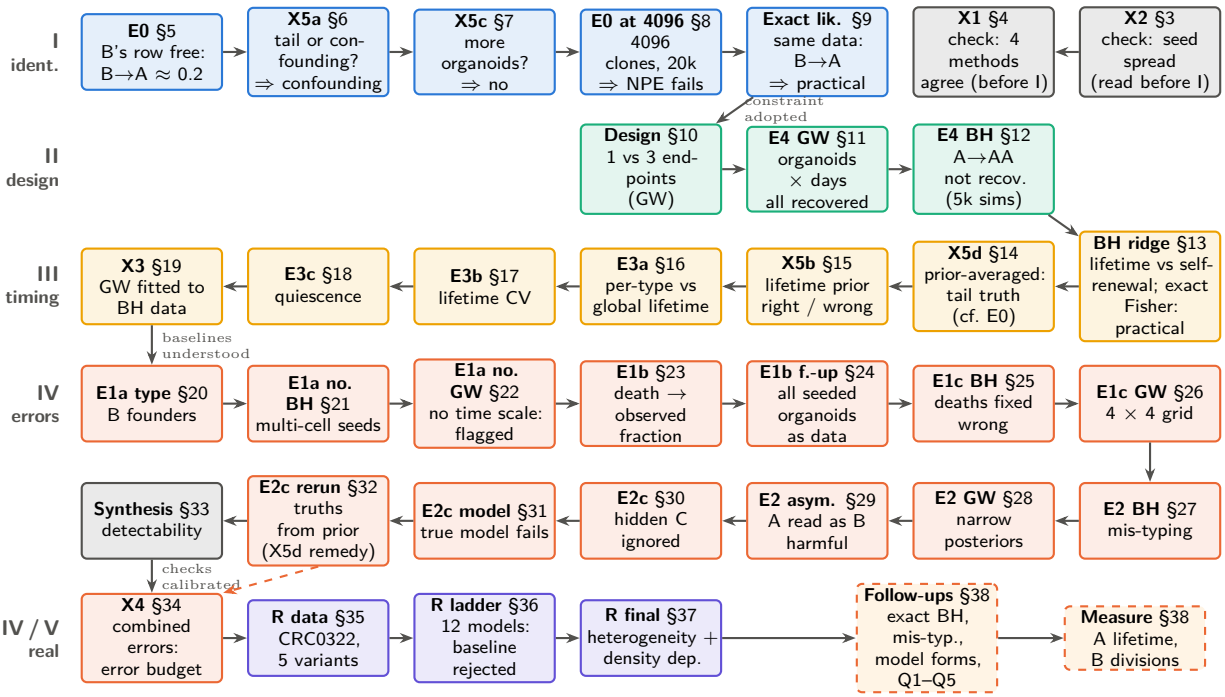


Figure 1: Dependency graph of the experiments. Solid arrows: the order of this report (each result raises the question of the next experiment). Two lessons recur outside the chain and are noted in the boxes: tail truths (E0 → X5d → E2c rerun) and the free time scale (BH ridge → X3 and the BH/GW contrasts of Part IV). Dashed arrow: the one-at-a-time errors (ending with the hidden type C) feed the combined-error experiment X4, the last section of Part IV. Colours = parts; grey = methodological checks (X2, X1, read before Part I) and synthesis; dashed orange boxes = planned follow-ups (Part VI; "Measure" = wet-lab measurements implied by E0 at 4096, the ridge, X4 and the real-data fit). Section numbers refer to this report.

Table 1: Experiment index. "Here": section and page in this report. "ID-ordered": section in `report/report.pdf` (same numbers, ordered by experiment ID). Folders under `results/`.

ID	Question (folder)	Here	ID-ord.	Motivated by	Leads to
Methodological check					
X2	seed spread (`x2`)	§3, p. 14	§22	5 seeds everywhere	how to read spreads
X1	4 inference methods, 5 settings (`x1`)	§4, p. 16	§21	NPE only everywhere	findings are the data's; NPE default
Part I — identifiability and the role of the prior					
E0	B's row identifiable? (`b_row`)	§5, p. 19	§2	first fits: B→A ≈ 0.2	X5a, X5c; constraint for all later runs
X5a	truth position × prior (`x5/a_*`)	§6, p. 20	§25.1	E0: tail or confounding?	X5c; constraint not prior
X5c	organoids vs prior (`x5/c_*`)	§7, p. 23	§25.3	E0 C2ST, X5a	E0 at 4096
E0 4096	full model, 4096 clones, 5k/20k (`e4/*_two_type_*`)	§8, p. 25	§2.1	X5c; E4 budget check	NPE fails; exact likelihood
XG	exact GW likelihood vs NPE, same data (`exact_gw`)	§9, p. 27	§26	E0 4096: data or NPE?	practical, not structural: fix/measure B div.; Part II
Part II — design and amount of data					
Design	1 vs 3 endpoints (`design_comparison`)	§10, p. 30	§3	E0/X5a: model fixed	E4 GW
E4 GW	organoids × endpoints (`e4/gw`)	§11, p. 31	§5	design comparison	E4 BH; GW = target
E4 BH	organoids × endpoints (`e4/bh`)	§12, p. 33	§4	E4 GW; BH closer to data	ridge (A→AA not recovered at 5k sims)
E4 20k	5k vs 20k sims, real-like to 4096 (`e4/*_20k`)	§12.1, p. 35	§4.1	E4 plateau, SBC growth	ridge: data-limited at real sizes
Part III — Bellman–Harris timing					
Ridge	lifetime vs self-renewal (`bh_ridge`)	§13, p. 38	§6	E4 BH bias	X5d, X5b; E1a/E1c GW contrasts
XB	exact BH Fisher information; preliminary posteriors (`exact_bh`)	§13.1, p. 39	§26	ridge; XG	ridge practical; lifetime information
X5d	prior-averaged (`x5/d_*`)	§14, p. 41	§25.4	ridge, E0: tail truths	X5b; E2c rerun
X5b	lifetime prior (`x5/b_*`)	§15, p. 42	§25.2	ridge, X5d	E3a
E3a	per-type vs global lifetime (`e3a`)	§16, p. 43	§18	ridge, X5b	E3b
E3b	lifetime CV (`e3b`)	§17, p. 44	§19	E3a: assumed shape	E3c
E3c	quiescence (`e3c`)	§18, p. 45	§20	ridge: other directions	X3
X3	GW fitted to BH data (`x3`)	§19, p. 46	§23	ridge, E3a: fixed time scale	Part IV; real-data time model
Part IV — misspecified biology and assay					
E1a type	founder type (`e1a/founder_type`)	§20, p. 48	§7	baselines set; organoid death	E1a number
E1a no. BH	founder number (`e1a/founder_number_bh`)	§21, p. 49	§8	E1a type	E1a no. GW
E1a no. GW	founder number (`e1a/founder_number_gw`)	§22, p. 50	§9	BH silent: time scale?	E1b
E1b	death → observed fraction (`e1b`)	§23, p. 51	§10	E1a: organoid loss	E1b follow-up
E1b f.-up	all seeded organoids as data (`e1b_all`)	§24, p. 52	§10.1	E1b: Q5 as a datum	E1c
E1c BH	deaths fixed wrong (`e1c/bh`)	§25, p. 53	§11	E1b: one constraint	E1c GW
E1c GW	deaths fixed wrong (`e1c/gw`)	§26, p. 55	§12	E1c BH silent	E2
E2a/b BH	mis-typing (`e2/*_bh`)	§27, p. 56	§13	clustering errors	E2 GW
E2a/b GW	mis-typing (`e2/*_gw`)	§28, p. 57	§14	BH: wide posteriors	E2 asym.
E2 asym.	asymmetric mis-typing (`e2_asym`)	§29, p. 59	§14.1	E2: one symmetric rate	E2c; real-data follow-up
E2c	hidden C ignored (`e2c/ignore_c`)	§30, p. 60	§15	E2: C a real type?	E2c model
E2c model	C modelled (`e2c/model_c_*`)	§31, p. 61	§16	E2c: attribution	E2c rerun
E2c rerun	truths from prior (`e2c_prior`)	§32, p. 62	§17	E2c model, X5d	synthesis, X4

Table 1 (continued).

ID	Question (folder)	Here	ID-ord.	Motivated by	Leads to
Part IV (continued)					
—	detectability synthesis	§33, p. 64	summary	all of Part IV	X4
X4	combined errors, 50k (`x4`)	§34, p. 65	§24	one-at-a-time errors	error budget; Part V
Part V — real data (CRC0322)					
R data	CRC0322 counts, 5 variants (`real`, `data/real`)	§35, p. 67	§29	end goal; Parts I–IV	ladder
R ladder	12 models (`real/<model>`)	§36, p. 68	§30	baseline of Parts I–IV	final model
R final	`bh_dd_hetero` (`real/bh_dd_hetero*`)	§37, p. 71	§31	ladder; E0, E1a	follow-ups, measurements
Part VI — planned follow-ups; Appendix					
Follow-ups	real-data model, Q1–Q5, measurements	§38, p. 73	§32	R final, X4, E0 4096	—
Bench	device / batch (`gpu_bench`)	§A, p. 74	§28	E0 run cost	run settings
P0	BH timing, idle machine (`p0_bh_timing`)	§A.1, p. 75	§28.1	sweep times under contention	BH cost = GW cost
Speed	exact likelihood vs NPE (`exact_benchmark`)	§B, p. 75	§27	XG, XB cost	exact = validation, not SBC

2 Methods overview

2.1 Model

- **Two cell types**: A = LGR5$^+$ (stem-like), B = KRT20$^+$ (differentiated).
- **One event per cell**: a cell is replaced by 0, 1 or 2 daughters. Outcomes of A: death, persist (stays, no division), A→B (switch), divisions A→AA, A→AB, A→BB. B mirrors A.
- **Parameters**: one probability simplex per type; Dirichlet(1,...,1) prior (uniform).
- **Laws**: `two_type` = all 12 outcomes free (10 free params); `reversible_ab` = B divisions fixed to 0 (B: death, persist, B→A; 7 free) — the baseline of every experiment after E0.
- **Founders**: a clone (organoid) starts from N cells ($P(N{=}n)$ = `founder_size`) of i.i.d. type (`founder`); baseline: one A cell.

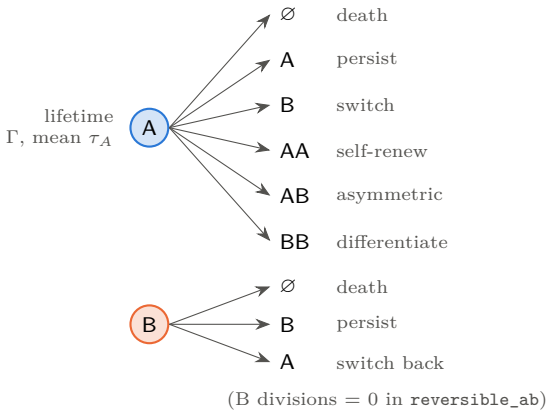


Table 2: Reference truth $\theta^\star$ used by every experiment unless stated (`law.reference`); one founder cell of type A. Prior probability of exceeding the truth under the uniform Dirichlet: A→AA 1.8%; B→A 0.03% (`two_type`) or 4% (`reversible_ab`).

Type	death	persist	switch	→AA	→AB	→BB	mean lifetime (BH)
A	0.03	0.03	0.05 (A→B)	**0.55**	0.05	0.29	τ_A = 20 h
B	0.10	0.10	**0.80** (B→A)	0	0	0	τ_B = 30 h

BH lifetimes: Gamma, shape 8 (CV 0.35). GW: one event per generation.

2.2 Two simulators

	Galton–Watson (GW)	**Bellman–Harris (BH)**
Time	synchronous generations; endpoints in *generations* (real-like: 6 and 10)	continuous; Gamma lifetimes per type; endpoints in *hours* (real-like: 144 and 240 h = days 6, 10)
Extra parameters	none	τ_A, τ_B (log-uniform prior); shape fixed (8)
Sampling	exact detection-conditioned sampler; exact observed fraction	forward simulation + rejection; observed fraction estimated
Cost (P0, idle machine)	real-like design, `dev`: 45 s per run	real-like design, `dev`: 42 s per run (training ≈ 85%)

GW and BH share the outcome law but not the time unit; a GW generation is *not* a day. X3 (§19) fits GW to BH data with a fixed generation length in hours (§2.5).

2.3 Observation, data model and inference model

- **Observation**: biology → capture (Binomial per true type) → labelling (confusion matrix true type × label; may be rectangular, e.g. hidden C read as A with prob. q) → selection of *recovered* clones (≥ 1 detected cell). Baseline: perfect capture and typing.
- **Data**: per endpoint, A and B counts of each recovered organoid. Endpoints are destructive: different organoids per endpoint.
- **Misspecification**: the reference data come from a separate *data model* (`data.simulator`, `data.law`, `data.observation`, with seed `data_seed`); the *inference model* defines prior, training simulations and checks. An error is tested by changing one side only.

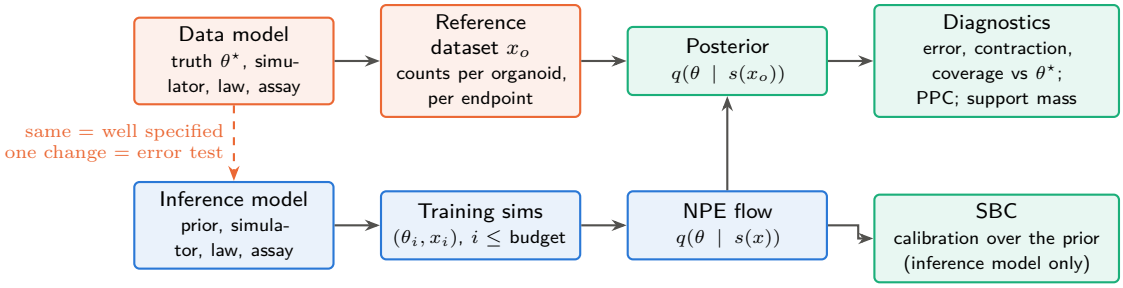
2.4 Inference

- **NPE** (`sbi` 0.27, neural spline flow, 5 transforms, width 64), amortised, 1 round; sequential NPE (SNPE) only in E0 and X1; NLE and NRE (MCMC) only in the methods check X1 (§4).
- **Summaries** (`moments`): per endpoint, mean, sd, quantiles of $\log(1+n)$ for A, B, total; fraction of clones containing each type; fraction empty; correlation of log counts.
- **Budgets**: `dev` = 5 000 simulations (all BH and most GW error experiments); 20 000 for E0/design comparison and the E4 budget checks; E0 also 50 000; X4 and the real-data fits use `full` (50 000, SBC on 1000 prior draws, PPC 500). E0 showed 5k→50k does not change recovery. Per run: 2 000 posterior samples, SBC on 200 prior draws, PPC with 200 datasets.
- **Parameterisation**: stick-breaking of each simplex (exact Dirichlet reparameterisation).
- **Exact likelihood** (§9): for GW (`inference=exact_gw`: pgf iterate + 2-D FFT, NUTS) and for BH with integer lifetime shape (`inference=exact_bh`: method of stages ODE + FFT, tempered SMC + adaptive MCMC) the posterior of the raw counts is sampled without simulations or summaries, with the same prior and coordinates as NPE — the reference for what the data contain.

2.5 Model extensions (X3, E1b follow-up, real data)

- **GW on an hour axis** (`simulator=galton_watson_hours`; X3): GW with a fixed generation length τ (`model.hours_per_generation`); an endpoint of h hours is simulated as $g = \mathrm{round}(h/\tau)$ generations, so designs and summaries stay in hours and a GW model can be fitted to BH data. τ is set, not inferred.
- **All seeded organoids and survivor summaries** (`observation=all_clones`, `summary=survivor_moments`; E1b follow-up): extinct organoids are recorded as (0, 0) rows; `survivor_moments` computes every `moments` statistic on the non-empty organoids only and adds the extinct fraction.
- **Organoid-level heterogeneity** (real data; `law=two_type_hetero`, law key `clone_heterogeneity`): each organoid draws its own outcome rows, $p_k^{\mathrm{org}} \sim m_k\,\mathrm{Dirichlet}(\kappa\, p_k/m_k)$ over the free entries of row k (mass m_k), independently per organoid and type; all cells of the organoid share them. $E[p^{\mathrm{org}}] = p$ and $\mathrm{Var}(p_{k,o}^{\mathrm{org}}) = p_{k,o}(m_k - p_{k,o})/(\kappa+1)$: small κ = strong heterogeneity, $\kappa \to \infty$ = the homogeneous model. One extra parameter `clone.kappa`, log-uniform prior on $[1, 1000]$. In the "fate-only" variant only the three divisions vary between organoids, while death, persistence and switching stay shared (option `outcomes=[div_AA,div_AB,div_BB]`).
- **Density-dependent lifetimes** (`simulator=bellman_harris_dd`; real data): a cell born when its organoid has N living cells lives Gamma(shape, $\tau_k(1+N/K)$/shape) hours, so the cycle lengthens as the organoid grows (decelerating growth, tighter sizes). N counts the living cells right after the birth; a lifetime is fixed at birth. The carrying capacity K (`density.capacity`) is shared by all organoids, log-uniform prior on $[10, 10^5]$; $K \to \infty$ is plain BH. τ_k is then the lifetime of a cell in a small organoid.
- **Measured-data fits** (`bio-sbi-infer <run> <data.csv> --ppc 500`): a trained network is conditioned on measured counts (one row per organoid: horizon, A, B); it reports posterior support mass and a PPC on the 48 summary features. Without a truth there is no error or coverage: identifiability is judged by the trained network's own synthetic recovery check.

2.6 Pipeline and experiment logic



Experiment logic: vary one error on the data side (or one assumption on the inference side), keep everything else at the baseline, and compare with the *well-specified control of the same sweep* (same seeds, same datasets; often the same trained network, so differences are paired by seed).

2.7 Diagnostics and how to read them

Metric	Definition	How to read
Posterior mean / error	mean of 2 000 posterior samples; |mean − truth|. MAE over probabilities only (`mae_probabilities`); lifetimes separately	compare misspecified vs control; "shift" = difference of posterior means, paired by seed, ± paired SE over 5 seeds
Contraction	$1 - \mathrm{Var}_{\mathrm{post}}/\mathrm{Var}_{\mathrm{prior}}$	0 = data uninformative; < 0 = wider than prior; > 0.5 = informative
90% coverage across seeds	number of seeds (of 5) whose 90% credible interval contains the truth	4–5/5 expected; 0/5 with high contraction = confident bias
SBC coverage error	mean |empirical − nominal| coverage of central intervals (50–95%) over 200 prior draws	≤ 0.03 = calibrated. Averages over the prior: blind to tail truths and to misspecification
PPC fraction outside 95%	share of summary features whose observed value lies outside the 95% posterior predictive interval	well specified: 0–0.03. Higher = the model cannot reproduce the data. Relative check
Posterior support mass	share of the NPE posterior inside the prior support	well specified ≈ 0.5–0.9 (not 1). ≈ 0 = model cannot explain the data; few/no samples, not an estimate
Recovered (TODO criterion)	contraction > 0.5, |error| < 0.05, coverage ≥ 4/5	used for "organoids needed" (E4)

Reading rules used throughout. (1) *BH baseline bias*: well-specified BH runs give A→AA ≈ 0.32 and $\tau_A \approx$ 16 h (truth 0.55, 20 h); BH misspecified arms are compared with their control, never with the truth. (2) With all transitions free, B's row is not identifiable from endpoint counts at feasible designs (E0; practically, not structurally: the exact likelihood recovers it at 4096 organoids per endpoint, where NPE does not, §9; §8). (3) A misspecification that raises neither PPC nor lowers support mass is *undetectable from the fit*. (4) 5 seeds per setting: effects below ≈ 2 paired SE are not resolved. Each seed changes both dataset and network; training noise is ≈ 20% of the posterior sd and about half of the seed-to-seed spread (X2, §3). (5) Wall times in run metrics were measured under heavy contention; only the device benchmark (§A) and P0 (§A.1) measure speed.

3 Methodological check: X2 — seed spread, dataset vs training (GW and BH)

Why this next. Placed before Part I although it was run last: it does not answer a biological question but tells how to read every comparison that follows. All of Parts I–IV rest on 5 seeds, each changing both the dataset and the trained network (reading rule 4), and single well-specified runs already vary widely in support mass and PPC (§11). It uses the configuration that Parts I–III arrive at (B divisions fixed, real-like design).

Question

Every experiment varies one `seed` per run, which changes both the reference dataset (`data_seed` = `seed`) and the network training and sampling. How much of the run-to-run spread of posterior means comes from the dataset and how much from training? Is training noise small compared with the posterior width?

Setup

Configuration	the final configuration: `reversible_ab` (B divisions fixed), real-like design (GW generations 6/10; BH 144/240 h; 165/101 clones), perfect observation, $\theta^\star$, NPE 5k (`budget=dev`), SBC 200, PPC 200; both simulators
Arms	*training*: `seed` 0–9 with `data_seed` 0 (one dataset, 10 networks); *data*: `data_seed` 0–9 with `seed` 0 (10 datasets, one network)
Seeds / runs	10 per arm and simulator (40); 10 rather than the planned 5, because an sd from 5 values is too noisy
Measures	post. sd = mean posterior sd; sd train, sd data = sd of the posterior mean over the 10 runs of each arm; training share = sd train^2 / sd data^2 (capped at 1). From 10 runs an sd ratio is uncertain by roughly ±25%
Folder	`results/x2/gw_training/`, `gw_data/`, `bh_training/`, `bh_data/` (preset `x2_seeds`, commit `503ec72`); `results/x2/x2_spread.csv` (`scripts/analyse_x2.py`); table `report/tables/x2.csv`

Result

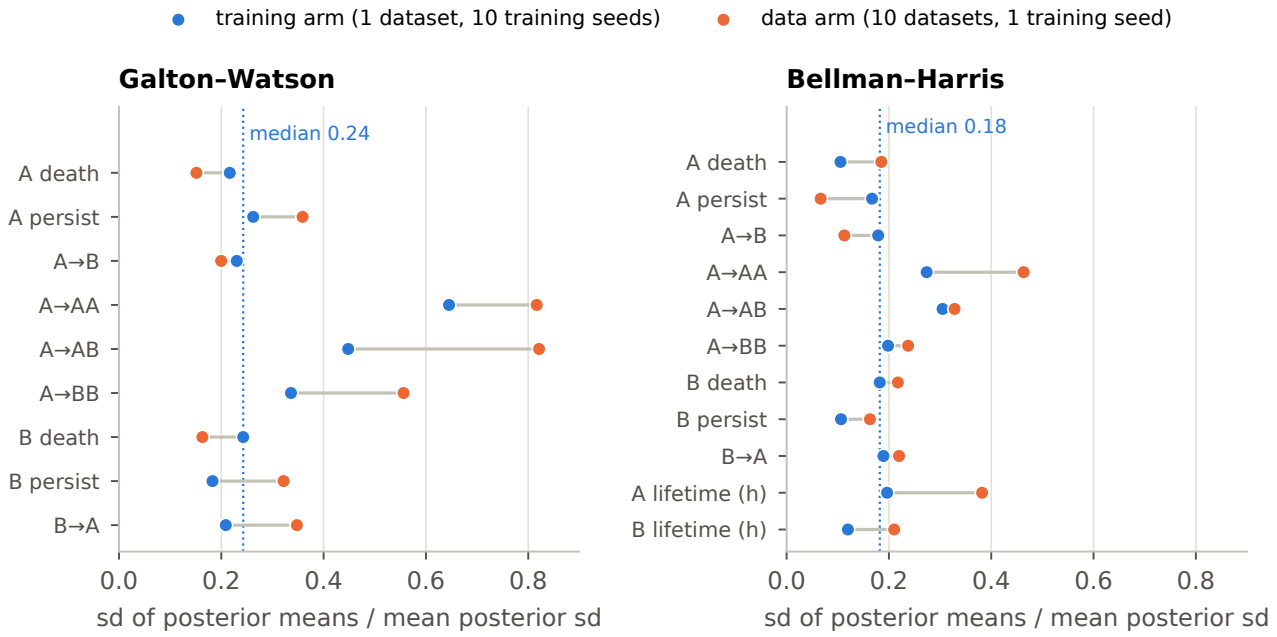


Figure 2: X2: sd of the posterior means over 10 runs, divided by the mean posterior sd, per parameter. Blue: training arm (one dataset, 10 training seeds); orange: data arm (10 datasets, one training seed; includes that network's training noise). Dotted: median of the training arm.

	Galton–Watson			Bellman–Harris		
Parameter	post. sd	sd train	sd data	post. sd	sd train	sd data
A death	0.019	0.004	0.003	0.047	0.005	0.009
A persist	0.034	0.009	0.012	0.089	0.015	0.006
A→B	0.025	0.006	0.005	0.049	0.009	0.006
A→AA	0.032	0.021	0.027	0.088	0.024	0.041
A→AB	0.047	0.021	0.038	0.107	0.033	0.035
A→BB	0.035	0.012	0.019	0.087	0.017	0.021
B death	0.084	0.020	0.014	0.137	0.025	0.030
B→A	0.123	0.026	0.043	0.194	0.037	0.043
B persist	0.106	0.019	0.034	0.178	0.019	0.029
τ_A (h)	—	—	—	1.88	0.37	0.72
τ_B (h)	—	—	—	11.5	1.38	2.42
Median sd train / post. sd		0.24			0.18	
Median training share		0.53			0.70	

Comment

- **Training noise is small relative to the posterior width**: ≈20% of the posterior sd (its variance ≈4% of the posterior variance). One trained network gives a reliable posterior, and differences between arms larger than ≈0.3 posterior sd are not training artefacts. Exception: GW A→AA, whose posterior is narrow (sd 0.032), so training noise is 65% of it.
- **At this design the dataset spread is not much larger than the training spread** (training ≈ half of the between-run variance in GW, ≈70% in BH): the seed-to-seed sd reported in the other experiments

is roughly half training noise, half data noise.

- **Posterior sds are wider than the spread of posterior means across datasets** (GW B→A 0.123 vs 0.043), consistent with coverage at or near 1 for most parameters: at the real-like design the posteriors are conservative (prior-shrunk), and the seed-to-seed sd is not a measure of data uncertainty — the posterior sd is.
- **The BH ridge bias is not noise**: the means agree between arms (training vs data arm: A→AA 0.38 vs 0.36, covered in 0/10 and 1/10 runs; τ_A 16.4 vs 16.5 h), i.e. the same for every dataset and every training seed.
- Consequence: keep 5 seeds (varying both) for averaging; quote the posterior sd as uncertainty and the seed sd only as a stability check. A larger simulation budget mainly reduces the training part (§12.1).
- Caveats: one configuration (real-like, `reversible_ab`, NPE 5k); the data arm includes the training noise of seed 0, so the data spread is not "pure"; SBC and PPC also depend on `seed` and were not analysed here.

Leads to. Differences between arms are read against the posterior sd, not the seed sd. The second check (X1, §4) asks whether the method itself shapes the results.

4 Methodological check: X1 — inference methods: NPE, sequential NPE, NLE, NRE (GW and BH)

Why this next. Placed next to X2, before Part I, although it was run last and uses settings from later parts: every result in Parts I–V comes from amortised NPE, so the reader should know before the first result whether its failures — B's row (Part I), the BH ridge (Part III), silent and flagged errors (Parts III, IV) — could be artefacts of that one method. Like X2 it answers no biological question; it decides how far the conclusions can be trusted and which method the real-data fits use. Its five settings are the decisive ones of E0 (§5), the ridge (§13), E3a (§16) and X5b (§15); they are described here in enough detail to be read first.

Question

Every other experiment used amortised NPE (SNPE only in E0). Are the identifiability findings — B's row with B divisions free (§5), the BH lifetime/self-renewal ridge (§13) — and the flagged/silent pattern of misspecification — too-short fixed lifetimes silent (E3a, §16), a too-long lifetime prior flagged (X5b, §15) — properties of the data or of NPE? Which method should a real-data fit use?

Setup

Settings	five decisive settings, synthetic data from $\theta^\star$: *E0 free* (GW `two_type`, all transitions free) and *E0 fixed* (GW `reversible_ab`, B divisions fixed), both at generations 4/8/12 × 32 clones; *BH ridge* (BH `reversible_ab`, real-like design 144/240 h × 165/101, vague lifetime prior, preset `x5b_lifetime_prior`); *silent* (BH, lifetimes fixed to 18.75 h, 25% below the pooled mean; truth A 20 h, B 30 h; preset `e3a_lifetime_fixed`); *loud* (BH, lifetime prior 50% too long, truth outside it)
Methods	NPE (amortised, as in every other experiment); SNPE (NPE-C, 4 rounds, no SBC); NLE and NRE-C (MCMC with slice sampling; SBC on 50 cases)
Budget	`budget=dev`: 5k simulations for every method
Key parameter	B→A (truth 0.80) in the E0 settings; A→AA (truth 0.55) in the BH settings
Seeds / runs	5 per setting and method: 100 planned, **94 finished**; 6 SNPE runs never finish (loud: 5/5; silent: seed 3), stopped after 45 min
Run time	minutes per run with the four method lanes running in parallel: a relative comparison, not idle-machine timings (cf. P0, §A.1)
Folder	`results/x1/<setting>/<method>/` (commit `fa3e96a`; four NPE/NRE E0-free seeds rerun at `a64d17e`); `x1_groups.csv`, `x1_runs.csv` (`scripts/analyse_x1.py`); `snpe_failures.md`; table `report/tables/x1.csv`

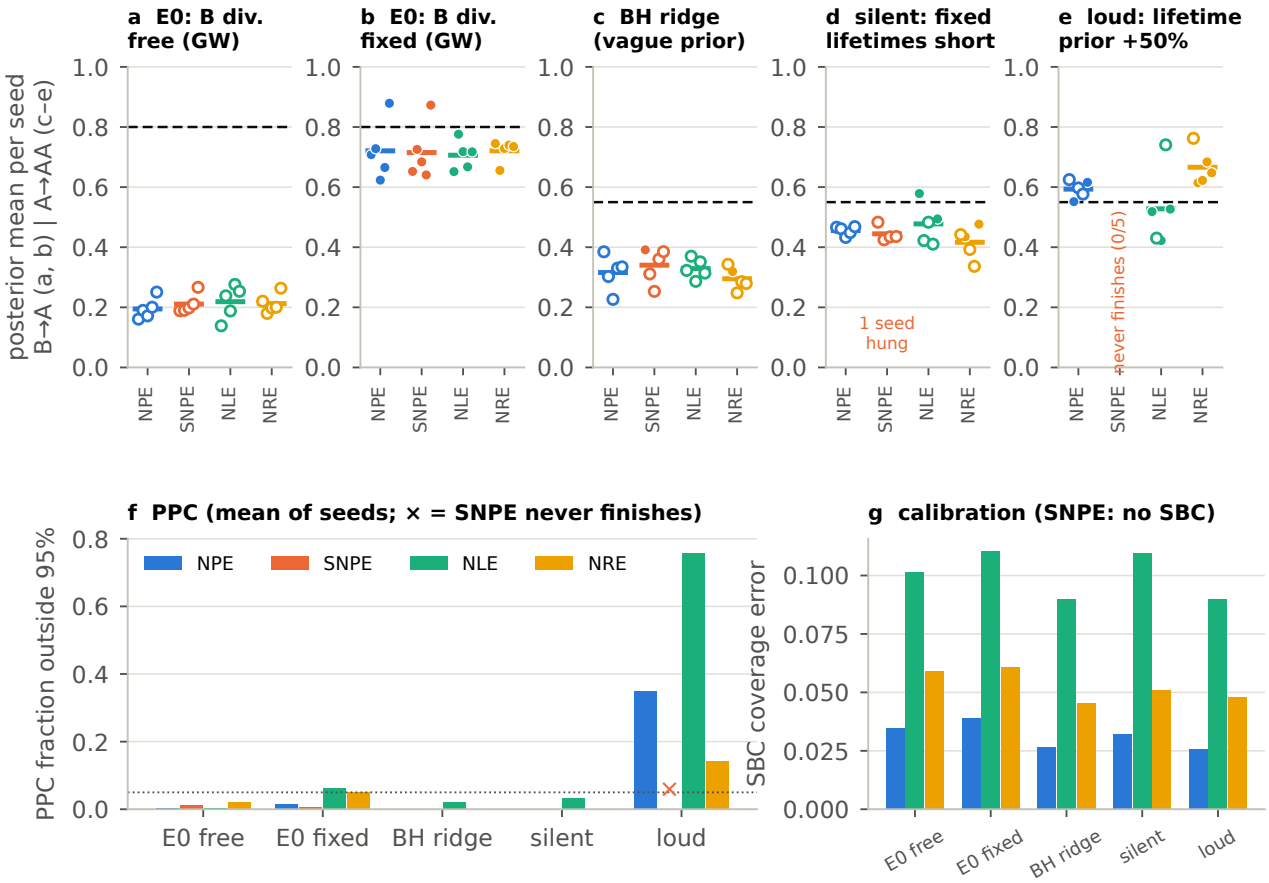


Figure 3: X1: four inference methods on five settings. (a–e) Posterior mean of the key parameter per seed (filled = truth inside the 90% interval, open = not; bar = mean over seeds); dashed = truth. SNPE never finishes in the loud setting and on one silent seed. (f) PPC fraction of summary features outside the 95% predictive interval (misspecification signal; dotted = 0.05); × = SNPE never finishes. (g) SBC coverage error (SNPE has no SBC).

Setting	Method	MAE	90% cov.	contr.	PPC	SBC	key parameter [covered]	min/run
E0 free	NPE	0.144	0.76	0.50	0.003	0.035	B→A 0.19 [0/5]	1.4
	SNPE	0.139	0.73	0.47	0.011	—	0.21 [0/5]	12.7
	NLE	0.145	0.80	0.53	0.003	0.102	0.22 [0/5]	43.8
	NRE	0.136	0.80	0.41	0.019	0.059	0.21 [0/5]	3.0
E0 fixed	NPE	0.048	0.93	0.85	0.014	0.039	B→A 0.72 [5/5]	1.6
	SNPE	0.052	0.93	0.83	0.006	—	0.72 [5/5]	7.7
	NLE	0.059	0.89	0.80	0.061	0.110	0.71 [5/5]	35.8
	NRE	0.060	0.89	0.75	0.050	0.061	0.72 [5/5]	3.3
BH ridge	NPE	0.116	0.75	0.63	0.000	0.026	A→AA 0.32 [0/5]	3.2
	SNPE	0.075	0.85	0.71	0.000	—	0.34 [1/5]	11.8
	NLE	0.085	0.87	0.67	0.021	0.090	0.33 [0/5]	30.2
	NRE	0.112	0.80	0.54	0.000	0.045	0.30 [1/5]	4.3
silent (lifetimes fixed 25% short)	NPE	0.075	0.76	0.84	0.000	0.032	A→AA 0.46 [0/5]	2.0
	SNPE (4/5)	0.079	0.81	0.80	0.000	—	0.44 [0/4]	9.1
	NLE	0.078	0.78	0.83	0.033	0.110	0.48 [2/5]	18.9
	NRE	0.068	0.91	0.73	0.000	0.051	0.42 [2/5]	2.9
loud (lifetime prior +50%)	NPE	0.077	0.38	0.81	**0.350**	0.025	A→AA 0.59 [2/5]	2.3
	SNPE	**never finishes** (0/5 runs; proposal rejection accepts nothing)						
	NLE	0.129	0.55	0.78	**0.758**	0.090	0.53 [3/5]	19.8
	NRE	0.072	0.71	0.72	**0.142**	0.048	0.67 [4/5]	4.2

Means over seeds. NPE support mass in the loud setting 0.017 (0.64–0.82 in the other settings). SNPE "support" (0.03–0.09 even when well specified) is not a misspecification signal: later rounds train on a truncated proposal. Numbers: `results/x1/info.md`, `x1_groups.csv`.

Comment

- **The identifiability findings are method-independent.** With B divisions free every method leaves B→A at 0.19–0.22 (prior-dominated, 0/5 covered); with them fixed every method gets 0.71–0.72 (5/5). The BH ridge (A→AA 0.30–0.34 vs 0.55) appears for all four methods. At this design these are properties of the data, not of NPE: the exact GW likelihood agrees (B→A 0.27 free, 0.77 fixed, §9). At 4096 organoids per endpoint this no longer holds (§8).
- **So is the flagged/silent pattern.** Too-short fixed lifetimes stay silent for every method (PPC $\leq 3\%$) while biasing A→AA (0.42–0.48). The wrong lifetime prior is flagged by every method that finishes — most strongly by NLE (76% of PPC features outside), weakest by NRE (14%), which also stays best calibrated there (coverage 0.71 vs 0.38 for NPE).
- **SNPE has no misspecification signal of its own: it hangs.** Between rounds the next proposal is drawn from the current posterior by rejection inside the prior support; when the posterior lies outside the prior (the loud setting, and one silent seed) acceptance is 0% and the run never returns (log: `snpe_failures.md` in `results/x1/`). Where it finishes it helps a little on the ridge (MAE 0.075 vs 0.116) at 4–9× the cost of NPE.
- **NPE is the best calibrated and the fastest**: SBC error 0.025–0.039 vs NRE 0.045–0.061 and NLE 0.090–0.110; 1.4–3.2 min per run vs NRE 2.9–4.3 (1.4–2×) and NLE 19–44 (9–31×). NLE and NRE are less well calibrated at 5k simulations. At these single truths NPE's run coverage is not higher than NRE's (0.75–0.76 vs 0.80 in E0 free, BH ridge, silent): the truths sit in prior tails (X5d, §14).
- **Recommendation**: keep amortised NPE as the default (fast, best calibrated, with a built-in support-mass check); confirm key real-data conclusions with NRE, a different approximation at about twice the cost; use SNPE only on models that have passed NPE's support and PPC checks.
- Caveats: `budget=dev` (5k simulations) for every method; NLE/NRE SBC on 50 cases only; run times are relative (four lanes in parallel).

Leads to. The findings below belong to the data, not to the method, and amortised NPE is used throughout. With both reading keys in place, Part I starts from the first fits (E0, §5).

Part I

Identifiability and the role of the prior

Motivation. Every later question (how many organoids, which errors matter) presupposes that the parameters can be learned from endpoint counts at all. The very first fits failed on one parameter, B→A. This part establishes why: whether the prior or the model is at fault decides what the remedy is (a different prior, more data, or a biological constraint). It ends with a test at 4096 organoids per endpoint with a well-trained network, which NPE fails, and with the exact likelihood on the same data, which shows that this last failure was NPE's: the B row is practically, not structurally, non-identifiable. Feasible designs cannot decide it, so all later experiments adopt the constraint B divisions = 0 (`reversible_ab`).

5 E0 — Identifiability of B's row (GW)

Why this next. This is the starting point. With the general two-type law (all 12 outcomes free) the first well-specified fits returned B→A ≈ 0.2 for a truth of 0.8. Nothing downstream can be interpreted until this is understood.

Question

With all transitions free, B→A (true 0.80) came out at ≈0.2. Is this under-training, one bad dataset, miscalibration, a prior-tail effect, or non-identifiability (switching vs division)?

Setup

Simulator	Galton–Watson, perfect observation
Law (free)	`two_type` (10) vs `reversible_ab` (7; B divisions fixed to their true 0)
Data vs inference	well specified (same $\theta^\star$, Table 2)
Design	generations 4, 8, 12 × 32 clones (96 organoids)
Inference / budget	NPE amortised 5k, 20k, 50k; SNPE 4 rounds × 5k; `two_type` with flat prior on B→A (Beta(1,1)), 20k
Seeds / runs	5 / 40 (+ direct checks: C2ST of mean-preserving alternatives, 500 datasets per side)
Varied	law, budget, method, prior
Folder	`results/b_row/` (`distinguish.csv`, `truth_vs_posterior_20k*.csv`)

Result

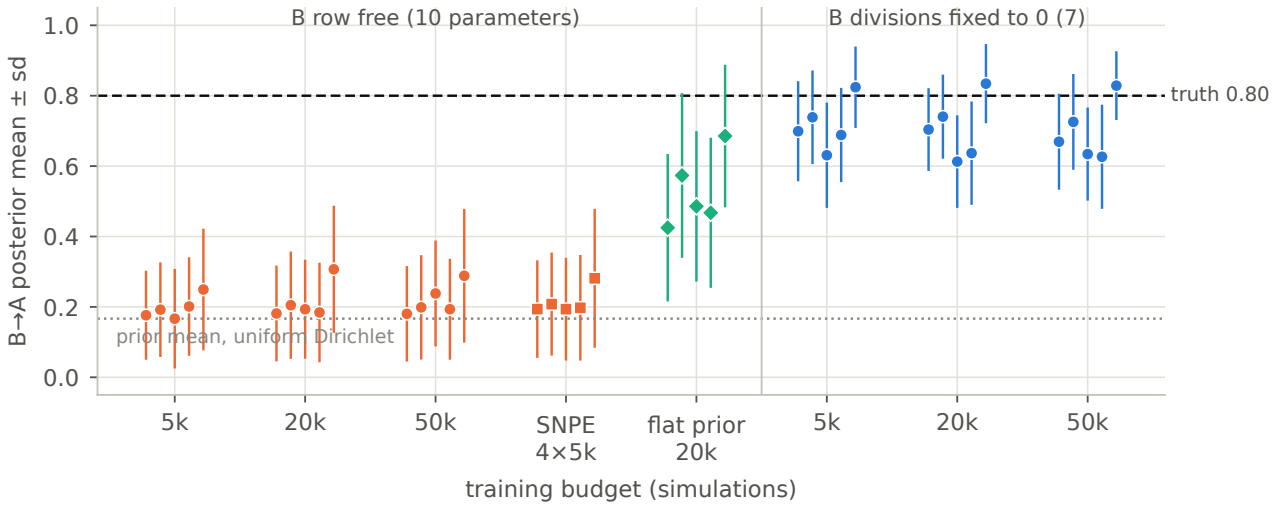


Figure 4: B→A posterior mean ± sd per seed. Orange: all B outcomes free (uniform Dirichlet; squares: SNPE). Green: flat prior on B→A. Blue: B divisions fixed to 0.

Law	Training	B→A mean	post. sd	contraction	covered	MAE (all)
`two_type`	5k / 20k / 50k	0.20 / 0.21 / 0.22	0.14–0.15	−0.03 / −0.13 / −0.18	0/15	0.14
`two_type`	SNPE 4×5k	0.21	0.16	−0.22	0/5	0.14
`two_type`, flat prior	20k	0.53	0.21	0.46	4/5	0.10
`reversible_ab`	5k / 20k / 50k	0.72 / 0.71 / 0.70	0.13–0.14	0.67 / 0.71 / 0.69	15/15	0.05

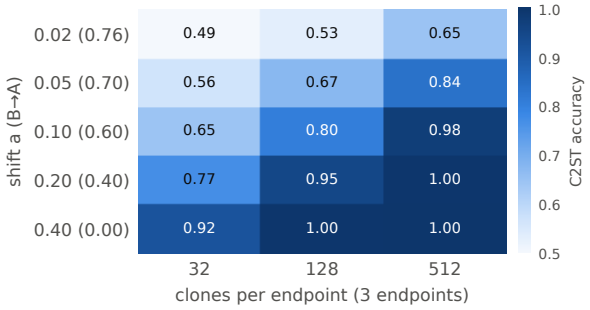


Figure 5: C2ST accuracy separating the truth from a mean-preserving alternative (B→AA = a, B→A = $0.8 - 2a$, B death = $0.1 + a$). 0.5 = indistinguishable.

- SBC coverage error 0.02–0.04 in every arm (looks calibrated).
- PPC ≤ 0.014 in every arm.
- Flat prior: corr(B→A, B→AA) = −0.67; SBC fails (0.069) for B's other coordinates (concentration 0.2), not for B→A.
- Truth-vs-posterior C2ST: inconclusive (0.62–0.95 in both laws).

Comment

- Under-training, dataset and calibration are ruled out: 10× more simulations, 5 datasets, SBC all unchanged.
- Main cause = prior tail: B→A ∼ Beta(1,5), P(> 0.8) = 0.03% (≈15 of 50k training simulations). SNPE cannot help: round 0 is already prior-like. *Revised by X5 (§6, §7) and tested at 4096 clones (§8)*: with B divisions free B→A is not recovered at any truth ≥ 0.5 (bulk included), nor with 1024 organoids, nor with 4096 per endpoint on three endpoints and a 20k network; the confounding below is the root cause and the tail only makes it visible. *Revised again by the exact likelihood (§9)*: the confounding is practical, not structural — on these datasets the exact posterior is as prior-dominated as NPE's (B→A 0.27 ± 0.17; Cramér–Rao sd 0.27, more than the prior sd), but at 4096 clones per endpoint the data decide it (0.78 ± 0.02).
- Data weakly confound switching with division: even with a flat prior the posterior is wide (sd 0.21) and anti-correlated with B→AA. They differ only in clone-size variability; in a direct test 128–512 clones/endpoint make moderate differences detectable, NPE's posterior does not separate them even at 4096, the exact posterior does (§8, §9).
- Consequence: fix what biology allows (B divisions = 0 ⇒ B→A ≈ 0.70, contraction 0.7), or measure it; a free B row cannot be rescued by realistic amounts of data. All later experiments use `reversible_ab`.

Leads to. Two explanations remain open: the prior tail (B→A = 0.8 has 0.03% prior mass) and confounding of switching with division (the flat-prior posterior is wide and anti-correlated with B→AA). X5a (§6) separates them by moving the truth through the prior; X5c (§7) asks whether more organoids, which the C2ST suggests would help, can overrule the prior.

6 X5a — Role of the prior I: truth position × prior shape (GW)

Why this next. E0 (§5) attributed the B-row failure mainly to the prior tail, with weak confounding as a second cause. If the tail were the cause, a truth in the prior bulk or a flatter prior should be recovered; if the confounding were the cause, no truth position or prior should help unless B divisions are fixed.

About X5. X5 ("role of the prior") is one experiment in four parts, placed here where each is motivated: X5a (this section), X5c (§7), X5d (§14), X5b (§15). Common to all: synthetic data, real-like design (GW generations 6/10 or BH 144/240 h; 165/101 clones unless swept), perfect observation, one A founder, NPE 5k (`budget=dev`), SBC 200, PPC 200; 355 runs in total (X5a 250, X5b 15, X5c 30, X5d 60). Folder `results/x5/` (overview `info.md`, per-part tables `x5*_findings.md`).

Question

Does recovery of B→A depend on where the truth sits in the prior (0.1–0.9) and on the prior's shape, with B divisions free vs fixed? Is E0's failure a tail-truth effect?

Setup

Data model	GW; B→A $\in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, B death = B persist = $(1 - \text{B} \to \text{A})/2$; other parameters $\theta^\star$
Laws (data = inference)	`two_type` (B divisions free, 10 free) and `reversible_ab` (B divisions fixed to 0, 7 free). Prior P(B→A > 0.9): 10^{-5} resp. 0.01
Priors (inference only)	uniform Dirichlet; Jeffreys (all concentrations 0.5); flat marginal on B→A (Beta(1,1): concentration 0.2 on B's 5 other outcomes in `two_type`, 0.5 on death/persist in `reversible_ab`). *Biological* (Dirichlet pseudo-counts per type, each row sums to 1; total ≈5–6 = weakly informative): **bio-B** = B row only, death/persist/to_A 1.5 and B divisions 0.2 each (`two_type`; fixed in `reversible_ab`), A row uniform; **bio** = bio-B plus A row death/persist/to_B 0.5, div_AA 2, div_AB 1, div_BB 1.5. Rationale: encode the biology without the answer — A (stem-like) mostly divides, self-renewal most likely; B (differentiated) rarely divides but may die, persist or switch back. bio-B vs bio separates B's prior from A's
Seeds / runs	5 / 125 per law (250: 150 reference + 100 biological)
Folder	`results/x5/a_two_type/`, `results/x5/a_reversible_ab/`, biological priors in `a_two_type_bio/`, `a_reversible_ab_bio/` (pooled `x5a_groups.csv`, `x5a_runs.csv` in the first two)

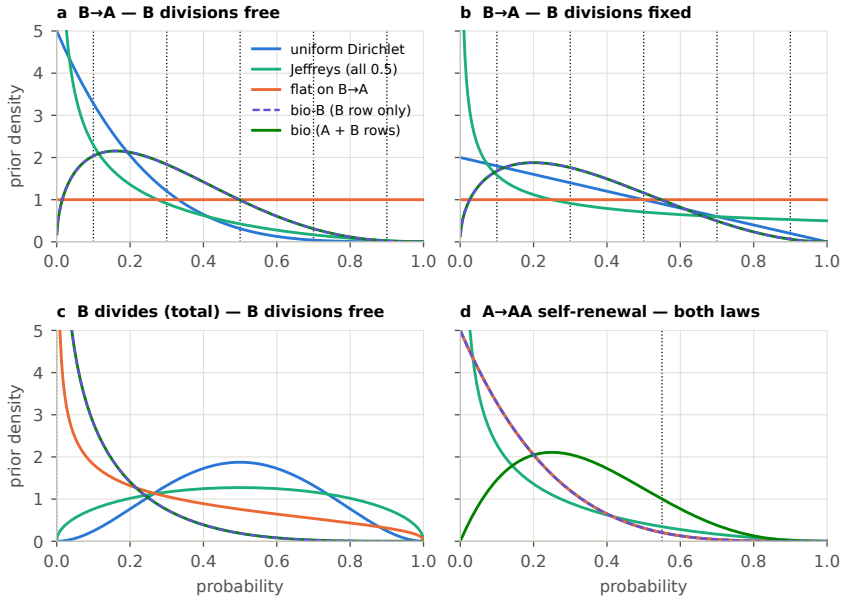


Figure 6: The five X5a priors compared (inference model; exact Beta marginals of the Dirichlet priors). (a, b) B→A with B divisions free / fixed; (c) total probability that B divides (B→AA + AB + BB), B divisions free; (d) A→AA, identical for both laws. Curves that coincide are drawn on top of each other: bio-B (dashed) = bio in (a–c); uniform = flat B→A = bio-B in (d). Dotted: truths used in X5a (B→A 0.1–0.9; B divisions 0; A→AA 0.55). Densities clipped at 5. Beta parameters, quantiles and tail probabilities: `tables/x5a_priors.csv`.

Result

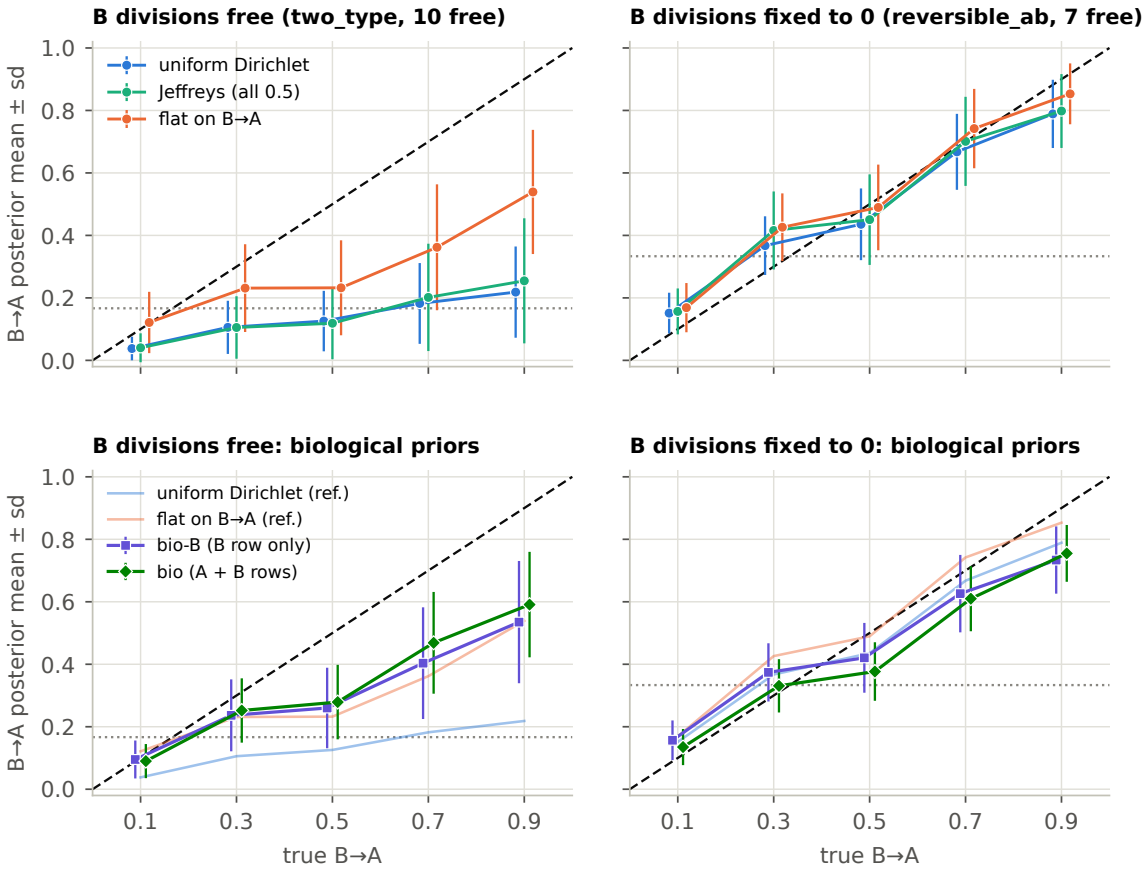


Figure 7: X5a: B→A posterior mean (mean over 5 seeds) ± mean posterior sd against the true B→A, one line per prior (slightly offset). Top: reference priors; bottom: biological priors bio-B and bio, with the uniform and flat-B→A means as thin reference lines. Dashed: identity (perfect recovery). Dotted: uniform-prior mean (1/6 left, 1/3 right).

	B divisions free (`two_type`)			B divisions fixed (`reversible_ab`)		
Truth	uniform	Jeffreys	flat B→A	uniform	Jeffreys	flat B→A
0.1	0.04 (3/5)	0.04 (5/5)	0.12 (5/5)	0.15 (5/5)	0.16 (4/5)	0.17 (5/5)
0.3	0.11 (1/5)	0.11 (2/5)	0.23 (5/5)	0.37 (5/5)	0.42 (5/5)	0.43 (4/5)
0.5	0.13 (0/5)	0.12 (0/5)	0.23 (2/5)	0.44 (5/5)	0.45 (5/5)	0.49 (5/5)
0.7	0.18 (0/5)	0.20 (0/5)	0.36 (3/5)	0.67 (5/5)	0.70 (5/5)	0.74 (5/5)
0.9	0.22 (0/5)	0.26 (0/5)	0.54 (1/5)	0.79 (5/5)	0.80 (5/5)	0.85 (5/5)
SBC error	0.037	0.054	0.080	0.032	0.056	0.037

	B divisions free (`two_type`)		B divisions fixed (`reversible_ab`)	
Truth	bio-B	bio (A+B)	bio-B	bio (A+B)
0.1	0.10 (5/5)	0.09 (5/5)	0.16 (4/5)	0.14 (5/5)
0.3	0.24 (5/5)	0.25 (5/5)	0.37 (4/5)	0.33 (5/5)
0.5	0.26 (1/5)	0.28 (1/5)	0.42 (5/5)	0.38 (4/5)
0.7	0.40 (2/5)	0.47 (3/5)	0.63 (5/5)	0.61 (4/5)
0.9	0.54 (1/5)	0.59 (1/5)	0.73 (1/5)	0.75 (3/5)
run MAE	0.05–0.09	0.03–0.07	0.03–0.06	0.03–0.05
SBC error	0.052	0.055	0.033	0.041
B-division mass	0.21	0.11	—	—

B→A posterior mean (seeds covered by the 90% interval). Run MAE = range over truths of the mean per-run MAE (all free parameters); B-division mass = posterior total of B→AA/AB/BB (truth 0; 0.54 under the uniform and flat priors). Reference priors, run MAE: free 0.12–0.15 (uniform), 0.10–0.16 (flat); fixed 0.02–0.06 (uniform). Numbers: `tables/x5a.csv`.

Comment

- **B divisions free: the estimate hardly follows the truth** (0.04 → 0.22 as the truth goes 0.1 → 0.9, uniform prior); no truth ≥ 0.5 is covered under the uniform or Jeffreys prior, and B→A contraction falls to ≤ 0 at 0.9. At this design the data cannot separate switching from division, so the prior sets the answer — at *every* truth position, not only in the tail.
- A flat prior on B→A tracks better (0.12 → 0.54) but still shrinks heavily and costs calibration (SBC 0.080) and coverage of the other parameters (run coverage 0.62–0.87).
- **B divisions fixed: every truth is recovered under every prior** (B→A contraction 0.73–0.94; the three priors differ by ≤ 0.06, mild shrinkage at the extremes). The prior only matters where the data cannot separate switching from division.
- Conclusion for E0 (§5): the B-row failure is the switching/division confounding, not a bad truth position; a prior cannot substitute for the constraint. The confounding is practical (Cramér–Rao sd of B→A 0.28 at this design, more than the prior sd): the exact likelihood is as prior-dominated here (B→A 0.25 ± 0.16 at truth 0.8) but recovers B→A at 4096 organoids per endpoint, where NPE did not (§9, §8).
- **Biological priors.** *B divisions free*: the soft "B rarely divides" prior cuts the posterior B-division mass from 0.54 to 0.21 (bio-B) and 0.11 (bio); B→A tracks like the flat prior (0.54–0.59 at truth 0.9) but with better calibration (SBC 0.05 vs 0.08) and the lowest error of any free-B prior (run MAE 0.03–0.07 for bio; the A-row prior gives most of this gain). It is still far from the hard constraint (0.79, 5/5 at truth 0.9): truths ≥ 0.5 are mostly not covered. *B divisions fixed*: the symmetric 1.5 prior on B's three outcomes (mean 1/3) shrinks B→A toward 1/3, and truth 0.9 is covered in only 1/5 (bio-B) and 3/5 (bio) seeds; the A-row prior lowers the error and raises support mass (0.79–0.89 vs 0.64–0.79 uniform). A soft biological prior helps but does not replace the constraint; with the constraint, keep B's prior near-uniform.

Leads to. The constraint, not the prior, decides. Remaining question: is it only a matter of sample size? (X5c, §7).

7 X5c — Role of the prior II: data vs prior at a tail truth (GW)

Why this next. X5a (§6) showed the confounding at the real-like size (165/101 clones). E0's C2ST (§5) predicted that 128–512 clones per endpoint make moderate switching/division differences detectable, so a large enough experiment might overrule the prior without a constraint.

Question

With all transitions free and a tail truth (B→A = 0.9, prior P = 10^{-5} under Beta(1,5)), how many organoids overrule the prior?

Setup

Data model	GW, `two_type`; B: death 0.05, persist 0.05, B→A 0.9 (B divisions 0); A row $\theta^\star$
Inference	`two_type` (10 free), uniform Dirichlet vs flat marginal on B→A (concentration 0.2 on B's other outcomes)
Design	generations 6/10 × {32, 128, 512} clones per endpoint
Seeds / runs	5 / 30
Folder	`results/x5/c_data_vs_prior/` (`x5c_groups.csv`, `x5c_runs.csv`)

Result

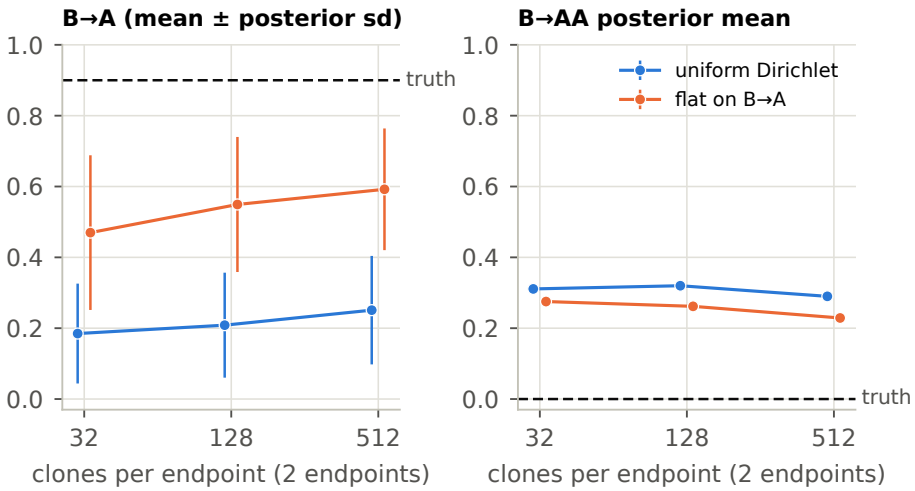


Figure 8: X5c: B→A (mean over 5 seeds ± mean posterior sd) and B→AA posterior mean against clones per endpoint; dashed = truth.

Prior	clones	B→A (0.9)	post. sd	cov.	contr.	B→AA (0)	MAE	support	SBC
uniform	32	0.19	0.14	0/5	0.01	0.31	0.161	0.79	0.031
	128	0.21	0.15	0/5	−0.10	0.32	0.151	0.72	0.037
	512	0.25	0.15	0/5	−0.17	0.29	0.139	0.55	0.044
flat B→A	32	0.47	0.22	2/5	0.44	0.28	0.119	0.49	0.078
	128	0.55	0.19	1/5	0.57	0.26	0.101	0.42	0.081
	512	0.59	0.17	1/5	0.65	0.23	0.086	0.46	0.083

PPC ≤ 0.03 in every group. Numbers: `tables/x5c.csv`.

Comment

- **1024 organoids do not overrule the uniform prior**: B→A moves 0.19 → 0.25 (truth 0.9), never covered, contraction ≤ 0; the mass goes to B→AA (≈ 0.3, truth 0) — the switching/division confounding of E0.
- A flat prior helps and improves with data (0.47 → 0.59, contraction up to 0.65) but is still far from 0.9 at 1024 organoids and costs calibration (SBC 0.08).
- Endpoint counts of this kind cannot resolve the confounding by sample size at realistic scales; the constraint (B divisions fixed, X5a) is what works. Here the limit is mainly the data's: 1024 organoids at {6,10} give only ≈ 1.4 nats of expected evidence against "B divides half the time" (note §5). Whether a far larger design with a better-trained network would succeed (a larger simulation budget does matter at very large designs, §12.1) is tested in §8 (NPE: no) and §9 (exact likelihood: yes).

Leads to. Does an experiment far beyond realistic sizes, with a better-trained network, close the loophole? (§8).

8 E0 revisited at 4096 clones: full transition model, 5k vs 20k simulations (GW and BH)

Why this next. X5a and X5c (§6, §7) found the B row confounded, but only up to 1024 organoids and with a 5k-simulation network, while E0's C2ST (§5) suggested larger designs could separate switching from division. E4 (Part II, §12.1) later showed that at very large designs a 5k network can be the limit. The remaining loophole: a far larger design with a better-trained network. Run after E4, placed here because it closes the identifiability chain E0 → X5a → X5c.

Question

With all 10 transitions free (E0's setting), does B's row become identifiable once neither the data (4096 clones on each of three endpoints, 12 288 organoids) nor the network (20k simulations, epoch cap 500) are limiting? E0's C2ST suggested that larger designs separate switching from division (data limit); X5a and X5c pointed to structural non-identifiability, but only up to 1024 organoids and with a 5k network.

Setup

Simulator / design	GW, generations {4,6,10}; BH, days {4,6,10} = {96,144,240} h; 4096 clones per endpoint; perfect observation
Law (data = inference)	`two_type` (10 free probabilities; BH also τ_A, τ_B), truth $\theta^\star$ (B divisions 0), uniform Dirichlet
Inference / budget	NPE amortised, 5k (epoch cap 200) vs 20k (`budget.simulations=20000 budget.max_epochs=500`)
Reference	B divisions fixed (`reversible_ab`), 20k, same truth, design and seeds (E4 budget check, §12.1)
Seeds / runs	5 / 20 (2 simulators × 2 budgets)
Folder	`results/e4/gw_xxl_two_type_5k/`, `gw_xxl_two_type_20k/`, `bh_xxl_two_type_5k/`, `bh_xxl_two_type_20k/` (commit `503ec72`; analysis in `results/e4/info.md`); table `report/tables/e0_large.csv`

Result

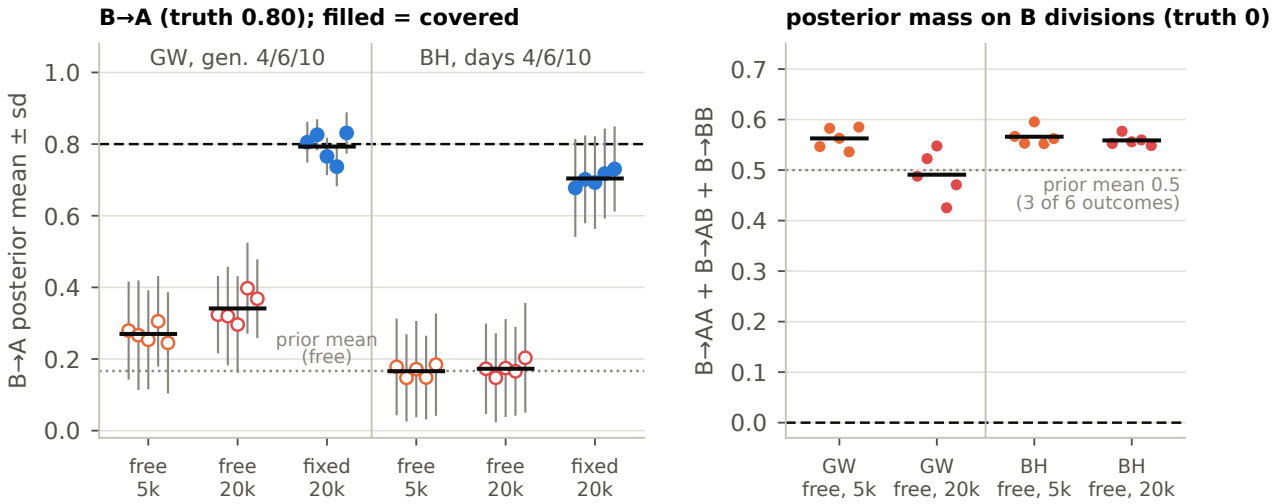


Figure 9: Full transition model at 4096 clones per endpoint, three endpoints. Left: B→A posterior mean ± sd per seed (filled = truth in the 90% interval; bars = mean over seeds); B divisions free (`two_type`, 5k and 20k simulations) vs fixed to 0 (`reversible_ab`, 20k). Right: posterior mass on B divisions (sum of the posterior means of B→AA, AB, BB) per seed; dotted = its prior mean 0.5. Dashed = truth.

	GW, B div. free		BH, B div. free		B div. fixed, 20k
	5k	20k	5k	20k	GW / BH
B→A (0.80): mean, covered	0.27, 0/5	0.34, 0/5	0.17, 0/5	0.17, 0/5	0.79, 5/5 / 0.70, 5/5
B→A contraction	0.04	0.24	0.15	0.12	0.95 / 0.71
Mass on B divisions (truth 0; prior 0.5)	0.56	0.49	0.57	0.56	0 (fixed)
of which B→AA	0.25	0.21	0.29	0.30	—
A persist (0.03)	0.17	0.16	0.12	0.13	0.05 / 0.07
A→BB (0.29)	0.20	0.21	0.23	0.23	0.28 / 0.32
A→AA (0.55)	0.45	0.47	0.34	0.36	0.55 / 0.49
MAE (free probabilities)	0.135	0.118	0.152	0.147	0.017 / 0.049
Coverage (run, 90%)	0.42	0.40	0.71	0.71	0.93 / 0.93
Support mass	0.32	0.53	0.75	0.67	0.78 / 0.51
PPC outside 95%	0.000	0.006	0.000	0.003	0.017 / 0.003
SBC coverage error	0.066	0.050	0.040	0.045	0.044 / 0.040
Epochs, mean	130	182	147	235	190 / 201

Means over 5 seeds. MAE over the free probabilities (10 with B divisions free, 7 fixed). B-division coverage is 0/5 by construction (truth on the boundary): read the mass.

Comment

- **Original reading (superseded): structural non-identifiability.** With B divisions free, NPE leaves B→A at 0.17–0.34 (truth 0.80, 0/20 runs covered, contraction ≤ 0.24) even with 4096 organoids per endpoint on three endpoints, and quadrupling the training budget changes it by ≤ 0.07 (GW) or not at all (BH). We concluded it was neither a data nor a network limit.
- **Corrected by the exact likelihood (§9): an NPE failure, not a data limit (GW).** On the same five GW datasets the exact posterior gives B→A 0.78 ± 0.02 (4/5 covered), B-division mass 0.04 and A persist 0.02, at the Cramér–Rao scale (sd 0.023); the exact likelihood rejects NPE's posterior means by 116–178 nats, and the moment summaries still separate the two answers (97% classifier accuracy): the information is in the data and in the summaries, the amortised network does not use it. The Fisher information is positive definite: the B row is *practically* non-identifiable at realistic sizes (CR sd 0.27–0.28 at 96–266 organoids, more than the prior sd) and identified at 4096 per endpoint. For BH the exact Fisher information is also nonsingular (§13.1); whether BH at 4096 is a data or an NPE limit is untested (scaling the real-like CR sd of 0.99 to 12 288 clones gives ≈ 0.15: still weak).
- **NPE's posterior does not say whether B divides**: its mass on B divisions stays at 0.49–0.57, its prior mean (0.5), with B→AA $\approx$ 0.2–0.3 for a truth of 0 (exact: 0.04 in total). Yet the PPC is clean ($\leq 0.6\%$): within the 95% bands the summaries are reproduced by "B switches back" and by "B divides, rarely switches". A clean fit is no evidence that the B row is right.
- **NPE's B-row error leaks into A's row in GW**: A persist 0.16 (truth 0.03, 0/5), A→BB 0.21 (0.29), run coverage 0.40; the exact posterior has A persist 0.02, A→BB 0.28 and MAE 0.011 on the same data.
- **Fixing B divisions recovers B→A on the same design** (0.79 GW / 0.70 BH, 5/5 covered). Fixing them — or measuring them, e.g. by live imaging of KRT20$^+$ cells or lineage tracing — remains the practical recommendation: realistic designs cannot make this decision from endpoint counts, and amortised NPE did not make it even at 4096 per endpoint.
- Side note (BH): with B divisions free, A's lifetime is recovered (19.4–19.6 h, 5/5) while A→AA is not (0.34–0.36); B's lifetime is not (17.6–19.2 h, truth 30 h). The ridge shifts to involve B's row; not interpreted further.
- Caveat: one truth, one design per simulator, 5 seeds.

Leads to. Is this a property of the data or of NPE? The exact GW likelihood answers it without simulations, on the same datasets (§9).

9 Exact likelihood (XG): is the B row a data limit or an NPE limit? (GW)

Why this next. E0 at 4096 (§8) was read as the end of the argument: if NPE with 12 288 organoids and a 20k network cannot find B→A, the model must be structurally non-identifiable. But every result of this part came from one approximation (amortised NPE on moment summaries), and X1 (§4) compared approximations only at realistic sizes. For Galton–Watson the likelihood of the raw counts is computable exactly, so "is the information in the data?" can be answered without any network, on the same datasets.

Question

Every result so far comes from amortised NPE on moment summaries. Are the failures to recover B's row (E0, X5a, X5c, E0 at 4096 clones) and the BH lifetime/self-renewal ridge properties of the data, or of the approximation? For Galton–Watson, and for Bellman–Harris with an integer lifetime shape, the law of a clone's counts can be computed exactly, so the posterior of the *raw counts* can be sampled without simulations or summaries — on the same datasets as the NPE runs.

Setup

Exact GW law	pgf of a clone at generation n: the n-fold iterate $F_n = f(F_{n-1})$ of the offspring pgfs, evaluated on an $L_x \times L_y$ grid of damped roots of unity and inverted by a 2-D FFT; grid sized from the data; exact gradients by forward mode through the recursion (`inference=exact_gw`, Numba kernel)
Exact BH law	method of stages: a Gamma(m) lifetime is m exponential phases, so the pgf solves $2m$ backward Kolmogorov ODEs per grid point (fixed-step DOP853, dt = 1–2 h; 3 h in the SMC presets, within 10^{-7} nats of 1 h), then a 2-D FFT; one ODE trajectory serves every endpoint; 14 forward tangents give the Jacobian (`inference=exact_bh`)
Likelihood	product over clones and endpoints of $p(A, B \mid \theta)$, conditioned on detection (clone not empty); same prior and stick-breaking coordinates as NPE
Samplers	GW: NUTS, 4 chains, 300 warmup + 500 draws each (E0 × 32: 200 + 250), dense mass matrix, started at the best of several optima. BH: first look with Laplace-initialised adaptive multiple importance sampling (AMIS, ≈1 000–1 300 likelihood evaluations plus 20–80 gradients; weights unreliable); then tempered SMC from a widened Laplace start + 64 adaptive-Metropolis chains stopped by nested $\hat{R}$ (`sampler=smc`, merged `c2b639c`)
Validation	GW: law at the posterior median vs an undamped grid twice as large, max relative error $\leq 1.1 \times 10^{-6}$ in every run. BH (`scripts/b_row_theory/bhexact_validation.md`): normalisation to 10^{-16}; means and the full 2-D law vs 2×10^5 simulated clones at 96/144/240 h (TV and χ^2 within sampling noise); shape 1 vs the closed form (5×10^{-12}); halving dt changes p by $< 10^{-15}$; derivatives vs finite differences
Theory	Cramér–Rao (CR) bounds from the exact Fisher information and profile likelihoods: `docs/b_row_identifiability/note.pdf`
GW designs	E0 at 4096 (gens {4,6,10} × 4096); E0 ({4,8,12} × 32); real-like ({6,10} × 165/101); early ({3} × 266, {2,3} × 133, the real organoid budget); `two_type` (B divisions free) and `reversible_ab` (fixed); reference truth; 5 seeds; reference datasets *identical* to the NPE runs (`data_seed` = seed)
Runs / folder	35 GW runs: `results/exact_gw/` (`info.md`, `exact_vs_npe.csv`, `scripts/analyse_exact_gw.py`; commit `41d1ae4`); BH: Fisher information (`bhfisher.py`), preliminary posteriors (5 AMIS runs, 1 SMC run; not converged) in `results/exact_bh/`; table `report/tables/exact_gw.csv`

Result

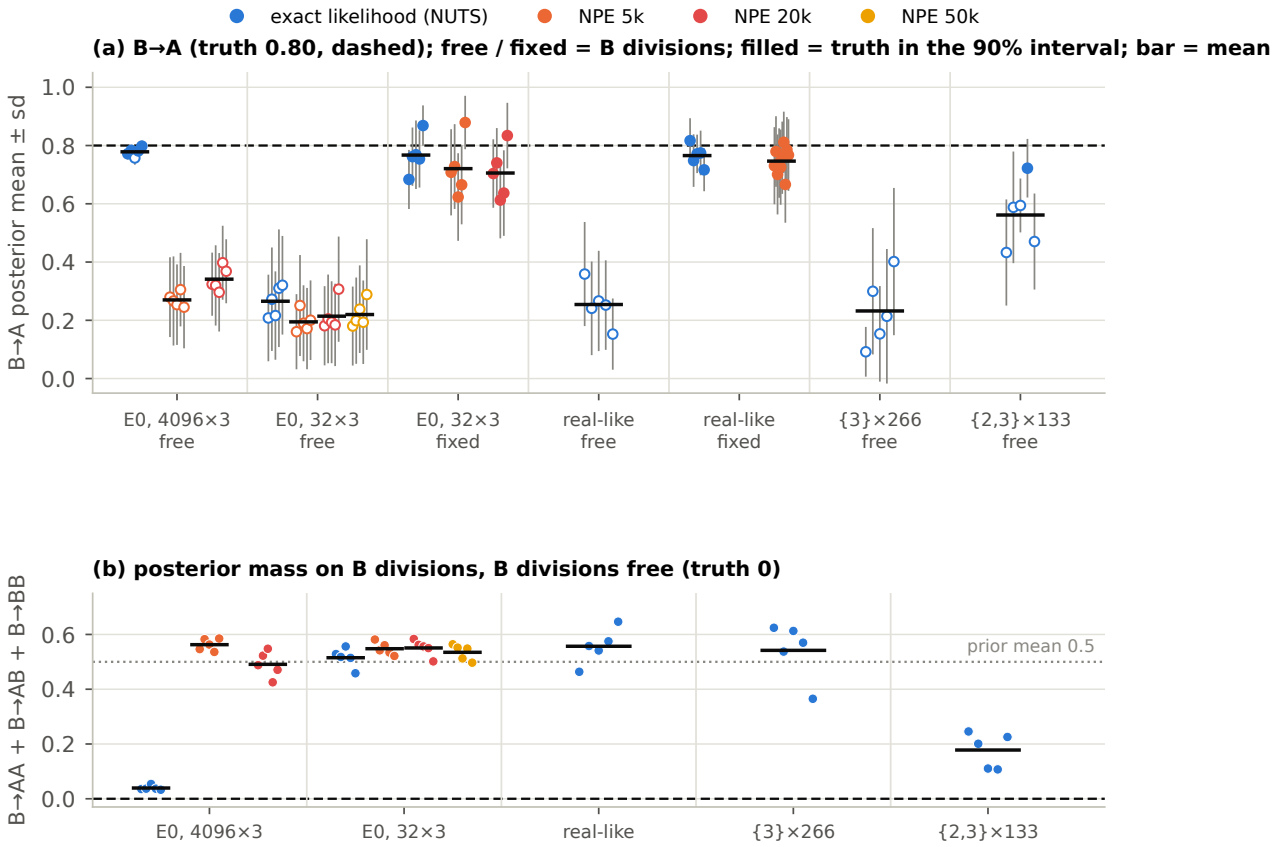


Figure 10: Exact GW likelihood vs NPE on identical datasets. (a) B→A posterior mean ± sd per run (filled = truth in the 90% interval; bar = mean over runs); free / fixed = B divisions free (`two_type`) or fixed to 0 (`reversible_ab`); E0 × 32 NPE 20k/50k from `results/b_row`; real-like NPE: 10 datasets (`results/x2/gw_data`). (b) Posterior mass on B divisions (sum of the posterior means of B→AA, AB, BB), B divisions free; dotted = prior mean. Plotted numbers: `tables/exact_gw.csv`.

GW design, B divisions	Method	B→A (0.8)	post. sd	covered	B-div. (0)	MAE	CR sd
E0 at 4096, {4,6,10}, free	**exact**	**0.78**	0.020	4/5	**0.04**	0.011	0.023
	NPE 20k / 5k	0.34 / 0.27	0.12 / 0.14	0/5	0.49 / 0.56	0.118 / 0.135	
E0, {4,8,12} × 32, free	exact	0.27	0.17	0/5	0.52	0.126	0.27
	NPE 5k / 20k / 50k	0.20 / 0.21 / 0.22	0.14–0.15	0/5	0.54–0.55	0.14	
E0, {4,8,12} × 32, fixed	exact	0.77	0.10	5/5	—	0.031	
	NPE 5k / 20k	0.72 / 0.71	0.13	5/5	—	0.048 / 0.052	
real-like, free	exact	0.25	0.16	0/5	0.56	0.130	0.28
real-like, fixed	exact	0.77	0.08	5/5	—	0.022	
	NPE 5k (10 datasets)	0.75	0.12	10/10	—	0.034	
early {3} × 266, free	exact	0.23	0.19	0/5	0.54	0.127	0.093
early {2,3} × 133, free	exact	**0.56**	0.15	1/5	**0.18**	0.059	0.077

Means over runs. B-div.: posterior mass on B divisions. MAE over the free probabilities. CR sd of B→A: local Cramér–Rao bound at $\theta^\star$ with all other parameters free (note §3); uniform-prior sd of B→A is 0.14 (free) / 0.24 (fixed). No NPE run exists at the reference truth for real-like `two_type` (X5a: 0.18 / 0.22 at truths 0.7 / 0.9) or for the early designs. Max split $\hat{R} \leq 1.06$ except E0 × 32 free seed 4 (1.12), real-like fixed seed 1 (1.18) and early {3} seed 3 (1.11): several modes, chain means consistent with the other seeds. PPC 0–4%.

Comment

- **At 4096 clones × 3 endpoints the B row is identified — the failure reported in §8 was NPE's, not the data's.** On the same five datasets the exact posterior gives B→A 0.78 ± 0.02 (4/5 covered) and B-division mass 0.04, where NPE gave 0.27–0.34 with half its mass on divisions. The exact sd (0.020) is at the CR scale (0.023), and the exact likelihood rejects NPE's posterior means by 116–178 nats. A's row is recovered too (MAE 0.011, A persist 0.02 vs NPE 0.16): the A-row bias was also NPE's.
- **At realistic designs the exact posterior is as prior-dominated as NPE's**: B→A 0.27 (E0 × 32) and 0.25 (real-like) with B divisions free, mass on divisions ≈0.5. The CR sd of B→A there (0.27–0.28) exceeds the prior sd: no method can do better, so NPE's answers at these sizes are the data's. The B row is

practically non-identifiable at feasible designs; it is not *structurally* non-identifiable (positive definite Fisher information at every endpoint, note §3).

- **With B divisions fixed, exact and NPE agree** (0.77 vs 0.71–0.75, all covered); exact posteriors are 25–40% narrower and the MAE 30–40% lower: a moderate efficiency loss of NPE + summaries, no bias.
- **Early endpoints help, partly.** With the real budget of 266 organoids, $\{2,3\} \times 133$ moves B→A to 0.56 ± 0.15 and the division mass to 0.18; $\{3\} \times 266$ does not (0.23). Locally both have CR sd 0.08–0.09 (10–15× the information of {6,10}), but the likelihood flattens away from the truth (note §3), so early data alone do not decide the B row at 266 organoids; with a constraint or a measurement they help most.
- **BH (§13.1): the ridge and the B row are practical too, and the B row is much weaker than in GW.** The exact BH Fisher information is nonsingular for both laws. CR sd of A→AA 0.23 with B divisions fixed (NPE's 0.32 is one CR sd from the truth); B→A 0.82 fixed and 0.99 free (GW: 0.28). The weakest direction trades B→A against B persist and B's lifetime, so in BH fixing B divisions is not enough — B's lifetime has to be known as well. A→AA to sd 0.05 needs $\approx 6 \times 10^3$ clones (B divisions fixed), B→A $\approx 10^5$.
- **Consequences.** The practical recommendations stand — fix or measure B divisions; measure lifetimes — but for a different reason: at feasible sizes the data are too weak (not: the model cannot be identified). And NPE can fail silently at large data: at 4096 per endpoint it was calibrated (SBC 0.05–0.07) and passed the PPC while wrong by 0.45. Where the likelihood is tractable, validate NPE against exact inference.
- Caveats: one truth on the boundary (B divisions = 0: the mass is one-sided, coverage of B divisions meaningless); 5 seeds; the E0 × 32 `two_type` runs were rerun without SBC (to free the machine), SBC elsewhere uses 10 short-chain cases (100 for early); three runs are multimodal ($\hat{R}$ 1.11–1.18); CR bounds are local and above the simplex width at realistic sizes; BH posteriors preliminary (not converged).

Leads to. The practical conclusion of this part survives with a new reason: at feasible designs the data cannot say whether B divides, so B divisions are fixed to 0 (`reversible_ab`) in Parts II–IV and must be measured in the wet lab. The exact BH likelihood (§13.1) returns to the question for the lifetime ridge of Part III; its cost is in Appendix B.

Part II

Design and amount of data

Motivation. With an identifiable model (Part I), the planning question is the design: which days to measure and how many organoids per day. E0 showed that the information lies in clone-size variability across time, which suggests that the number of endpoints may matter as much as the number of organoids. We first compare one and three endpoints in GW, then map the full grid in GW and BH. The BH half produces the result that drives Part III.

10 Design comparison — one vs three endpoints (GW)

Why this next. Part I fixed the law (B divisions = 0, §6). E0 used 3 endpoints × 32 clones; the real data have 2 endpoints, and a wet-lab plan can trade endpoints for organoids. Which matters more, at equal total organoids?

Question

Which planned design identifies the model best, and is it the number of time points or the number of organoids that matters?

Setup

Simulator	Galton–Watson, perfect observation
Law (free)	`two_type` (10) and `reversible_ab` (7)
Data vs inference	well specified, $\theta^\star$
Design	`single_128`: gen. 12 × 128; `three_128`: gen. 4, 8, 12 × 128; `single_384`: gen. 12 × 384; reference `three_32` (from E0)
Inference / budget	NPE amortised, 20k
Seeds / runs	5 / 30 (+ 10 reference runs from E0)
Varied	design × law
Folder	`results/design_comparison/`

Result

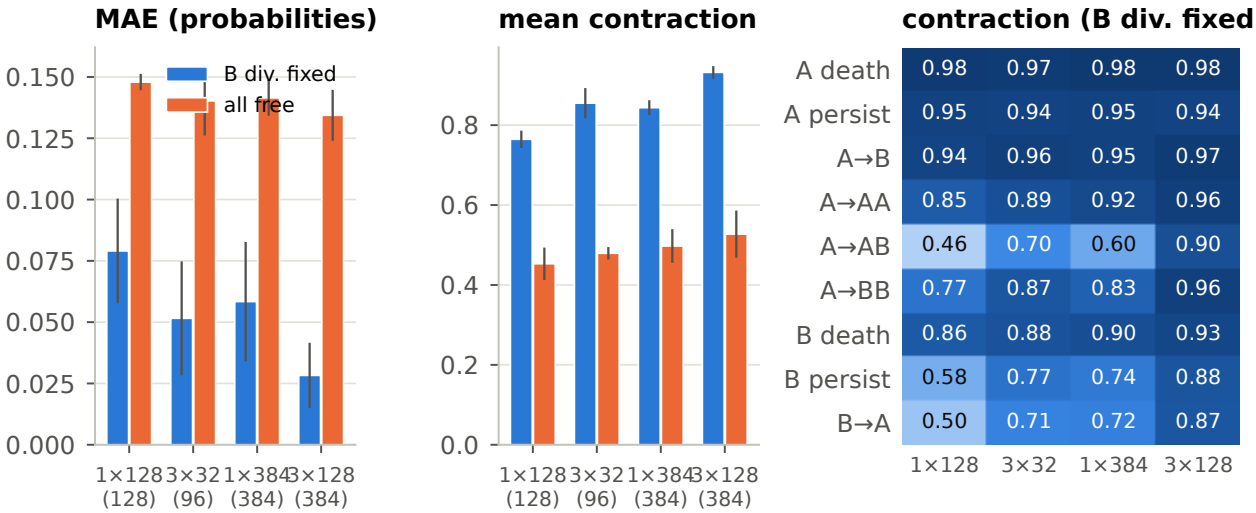


Figure 11: Left, centre: MAE and mean contraction per design (mean ± sd over 5 seeds); total organoids in brackets. Right: contraction per parameter, B divisions fixed.

Design (`reversible_ab`)	organoids	MAE	contraction	90% cov.	B→A	B→A contr.	PPC
1 × 128	128	0.079	0.77	0.87	0.61 ± 0.17	0.50	0.000
3 × 32	96	0.052	0.86	0.96	0.71 ± 0.13	0.71	0.008
1 × 384	384	0.058	0.84	0.93	0.67 ± 0.12	0.72	0.000
3 × 128	384	**0.028**	**0.93**	0.93	**0.78 ± 0.09**	**0.87**	0.031
`two_type`, any design	96–384	0.13–0.15	0.45–0.53	0.69–0.78	0.18–0.25	≤ -0.03	≤ 0.014

Comment

- At equal organoids, 3 endpoints halve the error of 1 (0.028 vs 0.058). Gains are in dynamics: A→AB contraction 0.90 vs 0.60, B→A 0.87 vs 0.72.
- 96 organoids over 3 endpoints ≈ 384 at one endpoint.
- With all transitions free no design recovers B's row (B→A contraction ≤ 0) — consistent with E0; not even 4096 per endpoint (§8).
- Calibration fine everywhere (SBC 0.02–0.03). Caveat: GW, one truth, generations not days.

Leads to. One slice (GW, generations 4/8/12, 20k simulations). E4 maps the full grid of organoids × endpoint sets, including the real-like {6, 10}: GW first (§11), then BH (§12).

11 E4 — Amount of data: organoids and endpoints (GW)

Why this next. The design comparison (§10) showed that three endpoints beat one at equal organoids. For a plan we need the minimum number of organoids per endpoint for each parameter and the best endpoint set, systematically. GW comes first: it has no lifetimes, so it shows what endpoint counts can deliver when the time scale is known.

Question

How do contraction and error depend on organoids per endpoint (16–4096) and on the measured generations? What is the minimum number to recover each parameter when there is no lifetime ridge?

Setup

Simulator	Galton–Watson (endpoints in *generations*; day-mapping nominal)
Law (free)	`reversible_ab` (7 probabilities)
Data vs inference	well specified, $\theta^\star$
Design	clones per endpoint $\in \{16, 32, \ldots, 512\}$ and extensions $\{1024, 2048\}$, $\{4096\}$ × generations {10}, {6,10}, {6,10,14}, {4,6,10}
Inference / budget	NPE amortised, 5k; budget check: 20k at 4096 clones, {4,6,10}, same 5 datasets (§12.1)
Seeds / runs	5 / 180 = 120 (16–512) + 40 (1024, 2048) + 20 (4096); + 5 at 20k; all with 2 000 posterior samples, 200 SBC cases
Folder	`results/e4/gw/` (16–512), `results/e4/gw_large/` (1024, 2048; commit `5cb6018`), `results/e4/gw_xxl/` (4096; `07fb7fe`), `results/e4/gw_xxl_20k/` (20k; `07fb7fe`, preset docs `503ec72`); pooled tables `results/e4/gw_all_summary_*.csv`

Result

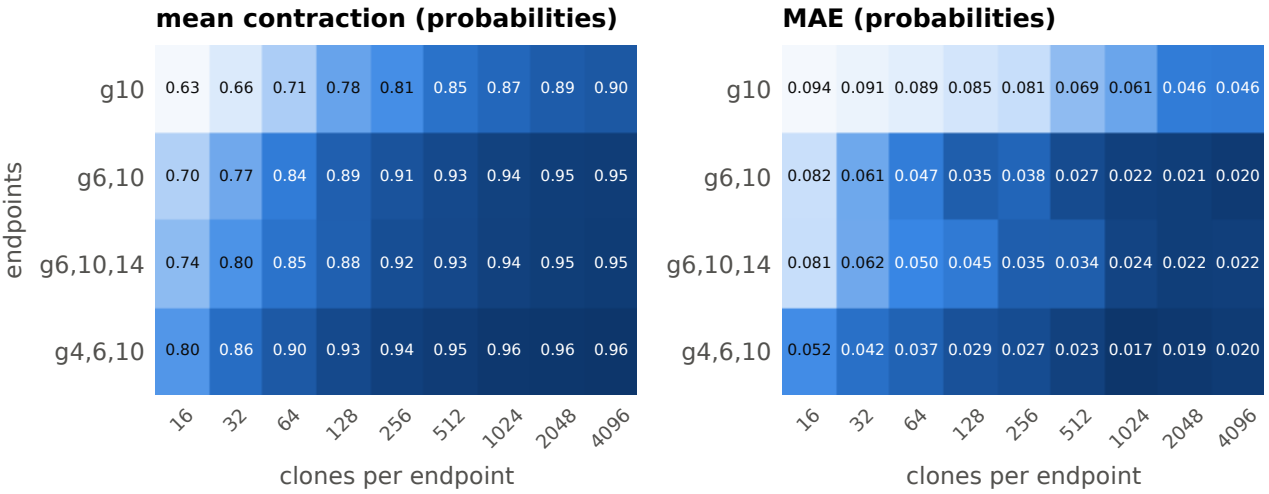


Figure 12: E4 GW: mean contraction and MAE of the outcome probabilities (mean over 5 seeds); endpoints in generations; NPE 5k; 1024–4096 from the extensions.

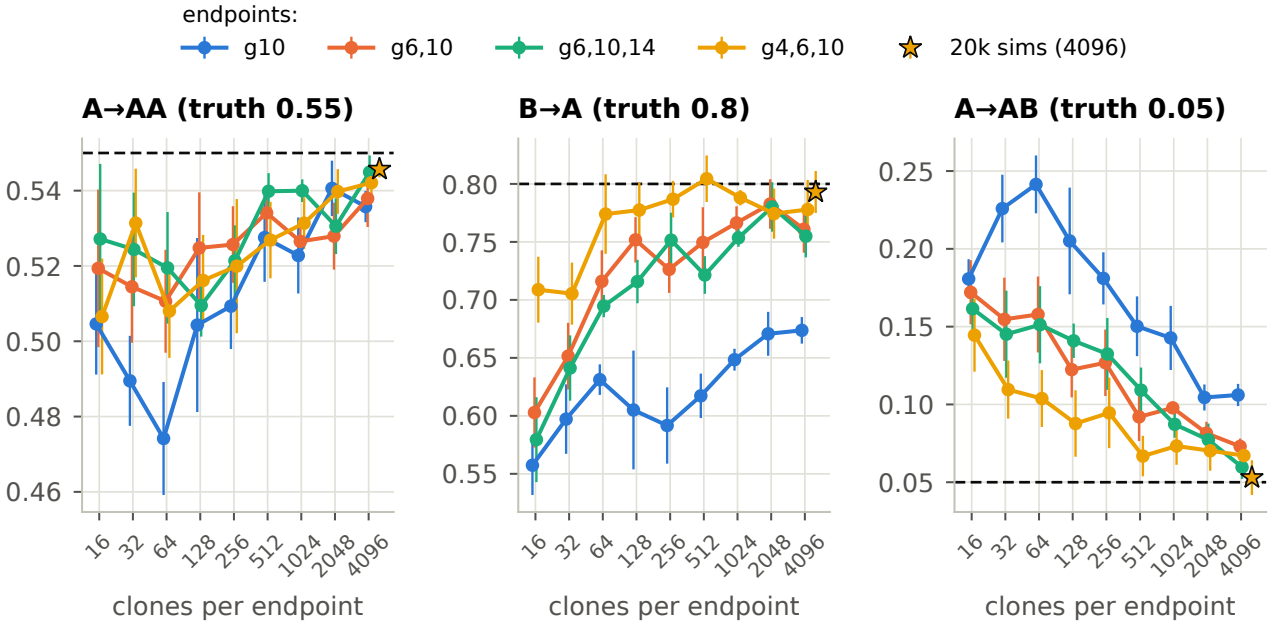


Figure 13: E4 GW: posterior means (mean ± SE over seeds) vs organoids per endpoint, NPE 5k; dashed = truth. Star: {4,6,10} at 4096 with 20k simulations (§12.1).

Minimum organoids per endpoint for "recovered" (contraction > 0.5, |error| < 0.05, ≥ 4/5 covered; must hold at every larger size up to 4096; — = not reached; NPE 5k; for BH see §12)

	GW, generations			
Parameter	{10}	{6, 10}	{6, 10, 14}	{4, 6, 10}
A death, A persist, A→B	16	16	16	16
B death	32	16	16	16
A→AA	256	16	16	2048[a]
A→BB	—	256	2048	**32**
B persist	—	512	—[c]	**64**
B→A	—	1024	—[c]	**128**
A→AB	—	512	2048	4096[b]

[a] error ≤ 0.04 at every size, but coverage 2–3/5 at 256–1024 (narrow posteriors, sd ≈ 0.02).
[b] also met at 128 and 512 (1024/2048: error 0.03, 3/5); at 4096 error 0.017, 4/5.
[c] 1024 up to 2048; at 4096 error 0.051 (B persist) and 0.053 (B→A), just above the 0.05 threshold.

Extensions: 1024, 2048, 4096 clones per endpoint (5 seeds, NPE 5k; MAE of probabilities / mean contraction / SBC coverage error)

Generations	512	1024	2048	4096
{10}	0.069 / 0.85 / 0.034	0.061 / 0.87 / 0.039	0.046 / 0.89 / 0.038	0.046 / 0.90 / 0.043
{6, 10}	0.027 / 0.93 / 0.037	0.022 / 0.94 / 0.044	0.021 / 0.95 / 0.045	0.020 / 0.95 / 0.050
{6, 10, 14}	0.034 / 0.93 / 0.041	0.024 / 0.94 / 0.047	0.022 / 0.95 / 0.055	0.022 / 0.95 / 0.058
{4, 6, 10}	**0.023** / **0.95** / 0.044	**0.017** / **0.96** / 0.052	**0.019** / **0.96** / 0.059	**0.020** / **0.96** / 0.064

Comment

- **Design ranking**: an early endpoint is worth most ({4,6,10} best at every size); a late extra endpoint ({6,10,14}) adds little over {6,10}; one endpoint plateaus and cannot separate A→AB from A→BB (0.15–0.24 vs 0.05; 0.15–0.20 vs 0.29 up to 512; still 0.11 and 0.24 at 4096).
- **Without lifetimes everything is recoverable**: with {4,6,10} and ≈128 organoids per endpoint all probabilities except A→AB (met at 128 and 512, not stably) meet the criterion; A→AA is right from 16 organoids. The GW numbers are the attainable target *if* division times are known (BH with lifetimes fixed: MAE 0.059, §13).
- **Plateau at ≈ 1024 per endpoint with NPE 5k** (extensions): multi-endpoint designs reach MAE ≈ 0.02 and contraction ≈ 0.95 and gain nothing from 2048 or 4096; with two or three endpoints every parameter meets the criterion at 1024–2048 except A→AB with {4,6,10} (intermittent). B→A contraction 0.87–0.93.
- **The plateau is the network's**: with 20k simulations at 4096 ({4,6,10}, same datasets) MAE 0.020 → 0.017, B→A error 0.047 → 0.032 (5/5 covered), SBC 0.064 → 0.044, support 0.59 → 0.78; training used 190 of 200 epochs on average (§12.1, after the BH grid).

- **A single endpoint never catches up**: it keeps improving, but at 4096 organoids (MAE 0.046) it is still worse than three endpoints at 32 (0.042); B→A, A→AB, A→BB and B persist never meet the criterion.
- **SBC error grows with size** (0.03 → 0.05–0.06 at 1024–4096) while coverage of the truth stays 0.87–0.98: at these sizes the amortised NPE (5k simulations) limits accuracy, as the 20k check confirms.
- Checks: single well-specified GW runs reach PPC 0.10 and support mass 0.07 ({4,6,10}, seed 4): borderline single values are not evidence of misspecification.
- Caveat: generations are not days; one truth, perfect assay, single A founders, 5k simulations (20k only for {4,6,10} at 4096). The full transition model (`law=two_type`) at 4096 clones is in §8: B's row stays unidentifiable.

Leads to. The same grid in BH (§12), the simulator closer to the real data (time in hours, free lifetimes). Does the ranking survive, and are the GW numbers reached?

12 E4 — Amount of data: organoids and endpoints (BH), and comparison with GW

Why this next. GW (§11) recovers every probability by ≈128 organoids with {4,6,10}. BH adds two free lifetimes and measures time in days; it is the model that will be fitted to real data.

Question

As §11 for Bellman–Harris (endpoints in hours): how do contraction and error depend on organoids per endpoint (16–4096) and on the measured days? What is the minimum number to recover each parameter, and are the GW numbers reached?

Setup

Simulator	Bellman–Harris, Gamma lifetimes shape 8
Law (free)	`reversible_ab`: 7 probabilities + τ_A, τ_B (9)
Data vs inference	well specified, $\theta^\star$
Design	clones per endpoint ∈ {16, 32, 64, 128, 256, 512} and extensions {1024, 2048}, {4096} × endpoints {240}, {144, 240}, {144, 240, 336}, {96, 144, 240} h = days {10}, {6,10}, {6,10,14}, {4,6,10}
Inference / budget	NPE amortised, 5k; budget check: 20k at 512 and 4096 clones (days 4/6/10) and at the real-like design, same datasets (§12.1)
Seeds / runs	5 / 180 = 120 (16–512) + 40 (1024, 2048) + 20 (4096); + 15 at 20k
Varied	clones × endpoint set; simulation budget (check only)
Folder	`results/e4/bh/` (16–512), `results/e4/bh_large/` (1024, 2048; commit `5cb6018`), `results/e4/bh_xxl/` (4096; `07fb7fe`), `results/e4/bh_xxl_20k/` (20k; `07fb7fe`, preset docs `503ec72`), `results/e4/bh_512_20k/`, `results/e4/bh_real_20k/` (20k; `503ec72`); pooled tables `results/e4/bh_all_summary_*.csv`

Result

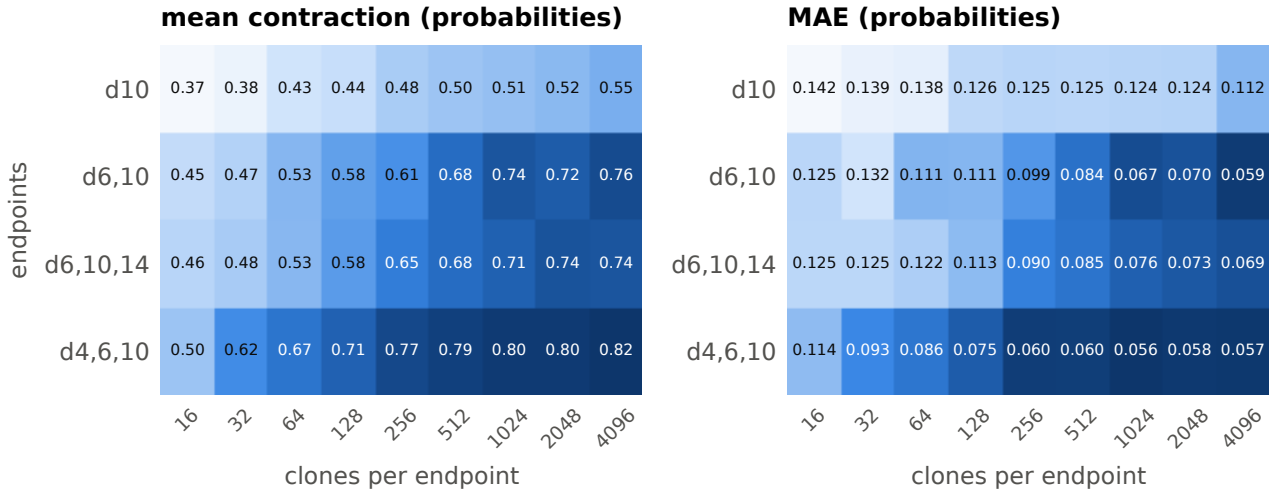


Figure 14: Mean contraction and MAE of the 7 outcome probabilities (mean over 5 seeds, NPE 5k); 1024–4096 from the extensions.

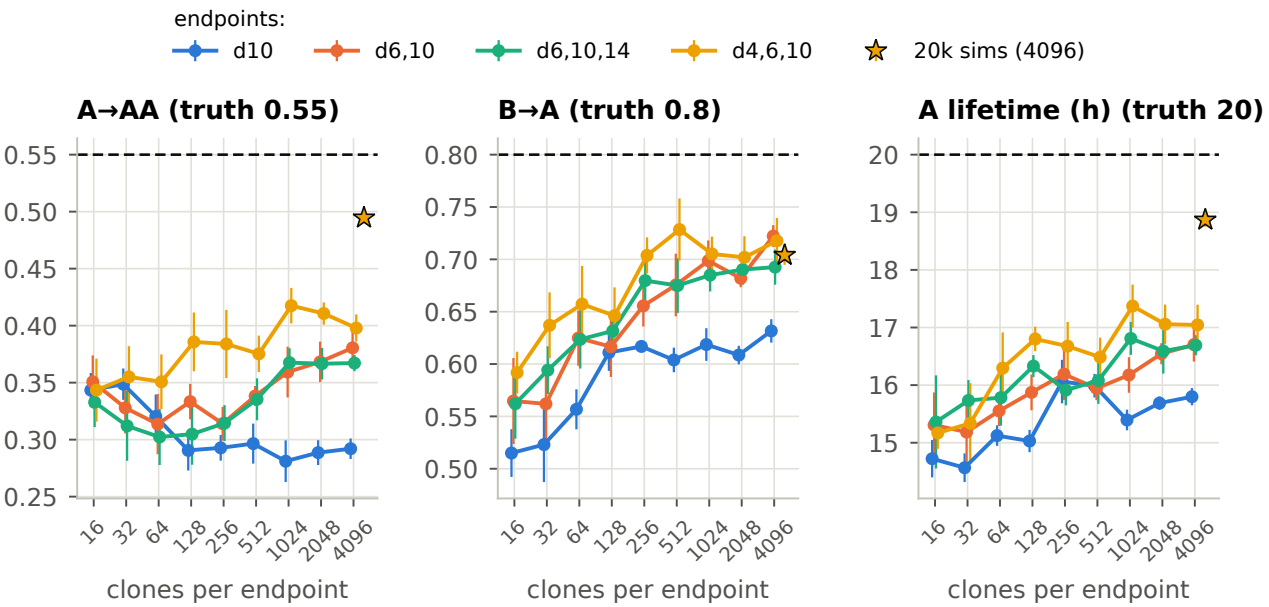


Figure 15: Posterior means (mean ± SE over seeds) vs organoids per endpoint, one line per endpoint set, NPE 5k; dashed = truth. Star: days 4/6/10 at 4096 with 20k simulations (§12.1).

Minimum organoids per endpoint for "recovered" (contraction > 0.5, |error| < 0.05, ≥ 4/5 covered); — = not reached at 4096 (NPE 5k; the 1024–4096 extensions change one entry).

Parameter	day 10	days 6, 10	days 6, 10, 14	days 4, 6, 10
A death, A→B (small truths)	16	16	16	16
A→BB	—	512	256	**128**
B death	—	256	256	**128**
A→AB	—	4096[a]	—	**256**
A→AA, A persist, B persist, B→A	—	—	—	—
τ_A, τ_B (biased: τ_A 15–17 h, 0/5 covered)	—	—	—	—

[a] met only at the largest size (error 0.026, 5/5); not a stable threshold.

Extensions: 1024, 2048, 4096 clones per endpoint (5 seeds, NPE 5k; probabilities: MAE, contraction; lifetimes: mean relative error)

Endpoints (days)	MAE 1024 / 2048 / 4096	contraction 1024 / 2048 / 4096	rel. error τ_A, τ_B 1024 / 2048 / 4096	A→AA (0.55) 512 → 2048 → 4096	τ_A (20 h) 2048 / 4096
{10}	0.124 / 0.124 / 0.112	0.51 / 0.52 / 0.55	0.32 / 0.30 / 0.30	0.30 → 0.29 → 0.29	15.7 / 15.8
{6, 10}	0.067 / 0.070 / 0.059	0.74 / 0.72 / 0.76	0.29 / 0.27 / 0.24	0.34 → 0.37 → 0.38	16.5 / 16.7
{6, 10, 14}	0.076 / 0.073 / 0.069	0.71 / 0.74 / 0.74	0.25 / 0.27 / 0.27	0.34 → 0.37 → 0.37	16.6 / 16.7
{4, 6, 10}	**0.056** / **0.058** / **0.057**	**0.80** / **0.80** / **0.82**	0.24 / 0.27 / 0.27	0.38 → 0.41 → 0.40	17.1 / 17.0

A→AA and τ_A covered 0/5 at every size and design with NPE 5k.

Comparison with GW (§11; same grid, endpoints in generations)

	BH (days)	**GW** (generations)
Contraction, {4,6,10}, 16/512/2048/4096	0.50 / 0.79 / 0.80 / 0.82	0.80 / 0.95 / 0.96 / 0.96
MAE, {4,6,10}, same sizes	0.114 / 0.060 / 0.058 / 0.057	0.052 / 0.023 / 0.019 / 0.020
One late endpoint, contraction	0.37 / 0.50 / 0.52 / 0.55	0.63 / 0.85 / 0.89 / 0.90 (A→AB/BB conf.)
A→AA (0.55), NPE 5k	0.28–0.42, 0–2/5 covered	0.47–0.55 at every design
A→AA, 4096, {4,6,10}, 5k / 20k	0.40 (0/5) / 0.50 (3/5)	0.54 / 0.55 (5/5)
B→A (0.80), {4,6,10}, 128/512/2048/4096	0.65 / 0.73 / 0.70 / 0.72	0.78 / 0.80 / 0.77 / 0.78
SBC error, 16–512 / 1024–4096	0.025–0.034 / 0.029–0.039	0.024–0.044 / 0.038–0.064
Support mass / PPC (per run)	0.37–0.92 / ≤ 0.042	0.07–0.89 / ≤ 0.10 (groups ≤ 0.021)

Comment

- **Same design ranking as GW** (§11): the ranking is a property of the endpoint information, not of the time model. An early endpoint (day 4) helps most: days 4/6/10 are best on every metric at every size and the only design where A→AB is recovered from a moderate size (0.054 at 2048, 0.042 at 4096, truth 0.05; days 6/10 meet it only at 4096). Day 10 alone plateaus (contraction 0.37 → 0.50 → 0.52 → 0.55 from 16 to 512 to 2048 to 4096).

- With several endpoints and NPE 5k more organoids help up to ≈512–1024 (MAE 0.114 → 0.060 → 0.056 for days 4/6/10; 0.058 at 2048, 0.057 at 4096). B→A rises 0.59 → 0.73 and is covered at every size, but not recovered (0.61–0.72 at 1024–4096).
- **With NPE 5k the ridge does not yield to sample size.** At 4096 clones per endpoint (up to 12 288 organoids) A→AA is 0.29–0.40 (truth 0.55) and τ_A 15.8–17.0 h (truth 20), 0/5 covered in every design. The posterior creeps along the ridge (days 4/6/10: 0.34 → 0.41 from 16 to 2048, flat at 4096) with τ_A contraction 0.99: confidently on the wrong point.
- **At 4096 this is partly a training limit, at realistic sizes it is not** (§12.1): with 20k simulations on the same 4096-clone datasets A→AA moves 0.40 → 0.50 and τ_A 17.0 → 18.9 h (3/5 covered each); at 512 per endpoint only 0.38 → 0.44 (1/5), and at the real-like design not at all (0.32 → 0.33). The ridge yields gradually to data with a well-trained network; at realistic sizes lifetime information remains necessary (§13, X5b).
- Calibration at 1024–4096: SBC error rises mildly (0.028–0.039), PPC ≤ 1.4%, support mass 0.37–0.81.
- "Recovered" for A death / A→B is easy because the truths are small (|error| < 0.05 is loose there).
- GW, without lifetimes, reaches contraction 0.80 → 0.95 → 0.96 (16 → 512 → 2048, also 0.96 at 4096) and recovers A→AA at every design (table above).
- Caveat: one truth, perfect assay, single A founders, 5k simulations (20k only for days 4/6/10 at 512 and 4096, and at the real-like design).

12.1 Simulation budget check: 5k vs 20k simulations, from the real-like design to 4096 clones (BH and GW)

Question

Is the stuck BH ridge (above), and the GW large-design plateau (§11), a limit of the data or of the 5k-simulation network — and does the answer depend on the size of the experiment?

Setup

Designs	BH: real-like (144/240 h × 165/101 clones; same 5 datasets as the BH ridge control and X5b vague), 512 and 4096 clones per endpoint on days {4,6,10} = {96,144,240} h (same datasets as the E4 runs); GW: 4096 clones on generations {4,6,10}
Inference / budget	NPE amortised, 5k (from E4 / X5b) vs 20k (`budget.simulations=20000`); epoch cap 200 at 4096; 500 at real-like and 512 (`budget.max_epochs=500`; early stopping at 104–161 epochs, cap not binding)
Seeds / runs	5 paired seeds (same datasets) / 20 at 20k (5 per design)
Folder	`results/e4/bh_real_20k/`, `results/e4/bh_512_20k/` (commit `503ec72`); `results/e4/gw_xxl_20k/`, `results/e4/bh_xxl_20k/` (`07fb7fe`; preset docs `503ec72`); table `report/tables/e4_budget.csv`

Result

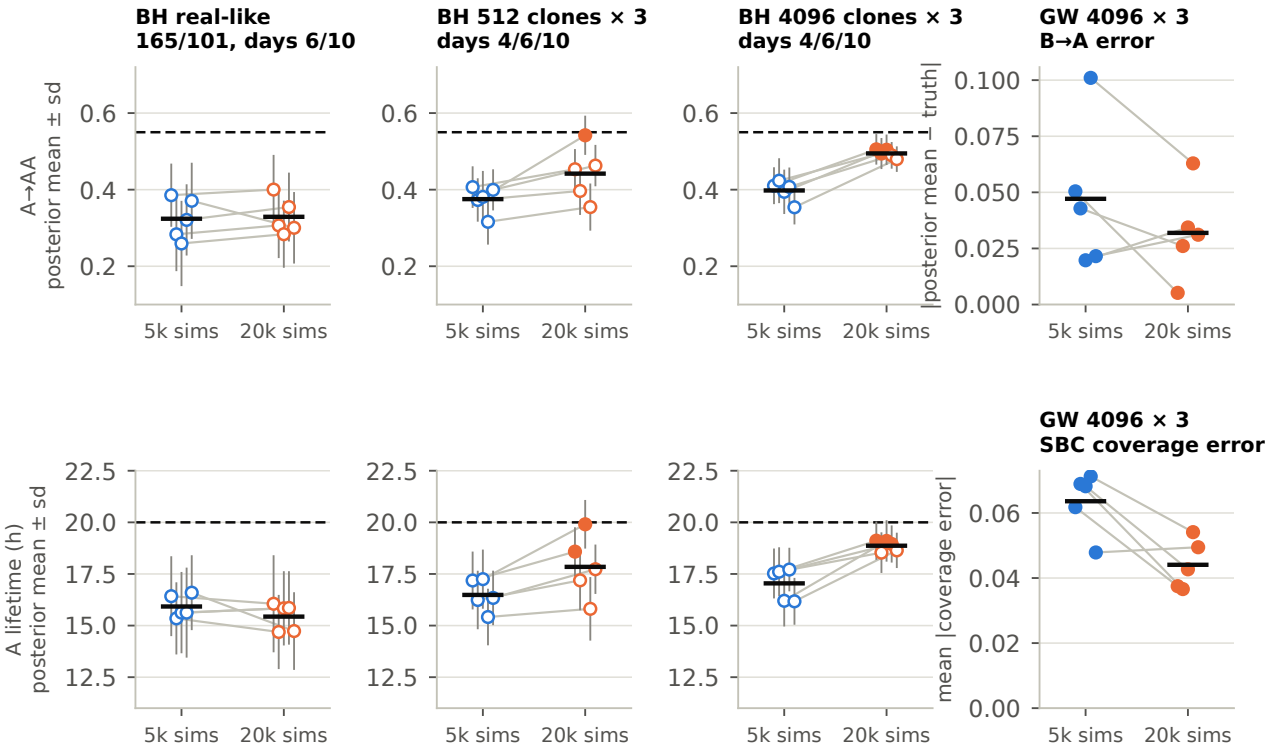


Figure 16: 5k vs 20k training simulations on the same 5 datasets (lines join seeds; bars = mean over seeds). Left three columns, BH: A→AA (top) and τ_A (bottom), posterior mean ± sd, at the real-like design, 512 and 4096 clones per endpoint (days 4/6/10); filled = truth in the 90% interval, open = not covered; dashed = truth. Right, GW at 4096 (generations 4/6/10): B→A absolute error and run SBC coverage error.

BH ridge across sizes (days 6/10 real-like; days 4/6/10 at 512 and 4096 per endpoint)

	real-like (165/101)		512 × 3		4096 × 3	
	5k	20k	5k	20k	5k	20k
A→AA (0.55): mean, covered	0.32, 0/5	0.33, 0/5	0.38, 0/5	0.44, 1/5	0.40, 0/5	**0.50, 3/5**
τ_A (20 h): mean, covered	15.9, 0/5	15.4, 0/5	16.5, 0/5	17.8, 2/5	17.0, 0/5	**18.9, 3/5**
τ_B (30 h): mean, covered	20.2, 4/5	17.8, 4/5	18.4, 3/5	21.2, 5/5	18.4, 2/5	22.5, 5/5
A→AB (0.05)	0.23	0.14	0.07	0.05	0.04	0.02
B→A (0.80)	0.63	0.70	0.73	0.69	0.72	0.70
MAE (probabilities)	0.109	**0.079**	0.060	0.058	0.057	**0.049**
Rel. error lifetimes	0.26	0.32	0.28	0.20	0.27	**0.15**
Coverage (run, 90%)	0.76	0.80	0.78	0.86	0.76	**0.93**
SBC coverage error	0.027	0.022	0.034	0.031	0.039	0.040
Epochs, mean (cap)	91 (200)	114 (500)	122 (200)	139 (500)	160 (200)	201 (200)

4096 clones per endpoint, both simulators

	GW {4,6,10}		BH {96,144,240} h	
	5k	20k	5k	20k
MAE (probabilities)	0.020	**0.017**	0.057	**0.049**
Mean contraction (all parameters)	0.964	0.977	0.842	0.893
Coverage (run, 90%)	0.93	0.93	0.76	**0.93**
SBC coverage error	0.064	**0.044**	0.039	0.040
Support mass	0.59	0.78	0.56	0.51
Epochs, mean (cap 200)	162	190	160	**201** (all 5)
B→A (0.80): mean, error, covered	0.78, 0.047, 4/5	0.79, 0.032, 5/5	0.72, 0.082, 5/5	0.70, 0.096, 5/5
A→AA (0.55): mean, covered	0.54, 5/5	0.55, 5/5	0.40, 0/5	**0.50, 3/5**

Comment

- **At the realistic design the ridge is a property of the data**: 20k simulations do not move A→AA (0.32 → 0.33) or τ_A (15.9 → 15.4 h), 0/5 covered in both. The gain (MAE 0.109 → 0.079) is in other parameters: A→AB 0.23 → 0.14, A→BB 0.19 → 0.26, B death 0.17 → 0.13.
- **With more data it resolves gradually, provided the network is trained enough**: at 512 per endpoint on three endpoints it starts to move (A→AA 0.38 → 0.44, 1/5; τ_A 16.5 → 17.8 h, 2/5; lifetime error 0.28 → 0.20), at 4096 it moves most of the way (0.40 → 0.50 and 17.0 → 18.9 h, 3/5 each; lifetime error 0.27 → 0.15). "The ridge does not yield to sample size" holds for NPE at 5k simulations, not for the data — but at realistic sizes independent lifetime information remains necessary (§13, §15).
- **The 5k network, not the data, sets the large-design plateau**: at 4096 four times the training simulations lower the error in both simulators and improve calibration (GW SBC 0.064 → 0.044) or coverage (BH 0.76 → 0.93). Those runs stopped at the 200-epoch cap (all 5 BH, GW 190 on average), so 20k is probably not the limit there; the realistic-size runs stopped early under a 500 cap.
- B→A in BH: no gain at 512 and 4096 (0.73 → 0.69, 0.72 → 0.70; still covered), a gain at the real-like design (0.63 → 0.70). BH support mass 0.56 → 0.51 at 4096.
- Caveat: one truth, 5 seeds; BH 512/4096 only on days 4/6/10, GW only at 4096.

Leads to. With the standard 5k network A→AA and τ_A are wrong at *every* size (16–4096) and endpoint set, with coverage falling as data grow. A 20k network moves them gradually with size — not at the real-like design, a little at 512, most of the way at 4096 per endpoint. At realistic sizes this is a bias of the data that neither more organoids nor more training removes. Part III opens with the hypothesis that explains it (§13).

Part III

Bellman–Harris timing: the lifetime ridge

Motivation. E4 BH (§12) left a bias that more organoids do not remove with the standard 5k-simulation network, not even 4096 per endpoint (a 20k network moves it gradually with size, most of the way only at 4096, and not at the real-like design, §12.1). This part identifies it as a ridge (§13), asks whether it is a property of the hand-picked truth (X5d), and tests the remedy suggested by the ridge, lifetime information, in its soft (literature prior, X5b) and hard (fixed lifetimes, E3a) form. Fixing lifetimes rests on timing assumptions: one value per type (E3a), the lifetime variability (E3b), and the absence of quiescence (E3c), which are tested last.

13 BH ridge — lifetime vs self-renewal

Why this next. E4 BH (§12): A→AA 0.28–0.42 and τ_A 15–17 h at every design up to 4096 organoids per endpoint with NPE 5k, 0/5 covered, whereas GW recovers A→AA at every design (§11). The only structural difference is the free time scale. Hypothesis: endpoint counts fix the growth rate, which faster cycles or more self-renewal produce equally.

Question

Is the BH baseline bias (A→AA ≈ 0.32, τ_A ≈ 16 h) a ridge on which endpoint counts fix only a combination of lifetime and self-renewal, with the prior picking the point?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	control: `reversible_ab` (9); lifetimes fixed: 7; flat prior: 9
Data vs inference	data always $\theta^\star$; only the inference model changes
Design	real-like: 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Arms	control (from E3b, shape 8); τ_A=20, τ_B=30 h fixed; flat prior on A→AA (concentration 1 vs 0.2 for A's other outcomes: A→AA ∼ Beta(1,1))
Seeds / runs	5 / 10 (+5 control)
Folder	`results/bh_ridge/` (control: `results/e3b/*lifetime_shape=8*`)

Result

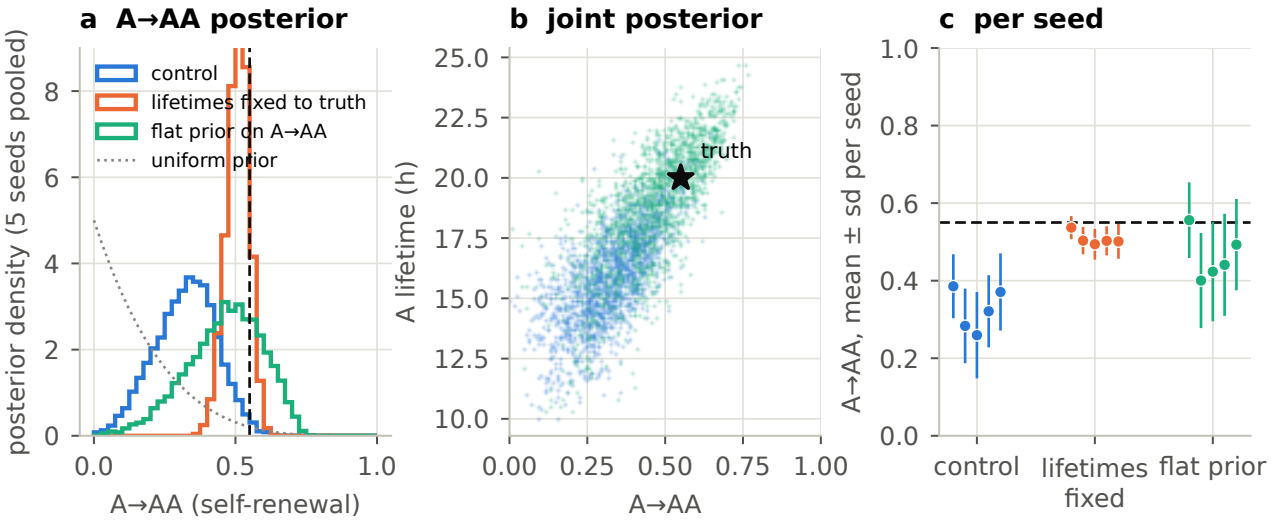


Figure 17: (a) A→AA posterior pooled over 5 seeds; dotted: uniform-Dirichlet prior Beta(1,5). (b) Joint posterior of A→AA and τ_A (blue: control; green: flat prior); star = truth. (c) Per-seed means ± sd; dashed = truth.

	Control	Lifetimes fixed	Flat prior
A→AA (0.55)	0.32 ± 0.10, 0/5, contr. 0.52	**0.51 ± 0.04, 5/5, 0.93**	0.46 ± 0.12, 5/5, 0.83
A→AB (0.05)	0.23, 3/5	0.17, 2/5	0.20, 5/5, contr. 0.00
B→A (0.80)	0.63, 5/5	0.68, 5/5	0.46, 3/5
τ_A (20 h)	15.9 ± 1.9, 0/5	fixed	18.9 ± 2.2, 5/5
MAE (probabilities)	0.109	**0.059**	0.117
SBC error / PPC / support	0.025 / 0.008 / n/r	0.031 / 0.025 / 0.65	0.079 / 0.000 / 0.51

Comment

- Ridge confirmed: posterior corr(A→AA, τ_A) ≈ +0.7 (E3c). Removing the lifetime degree of freedom recovers self-renewal and halves the error.
- The prior chose the point: only 1.8% of uniform-Dirichlet mass lies above 0.55. A flat prior re-centres (5/5 covered) but stays wide, degrades B→A and calibration: not a fix.
- GW has no lifetimes and no ridge: the GW control at the same design gives A→AA 0.52 (§20), and 0.47–0.55 at every E4 design (§11).
- Sample size alone, with the standard 5k-simulation NPE, does not leave the ridge: A→AA stays 0.29–0.41 and τ_A 15.7–17.1 h (0/5) up to 4096 clones per endpoint in every E4 design. With 20k simulations it moves gradually with size: not at the real-like design (A→AA 0.32 → 0.33, τ_A 15.9 → 15.4 h, 0/5), a little at 512 per endpoint on days 4/6/10 (0.44, τ_A 17.8 h, 2/5), most of the way at 4096 (0.50, 18.9 h, 3/5; §12.1). The ridge is a property of the data at realistic sizes, not a training artefact.
- Consequence: at realistic sizes measure A's division time (live imaging) or use it as an informative prior; otherwise report BH results with prior sensitivity. Train BH fits on ≥ 20k simulations anyway: it does not remove the ridge at the real-like design but lowers the error of the other probabilities (MAE 0.109 → 0.079). X5 tested both: a right ±25% lifetime prior gives A→AA 0.46, MAE 0.068 (§15); averaged over prior-drawn truths A→AA is learned well (contraction 0.82, §14), so the ridge bites at tail truths such as $\theta^\star$.
- **Exact BH likelihood: the ridge is practical, not structural** (§13.1). The Fisher information of the exact BH likelihood at $\theta^\star$ and this design is nonsingular (smallest eigenvalue 0.585, B divisions fixed). The local Cramér–Rao sd is 0.23 for A→AA and 0.19 on the log scale (3.8 h) for τ_A: NPE's 0.32 ± 0.10 lies one CR sd from the truth, i.e. the data at 266 clones are too weak to pin the point on the ridge, without the model being degenerate. About 6×10^3 clones (165:101 split) would give A→AA sd 0.05, $\approx 4 \times 10^3$ the log-lifetime — consistent with NPE 20k moving most of the way only at 12 288 clones. Preliminary exact posteriors on the same datasets (not converged, §13.1) sit closer to the truth along the ridge than NPE (A→AA 0.46–0.49 vs 0.33–0.39, τ_A 17.7–18.3 vs 15.7–16.6 h): part of NPE's bias here may be the network and summaries, as in GW; the CR sd allows either, so converged posteriors are needed to decide.

13.1 XB — Exact BH likelihood: the ridge is practical, not structural

Why this next. The exact GW likelihood (§9) showed that the B-row failure at 4096 clones was NPE's. Is the BH ridge, too, an artefact of NPE + summaries, or is the model itself degenerate along it? For integer lifetime shapes the BH clone law is exact (method of stages ODE + 2-D FFT; method and validation in §9), so its Fisher information at $\theta^\star$ answers the structural question, and exact posteriors the practical one (preliminary, not converged: box below).

Exact BH Fisher information at $\theta^\star$	`reversible_ab` (7 + 2)	`two_type` (10 + 2)
smallest eigenvalue of the Fisher information (total)	0.585	0.211
condition number (raw / correlation-scaled)	1.6×10^5 / 3.8×10^4	4.8×10^5 / 2.3×10^5
CR sd A→AA (truth 0.55)	0.23	0.54
CR sd A lifetime, log scale (hours)	0.19 (3.8 h)	0.40 (7.9 h)
CR sd B→A / B-division mass	0.82 / —	0.99 / 0.98
clones for sd < 0.05: A→AA / A log-lifetime	5.6×10^3 / 3.9×10^3	3.1×10^4 / 1.7×10^4
clones for sd < 0.05: B→A / B-division mass	7.1×10^4 / —	1.0×10^5 / 1.0×10^5

Real-like design (144 h: 165, 240 h: 101 detected clones), lifetime shape 8, τ_A 20 h, τ_B 30 h; grid 1024 × 512. Clone numbers in the same 165:101 split, $n = 266\,(\text{sd}/0.05)^2$ (local, large-n). Weakest direction: B→A against B persist and B's mean lifetime (`reversible_ab`); B's lifetime, persist and divisions (`two_type`). Source: `scripts/b_row_theory/bhexact_validation.md`.

Preliminary result — exact BH posteriors, not converged (`results/exact_bh/`)

Status. The planned sweeps (real-like design 144/240 h × 165/101 with B divisions fixed, `reversible_ab`, and free, `two_type`, seeds 0–4; the ridge at {96,144,240} h × 512; datasets identical to the NPE runs) were **stopped and will not be run now**: a converged real-like posterior needs $\approx 10^5$ likelihood evaluations, ≈3.5–4 h on 16 threads (7–8 h on 8), so the 15 runs would take 2–3 days. What exists:

- **AMIS first look** (5 runs, 4–6 h each on 3 threads under load): importance weights unreliable in every run (smoothed ESS 12–38 vs target 300, Pareto k 0.67–0.94). Importance sampling fails because the logit coordinates have long tails and the A→AA–lifetime ridge is curved.
- **SMC sampler** (`sampler=smc`, merged `c2b639c`): adaptive tempering from a widened Laplace start, then 64 adaptive-Metropolis chains stopped by nested $\hat{R}$ (Margossian et al. 2024). On a small design ({96,144} h × 64, `reversible_ab`) it matches a 128k-evaluation reference (means within 0.72 MC se, sds within 1.45 se), where AMIS was too narrow (A→AA sd 0.075 vs 0.090). Real-like seed 0: 40k evaluations in 91 min on 16 threads, **not converged** (min ESS 176 of 400, nested $\hat{R}$ 1.0101 > 1.01).

Dataset (seed)	Sampler	A→AA (0.55)	τ_A (20 h)	B→A (0.8)	NPE, same data: A→AA / τ_A
real-like, B div. fixed (0)	SMC, 40k	0.49 ± 0.06 (0.38–0.59)	18.3 ± 1.4	0.65 ± 0.11	0.385 / 16.6 h
	AMIS	0.47	17.7	0.67	
real-like, B div. fixed (3)	AMIS	0.46	18.2	0.71	0.33 / 15.7 h
real-like, B div. free (0 / 3)	AMIS	0.41 / 0.46	18.3 / 19.5	0.16 / 0.18	—
{96,144,240} h × 512 (0)	AMIS	0.49	18.5	0.83	0.45 / 18.6 h (20k)

SMC: mean ± sd (5–95%); B lifetime 24.3 ± 7.6 h (truth 30). AMIS: weighted means, indicative only. NPE: `results/x3/bh_control` (5k; `results/x5/b_vague` within 0.01 / 0.2 h) and `results/e4/bh_512_20k`; no NPE run at the reference truth for `two_type` real-like.

Reading (preliminary). With B divisions fixed, the exact posterior sits closer to the truth along the ridge than NPE on the same datasets (A→AA 0.46–0.49 vs 0.33–0.39; τ_A 17.7–18.3 h vs 15.7–16.6 h), so part of NPE's ridge bias at the real-like design may be the network and summaries — consistent with the GW finding (§9). At 512 clones per endpoint NPE 20k and exact agree. These posteriors are not converged and their widths are not reliable; the Fisher/CR result (practical, not structural ridge) stands. Follow-up: converged exact BH posteriors (2–3 days of compute) or a faster sampler (HMC with the population covariance as metric, a reparametrisation along the ridge).

Comment

- **The ridge is practical**: the Fisher information is nonsingular for both laws. The CR sd of A→AA (0.23) and of the log A lifetime (0.19, 3.8 h) at 266 clones are as large as NPE's bias (0.23, 4 h), so the data at the real-like design cannot pin the point on the ridge, without the model being degenerate. $\approx 6 \times 10^3$ clones would give A→AA sd 0.05 — consistent with NPE 20k moving most of the way only at 12 288 clones (§12.1).
- **B's row is much weaker in BH than in GW**: CR sd of B→A 0.82 (B divisions fixed) and 0.99 (free) vs 0.28 in GW; the weakest direction trades B→A against B persist and B's lifetime. In BH fixing B divisions is not enough: B's lifetime has to be known as well ($\approx 10^5$ clones for B→A sd 0.05 from endpoint counts).
- Consequence for this part: lifetime information (§15, §16) remains the practical lever, now justified as "the data are too weak" rather than "the model cannot tell". Caveat: local bounds, one truth; the exact posteriors that would show how far the bound is reached are preliminary (box above).

Leads to. Two questions: is the ridge bias a property of this truth, whose A→AA has only 1.8% prior mass above it (X5d, §14)? And can partial lifetime information — a literature prior — replace exact lifetimes (X5b, §15)?

14 X5d — Role of the prior III: performance averaged over the prior (GW and BH)

> **Why this next.** The three failures so far — E0's B row (§5), the BH ridge (§13), and later the original E2c (§31) — all occurred at truths in the prior tail. X5a (§6) showed that the B-row failure is not a tail effect. Is the BH ridge more than a tail effect, or does BH work for typical truths?

Question

Every other experiment uses one hand-picked truth, partly in the prior tail. Averaged over truths drawn from the prior, are GW and BH calibrated and informative?

Setup

Data = inference	`reversible_ab`, 30 truths drawn from the prior (`truth.source=prior`, `data_seed` = seed 0–29; no redraws needed); BH lifetime prior log-uniform [10, 80] h
Design	real-like: GW generations 6/10, BH 144/240 h; 165/101 clones
Seeds / runs	30 per simulator (60)
Prior-mean baseline	error of the uniform Dirichlet mean (1/6 for A's outcomes, 1/3 for B's)
Folder	`results/x5/d_gw/`, `results/x5/d_bh/` (`prior_average_parameters.csv`, `prior_average_runs.csv`)

Result

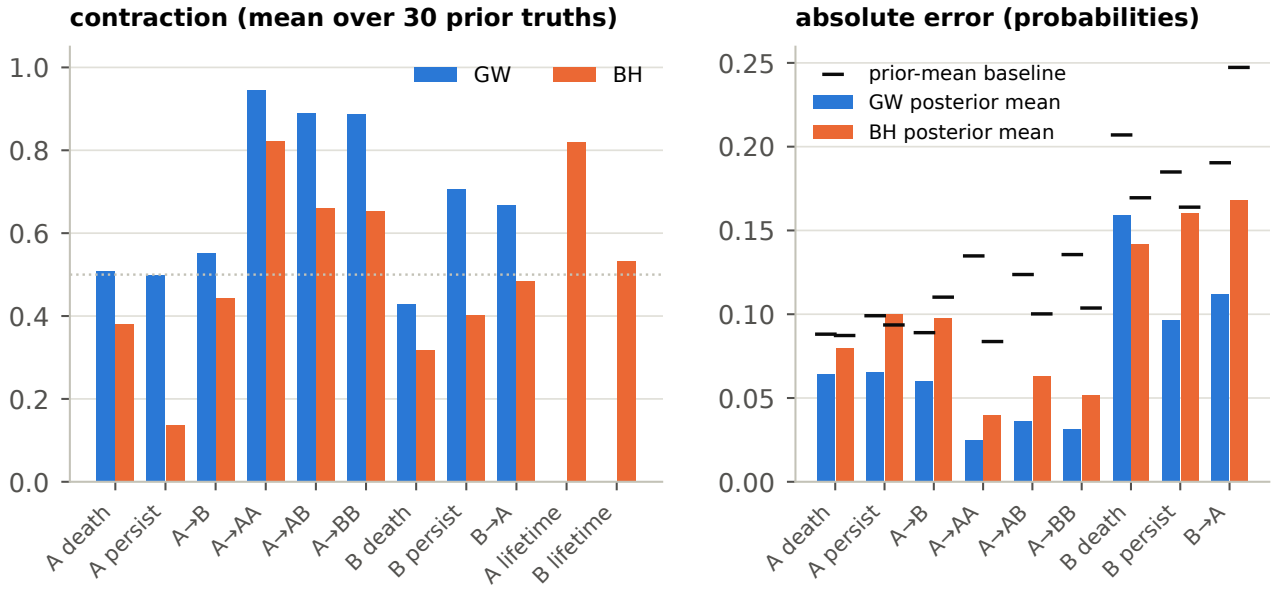


Figure 18: X5d: per-parameter contraction (left) and absolute error of the posterior mean (right; black ticks = prior-mean baseline), averaged over 30 truths drawn from the prior; GW vs BH.

	GW	BH
SBC coverage error	0.032	0.028
90% coverage across truths	0.90	0.90
Mean contraction	0.68	0.51
MAE probabilities (prior-mean baseline)	0.072 (0.139)	0.100 (0.129)
A→AA contraction	0.94	0.82
B→A contraction / coverage	0.67 / 0.90	0.48 / 0.80
Support mass median (min)	0.88 (0.54)	0.89 (0.29)
PPC outside 95%	0.006	0.003

BH mean contraction includes τ_A (0.82) and τ_B (0.53); relative lifetime error 0.21 / 0.35. Numbers: `tables/x5d.csv`.

Comment

- **The methods are calibrated on average**: with truths drawn from the prior both simulators give 90% coverage and SBC error ≈ 0.03. Single-truth failures are about specific truths and priors, not broken inference.
- GW learns A's divisions well (contraction 0.89–0.94), A's death/persist/switch about halfway (≈ 0.5), B's row moderately (0.43–0.71; B death weakest).
- **BH learns less** (contraction 0.51 vs 0.68; MAE 0.100 vs 0.072): the lifetimes absorb information; A persist barely moves (contraction 0.14, error no better than the prior mean) and B death / B persist stay close to the prior-mean baseline (0.142 / 0.160 vs 0.170 / 0.164).

- **The BH ridge is a tail-truth effect**: averaged over the prior A→AA is learned well (contraction 0.82), so the failure at $\theta^\star$ (A→AA 0.55, 0/5 covered, §13) is specific to a truth with 1.8% prior mass above it, where the prior picks the point on the lifetime/self-renewal ridge (at realistic sizes whatever the training budget — 20k simulations leave A→AA at 0.33; the data override the prior gradually, 0.44 at 512 and 0.50 at 4096 per endpoint with 20k, §12.1).
- Caveat: X5d uses real-like generations 6/10; the E0/design-comparison single-truth runs used 4/8/12, so they are not directly comparable.

Leads to. The ridge bites where the prior chooses the point. The natural fix is to give the prior information about lifetimes (X5b, §15).

15 X5b — Role of the prior IV: informative lifetime prior, right vs wrong (BH)

Why this next. Fixing lifetimes to the truth removes the ridge (§13), and X5d (§14) showed that the bias arises where the vague lifetime prior picks the point on it. Literature division times are available, but only approximately, and may be wrong.

Question

Fixing lifetimes resolves the BH ridge (§13). What does an informative (literature) lifetime prior do if it is centred on the truth, or 50% too long (truth outside the prior)?

Setup

Simulator / law	Bellman–Harris shape 8, `reversible_ab` (9 free); data $\theta^\star$ (τ_A 20 h, τ_B 30 h)
Lifetime priors (log-uniform)	vague: [10, 80] h for both (= BH ridge control, bit-identical to E3b); right (±25%): A [16, 25], B [24, 37.5] h; wrong (+50%): A [24, 37.5], B [36, 56.25] h
Seeds / runs	5 / 15
Folder	`results/x5/b_vague/`, `b_right/`, `b_wrong/` (`x5b_groups.csv`)

Result

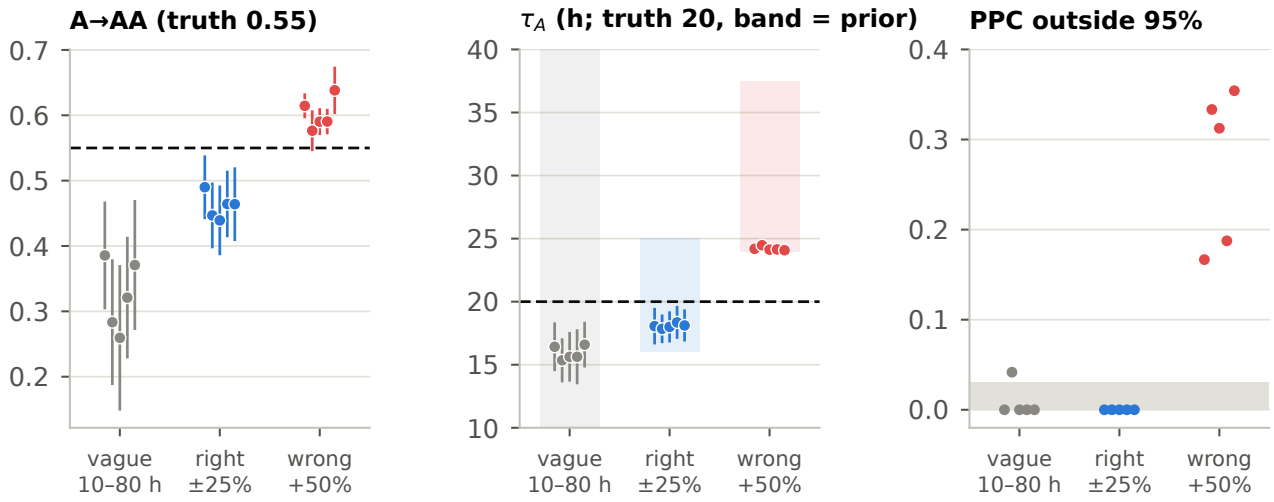


Figure 19: X5b: A→AA and τ_A per seed (posterior mean ± sd) and PPC per run for the three lifetime priors. Dashed = truth; shaded band = prior range of τ_A (vague prior extends to 80 h); grey band in the PPC panel = well-specified range.

Parameter (truth)	vague [10, 80] h	right (±25%)	wrong (+50%)
A→AA (0.55)	0.32 ± 0.10, 0/5	0.46 ± 0.05, 2/5	0.60 ± 0.03, 1/5
A→AB (0.05)	0.23 ± 0.12, 3/5	0.18 ± 0.08, 2/5	0.20 ± 0.06, 1/5
A→BB (0.29)	0.19 ± 0.09, 5/5	0.19 ± 0.06, 1/5	0.17 ± 0.04, 1/5
B→A (0.80)	0.63 ± 0.18, 5/5	0.69 ± 0.16, 5/5	0.68 ± 0.14, 5/5
τ_A (20 h)	15.9 ± 1.9, 0/5	18.1 ± 1.3, 4/5	24.2 ± 0.2 (prior edge), 0/5
τ_B (30 h)	20.2 ± 9.5, 4/5	29.5 ± 3.6, 5/5	49 ± 5.3, 0/5
MAE probabilities / rel. error lifetimes	0.109 / 0.26	**0.068** / 0.06	0.082 / 0.42
Support mass / PPC outside 95%	0.74–0.83 / 0.008	0.57–0.72 / 0.000	**0.00–0.02 / 0.27**
SBC coverage error	0.027	0.028	0.027

Mean over 5 seeds ± mean posterior sd, seeds covered. Lifetimes fixed exactly (§13): A→AA 0.51, MAE 0.059.

Comment

- **A right literature prior moves along the ridge towards the truth**: A→AA 0.32 → 0.46, lifetimes about right, MAE 0.109 → 0.068 — most of the way to fixing lifetimes exactly (0.51, 0.059), at the real-like design. More organoids alone reach a similar point only at 4096 per endpoint and with 20k simulations (A→AA 0.50); a 20k network does not help at the real-like design (0.33) and gives 0.44 at 512 per endpoint on three endpoints (§12.1). At realistic sizes lifetime information is the lever.
- **A prior 50% too long is clearly flagged**: support ≈ 0, PPC 0.27 (0.17–0.35), A's lifetime pinned at the prior's lower edge (the data push against it). Consistent with E3a, where too-long lifetimes fail visibly and too-short ones bias silently. Probabilities remain off (1/5 covered) despite high contraction.
- Caveat: one truth ($\theta^\star$), 5 seeds; a too-short wrong prior was not tested.

Leads to. Lifetime information works. In practice it will be a measured, possibly pooled value fixed in the model: E3a (§16) tests per-type vs global and wrong fixed values.

16 E3a — Per-type division time vs one global value (BH)

Why this next. The ridge (§13) and X5b (§15) make A's division time the key measurement. The real division time is estimated globally, but B may divide more slowly than A. What happens if one pooled or slightly wrong value is fixed?

Question

If B divides more slowly than A, can both lifetimes be inferred? What happens if both are fixed to one global (pooled) value, or to a wrong one (±25%)?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	free arm: `reversible_ab` (9); fixed arm: lifetimes fixed to T (7)
Data model	$\tau_A = 20$ h, $\tau_B \in \{20, 30, 45\}$ h; outcome probabilities $\theta^\star$
Inference model	(i) both lifetimes inferred; (ii) both fixed to pooled $T = (20 + \tau_B)/2$; (iii) pooled ±25%
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 60 (3 τ_B × (1 free + 3 fixed))
Control	free arm at the same τ_B
Folder	`results/e3a/free`, `results/e3a/fixed`

Result

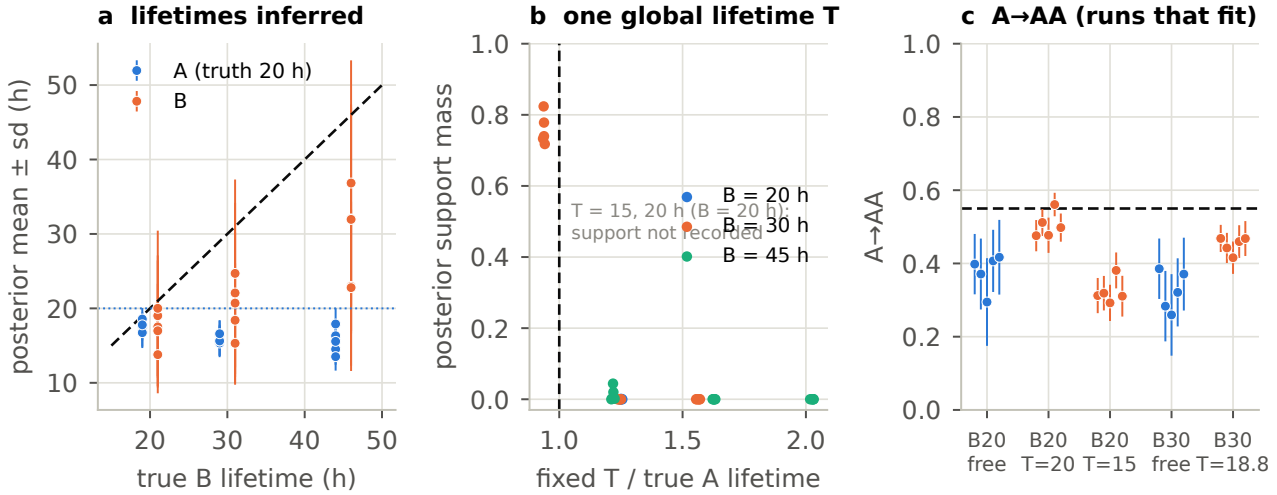


Figure 20: (a) Inferred lifetimes vs true τ_B (per run, ± sd); dotted = true τ_A. (b) Posterior support mass vs fixed T relative to the true τ_A (20 h). (c) A→AA in the settings that fit.

τ_B / T		A persist	A→AA	A→AB	B→A	MAE	PPC	support
20 / free	control	0.11	0.38 [1/5]	0.21	0.68	0.087	0.004	n/r
20 / **20**	T = truth	0.05	**0.50 [4/5]**	0.14	0.75	**0.042**	0.004	n/r
20 / 15	−25%	0.20 (+0.08)	0.32	0.15	0.79 (+0.10)	0.081	0.013	n/r
30 / 18.75	−25%	0.05	0.45 (+0.13)	0.13	0.59 [1/5]	0.072	0.000	0.76
any / ≥ 24.4	$T > \tau_A$	no estimate (0–2 000 samples; boundary values)				—	0.35–0.51	≈ 0

Free arm: τ_A 17.5 / 15.9 / 15.6 h and τ_B 17.5 / 20.2 / 29.3 h for true τ_B 20 / 30 / 45 (contraction of τ_B 0.84 / 0.75 / 0.49).

Comment

- Lifetimes fixed at the truth give the best BH recovery of all (A→AA 0.50, MAE 0.042, contraction 0.92) — direct evidence for the ridge.
- Inferred τ_A is biased short whatever τ_B; τ_B is compressed towards τ_A and weakly informed.
- T longer than A's lifetime ⇒ the model cannot fit (support ≈ 0, PPC 0.35–0.51): *detectable*. Depends on T vs τ_A, not on the pooled mean.
- T a little shorter than τ_A ⇒ fits normally but shifts outcomes by up to 0.10 (A persist absorbs the faster clock): *undetectable*.
- Consequence: measure A's division time per type; never use a pooled mean when B is slower. Caveat: coarse T grid (failure threshold between 20 and 24.4 h).

Leads to. Fixing the mean lifetime also fixes an assumed variability (Gamma shape 8, CV 0.35, in every BH run). E3b (§17) tests that assumption.

17 E3b — Desynchronisation: lifetime CV misspecified (BH)

Why this next. Every BH fit so far assumed lifetime shape 8 (CV 0.35); E3a (§16) showed that a wrong lifetime *mean* can bias silently. Real cells desynchronise; is a wrong *variability* equally harmful?

Question

How biased is the fit when the data's lifetime variability (Gamma shape 1–32, CV 1–0.18, or per type) differs from the assumed shape 8 (CV 0.35)? Does the PPC see it?

Setup

Simulator	Bellman–Harris
Law (free)	`reversible_ab` (9)
Data model	shape ∈ {1, 2, 4, 8, 32} and per type {A 4, B 16}, {A 16, B 4}
Inference model	shape 8 always (one network per seed: paired shifts)
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 35
Control	data shape 8 (= BH baseline)
Folder	`results/e3b/`

Result

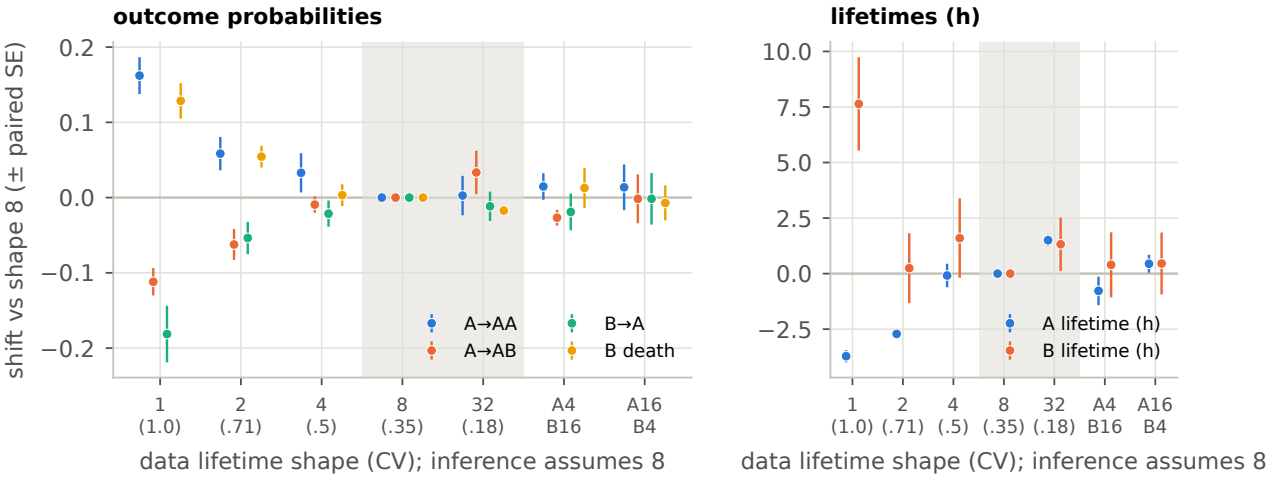


Figure 21: Shift vs shape 8 (± paired SE); grey band = CV 0.18–0.35, around the assumed value.

Data shape (CV)	A→AA	A→AB	B death	B→A	τ_A (h)	τ_B (h)	PPC
8 (0.35) control	0.32	0.23	0.17	0.63	15.9	20.2	0.008
1 (1.0)	+0.16 ± .02	−0.11	+0.13	−0.18± .04	−3.7± 0.3	+7.6 ± 2.1	0.004
2 (0.71)	+0.06	−0.06	+0.05	−0.05	−2.7	+0.2	0.000
4 (0.5)	+0.03	−0.01	+0.00	−0.02	−0.1	+1.6	0.000
32 (0.18)	+0.00	+0.03	−0.02	−0.01	+1.5 ± 0.2	+1.3	0.000
A 4 / B 16; A 16 / B 4	≤ 0.01	≤ 0.03	≤ 0.01	≤ 0.02	−0.8 / +0.4	≤ 0.5	≤ 0.004

Comment

- CV 0.18–0.5 (incl. per-type mixes) is harmless: shifts ≤ 0.035 and ≤ 1.6 h, small vs posterior sd (≈ 0.1, 2 h).
- Near-exponential cycle times are read as a faster clock with a different division split; bias is large and invisible to the PPC. (A→AA moves *towards* the truth only because the control is biased low.)
- Consequence: a rough CV estimate (live imaging) suffices; if CV $\gtrsim 0.7$ change the shape in the model. Shape was never inferred; support mass not recorded.

Leads to. One more parameter can slow growth without changing the clock: quiescence (A persist). E3c (§18) asks whether it lies on the same ridge.

18 E3c — Quiescence vs division time (BH)

Why this next. The ridge (§13) trades self-renewal against lifetime. Quiescent cells (A persist) also slow growth, so they may be a third direction on it; if so, fixing quiescence wrongly (e.g. to 0) would shift the lifetimes.

Question

Are quiescence (A persist) and A's lifetime confounded? What happens if quiescence is fixed to 0 while the data have A persist $\in \{0.03, 0.15, 0.30\}$?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	free arm `reversible_ab` (9); fixed arm A persist = 0 (8)
Data model	A persist 0.03 / 0.15 / 0.30, other A entries rescaled (A→AA 0.55 / 0.48 / 0.40)
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 30
Control	free arm at the same data
Folder	`results/e3c/persist_free`, `results/e3c/persist_fixed0`

Result

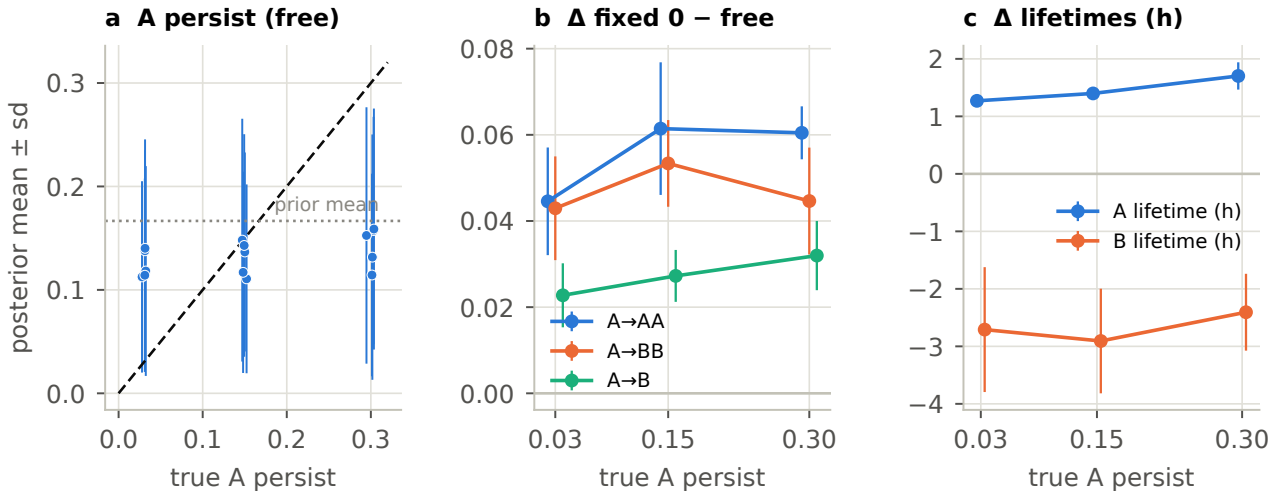


Figure 22: (a) A persist posterior (free) vs truth; dotted = prior mean 1/6. (b, c) Shift fixed-0 minus free (± paired SE).

True A persist	free: A persist (contr.)	free: τ_A	fixed 0: $\Delta\tau_A$	ΔA→AA	$\Delta\tau_B$	PPC
0.03	0.12 (0.51)	15.9	+1.3 ± 0.1	+0.05	−2.7	0.000
0.15	0.13 (0.47)	17.9	+1.4	+0.06	−2.9	0.000
0.30	0.14 (0.35)	19.3	+1.7	+0.06	−2.4	0.004

Posterior correlations (free arm): corr(A persist, τ_A) ≈ -0.6; corr(A→AA, τ_A) $\approx +0.7$.

Comment

- Quiescence is not identifiable from endpoint counts: the posterior stays near the prior whatever the truth.
- Quiescence, lifetime and self-renewal lie on one ridge (the growth rate).
- Fixing it to 0 gives a constant small shift (longer lifetimes absorb it), independent of the true value, undetected by the PPC.
- Consequence: fix A persist from independent evidence (Ki67) or report lifetimes and self-renewal conditional on it; measuring A's division time also breaks the confound. Q4 (meaning of "inattivazione") open.

Leads to. Lifetime information is what BH needs and what GW, with its fixed generation, has built in. Can the simpler GW model stand in for BH on BH data, and with which generation length (X3, §19)?

19 X3 — Galton–Watson fitted to Bellman–Harris data

Why this next. The ridge (§13) and E3a (§16): BH recovers self-renewal once lifetimes are fixed. GW has no lifetimes and no ridge, and Part IV will show that it flags errors BH absorbs. The real data must be fitted with one of the two, and GW is simpler — but the two do not share a time unit. X3 puts GW on an hour axis (a fixed generation length τ, §2.5) and fits it to BH data.

Question

Real-data fits need a time model. GW (synchronous generations) is simpler, has no lifetime ridge and flags more errors; BH (Gamma lifetimes) is closer to biology. If the data are BH, can GW with a fixed generation length of τ hours recover the outcome probabilities, which τ works, and do the checks flag a wrong τ?

Setup

Data model	BH `reversible_ab`, $\theta^\star$ (τ_A 20 h, τ_B 30 h, shape 8), perfect observation; the same 5 datasets in every arm (verified by dataset hash)
Inference model	GW `reversible_ab` on an hour axis (`simulator=galton_watson_hours`, §2.5): $\tau \in \{16, 20, 24, 30\}$ h, i.e. 9/15, 7/12, 6/10, 5/8 generations at 144/240 h (rounding ≤ 0.2 generation); control: BH, the true model
Design	real-like: 144/240 h × 165/101
Inference / budget	NPE amortised, 5k; SBC 200, PPC 200. Scored on the 9 shared outcome probabilities (GW has no lifetimes)
Seeds / runs	5 / 25 (20 GW + 5 BH)
Folder	`results/x3/gw`, `results/x3/bh_control`

Result

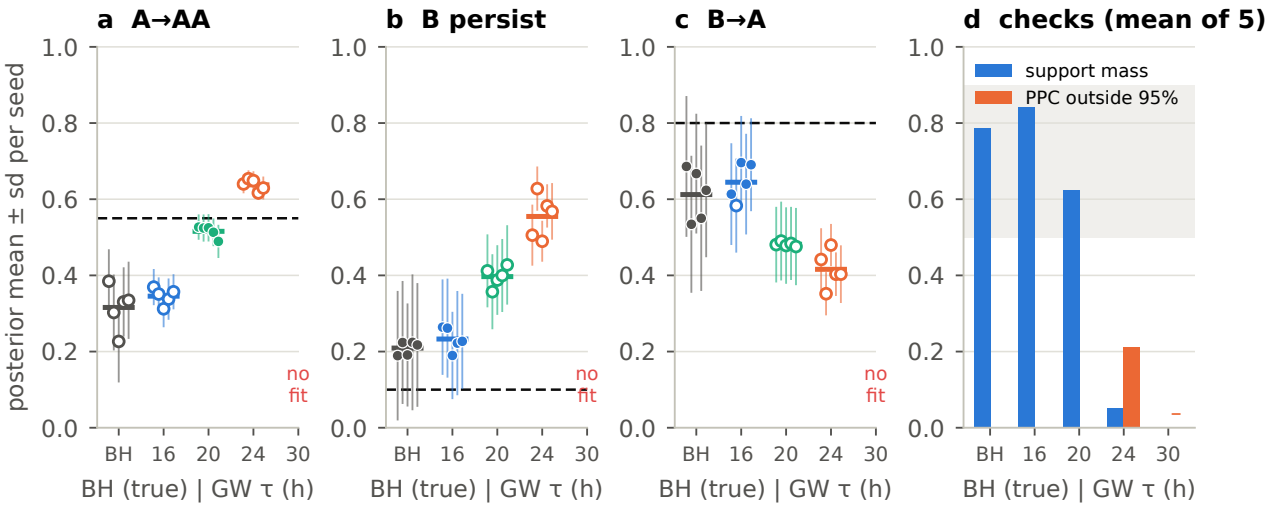


Figure 23: X3: GW with a fixed generation length τ fitted to BH data (τ_A 20 h, τ_B 30 h). (a–c) Posterior mean ± sd per seed (filled = truth in the 90% interval, open = not covered; bar = mean over seeds); dashed = truth; BH = the true model (control). τ = 30 h gives no posterior samples inside the prior. (d) Posterior support mass and PPC fraction outside 95% (mean of 5 seeds); grey band = well-specified support range.

Arm	MAE	90% cov.	contr.	support	PPC	SBC	A→AA (0.55)	B persist (0.10)	B→A (0.80)
BH (true)	0.116	0.82	0.58	0.79	0.000	0.026	0.32 [0/5]	0.21 [5/5]	0.61 [5/5]
GW $\tau = 16$ h	0.088	0.87	0.81	0.84	0.000	0.034	0.35 [0/5]	0.23 [5/5]	0.64 [4/5]
GW $\tau = 20$ h	**0.086**	0.78	0.89	0.62	0.000	0.034	**0.52 [5/5]**	0.40 [0/5]	0.48 [0/5]
GW $\tau = 24$ h	0.122	0.27	0.96	**0.05**	**0.21**	0.032	0.64 [0/5]	0.55 [0/5]	0.42 [0/5]
GW $\tau = 30$ h	—		—	**0.00**	—	0.032	no posterior samples inside the prior		

Means over 5 seeds; [covered/5].
Numbers: `results/x3/x3_groups.csv`, `x3_parameters.csv`; plotted: `tables/x3.csv`.

Comment

- **A GW model with τ at or below the fastest lifetime fits without any warning** (PPC 0, support 0.62–0.84). τ = A's lifetime (20 h) recovers A's row (A→AA 0.52; all A parameters 5/5 covered), but B divides every 30 h, not 20 h, so GW explains B's slower cycle as **persistence** (B persist 0.40 vs 0.10) and lowers B→A (0.48 vs 0.80): a silent bias in B's row. With $\tau = 16$ h the bias moves into A's row (A→AA 0.35).
- **τ longer than A's lifetime fails loudly**: GW cannot produce enough cells (τ 24 h: support 0.05, PPC 21%; τ 30 h: no posterior samples inside the prior). Same mechanism as a fixed global lifetime longer than τ_A in BH (E3a, §16).
- **GW with the right τ beats the true BH model on A's row** (MAE 0.086 vs 0.116): fixing the generation length is equivalent to fixing A's lifetime, which removes the BH lifetime/self-renewal ridge (cf. §13, lifetimes fixed: A→AA 0.51). The price is that differences in division time between types are absorbed into persistence.
- Consequence: GW is a usable approximation only with τ set from measured division times of the fastest type, and then B's persistence must be read as "slower cycle", not quiescence. Without division-time data τ is not identifiable from these checks inside 16–20 h, while $\tau \geq 24$ h is rejected. On the real data GW at 24 h per generation fails on the 2D counts and cannot reach the 3D sizes at all (§36).
- Caveats: one truth (A 20 h, B 30 h), 5 seeds, NPE 5k.

Leads to. Part III closes the baseline analysis: GW is the target, BH needs lifetime information, and GW stands in for BH only with a measured generation length. Part IV introduces the errors of the real data one at a time, each compared with these baselines.

Part IV

Misspecified biology and assay

Motivation. The real CRC0322 data will differ from the model in ways Parts I–III did not consider. This part introduces them one at a time, in the order of the experiment: what is seeded (founder type and number, E1a), what dies (E1b, E1c), how cells are labelled (mis-typing, E2a/b) and whether a third population exists (E2c). Each error is measured as a shift from the well-specified control of the same sweep (the BH control is biased, §13) and checked for detectability. Most errors are run in both simulators because the first BH result (§21) suggested that the free time scale hides errors; GW then shows the same error without that freedom. The hidden-type experiment ends by reusing the tail-truth lesson of X5. Section 33 collects detectability across the part, and X4 (§34) combines the realistic errors into an error budget.

20 E1a — Wrong founder type (organoid death as initial conditions; BH and GW)

Why this next. All runs so far assume each organoid starts from one A cell. Organoid loss at seeding is assumed (from the biology) to change only which clones start — their founder type and number — not the dynamics. The first test: B-founded organoids.

Question

If some organoids are founded by B cells (fraction $1 - f$) but the model assumes A founders, are the dynamics (A, B rows) biased? Does inferring the founder mix help?

Setup

Simulator	Bellman–Harris (shape 8) and Galton–Watson
Law (free)	`reversible_ab`: BH 9, GW 7; inferred arm adds `founder.A` (Dirichlet(1,1))
Data model	founder A with prob. $f \in \{1, 0.9, 0.7, 0.5\}$, else B; dynamics $\theta^\star$
Inference model	fixed arm: founder = A (wrong for $f < 1$); inferred arm: founder share free
Design	real-like: BH 144/240 h, GW generations 6/10; 165/101 organoids
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 80 (4 f × 2 arms × 2 simulators)
Control	$f = 1$ of the same arm (same trained network per seed: paired shifts)
Folder	`results/e1a/founder_type/{fixed,inferred}_{bh,gw}`

Result

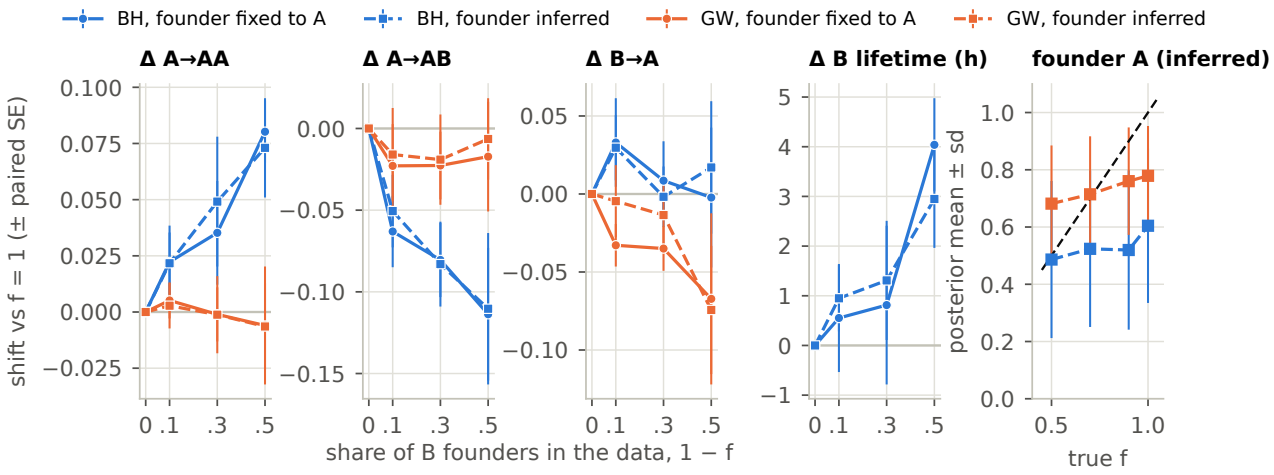


Figure 24: Shift of posterior means vs the $f = 1$ control (± paired SE over 5 seeds); blue BH, orange GW; solid = founder fixed to A, dashed = founder inferred. Right: inferred founder A share vs truth.

	BH, founder fixed				GW, founder fixed			
	control	$f = 0.9$	0.7	0.5	control	0.9	0.7	0.5
A→AA (0.55)	0.32 [0/5]	+0.02	+0.04	+0.08 ± .01	0.52 [3/5]	+0.01	−0.00	−0.01
A→AB (0.05)	0.23 [3/5]	−0.06	−0.08	−0.11± .04	0.12 [3/5]	−0.02	−0.02	−0.02
A→B (0.05)	0.07	+0.01	+0.01	+0.02 ± .006	0.035	+0.00	+0.01	+0.02 ± .006
B death (0.10)	0.17	−0.02	+0.00	−0.02	0.10	+0.01	+0.02	+0.03 ± .014
B→A (0.80)	0.63 [5/5]	+0.03	+0.01	−0.00	0.76 [5/5]	−0.03	−0.04	−0.07± .06
τ_B (30 h)	20.2	+0.6	+0.8	+4.0 ± 0.9			—	
MAE (probabilities)	.109	.090	.095	.089	.033	.033	.036	.047
PPC	.008	.008	0	0	.004	.008	.017	.017
support mass	n/r	n/r	n/r	n/r	0.67	0.69	0.69	0.72

Founder inferred: `founder.A` posterior mean (contraction) for $f = 1$ / 0.9 / 0.7 / 0.5:
BH 0.60 / 0.52 / 0.52 / 0.49 (0.06–0.12) GW 0.78 / 0.76 / 0.71 / 0.68 (0.49–0.62); GW support mass 0.39–0.59

Comment

- **Small shifts, PPC blind** in both simulators. BH: A→AB −0.11, A→AA +0.08 (towards the truth, by accident of the control's bias), τ_B +4 h at $f = 0.5$; all < 1 posterior sd. GW: shifts ≤ 0.07 (B→A, not resolved: ±0.06); A→B +0.02 and B death +0.03 resolved but ≈0.5 sd.
- **GW well-specified baseline is good**: MAE 0.033, contraction 0.89, A→AA 0.52 — the BH bias is specific to BH (ridge).
- **Founder mix**: not identifiable in BH (stays at prior 0.5); partly in GW (monotone but compressed: 0.68 at $f = 0.5$, 0.78 at $f = 1$). Inferring it removes none of the shifts in either simulator and costs τ_B (−3.9 h, BH).
- Plausible reason: B founders switch to A quickly (B→A = 0.8), so B- and A-founded clones look alike by day 6.
- Consequence: keep single-A founders unless measured at seeding; if measured, fix to the measured value. $f < 0.5$ not tested.

Leads to. Founder type is silent and small. The other initial condition is the number of founder cells (§21).

21 E1a — Wrong number of founder cells (BH)

Why this next. Wrong founder type was harmless (§20). Wells may also contain multi-cell seeds, which make clones larger from the start — an effect that could be mistaken for faster growth.

Question

If some organoids start from 2–3 A cells (P(1 cell) = p_1) but the model assumes single-cell seeding, are the dynamics biased? Can the founder size be inferred?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	`reversible_ab` (9); inferred arm adds `founder_size` (sizes 1–3, Dirichlet(8,1,1))
Data model	founder size $\{1, 2, 3\}$ with $p_1 \in \{1, 0.85, 0.6\}$ ($\{.85, .12, .03\}$, $\{.6, .3, .1\}$), all A
Inference model	sizes [1] (wrong for $p_1 < 1$) or [1,2,3] inferred
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 30
Folder	`results/e1a/founder_number_bh/` (GW: §22)

Result

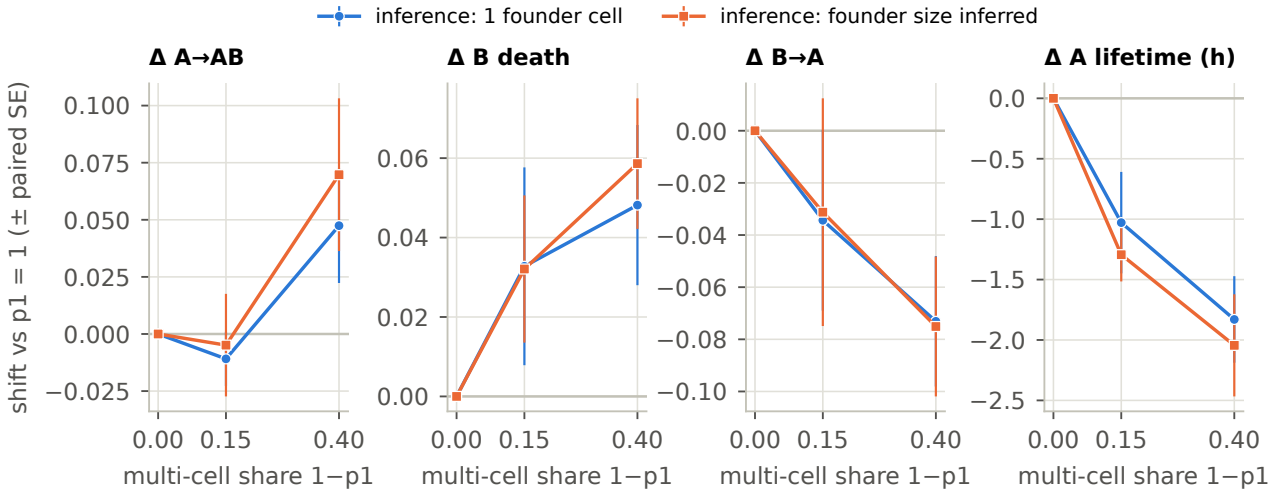


Figure 25: Shift vs the $p_1 = 1$ control of the same arm (± paired SE).

Arm, p_1	A→AB (.05)	B death (.10)	B→A (.80)	τ_A (20 h)	`founder_size.1`
[1], control	0.23	0.17	0.63 [5/5]	15.9	fixed
[1], 0.85 / 0.6	−0.01 / +0.05	+0.03 / +0.05 ± .02	−0.03 / −0.07 ± .03	−1.0 / −1.8 ± 0.4	fixed
[1,2,3], 0.85 / 0.6	−0.00 / +0.07	+0.03 / +0.06	−0.03 / −0.08	−1.3 / −2.0	0.81 / 0.78 (contr. ≤ 0.11)

Comment

- 40% multi-cell clones shift mostly lifetimes and B's row; coverage unchanged; PPC ≤ 0.008 (silent).
- Founder size stays at its prior (0.78–0.83); inferring it does not remove the shifts.
- Observed fraction rises with multi-cell seeding (0.95 → 0.97–0.98): fewer clones go extinct.
- Consequence: tolerable at $p_1 \geq 0.6$; do not infer founder size; measure it at seeding if needed. No support mass recorded (pre-`43e9e44` code).

Leads to. The shifts landed mostly on the lifetimes, and no check fired: does the free time scale absorb the error (as it absorbs self-renewal on the ridge, §13)? The same sweep in GW (§22) answers this.

22 E1a — Wrong number of founder cells (GW)

Why this next. In BH (§21), 40% multi-cell clones moved τ_A by −1.8 h and B→A by only −0.07, silently. GW has no lifetimes to absorb the extra growth.

Question

As §21 for Galton–Watson: are the dynamics biased by multi-cell founders, can the founder size be inferred, and does the answer depend on the time model?

Setup

Simulator	Galton–Watson
Law (free)	`reversible_ab` (7); inferred arm adds `founder_size` (sizes 1–3, Dirichlet(8,1,1))
Data model	founder size $\{1, 2, 3\}$, $p_1 \in \{1, 0.85, 0.6\}$, all A
Design	real-like generations 6/10 × 165/101
Inference / budget	NPE amortised, 5k (one network per arm and seed: paired shifts)
Seeds / runs	5 / 30
Folder	`results/e1a/founder_number_gw/`

Result

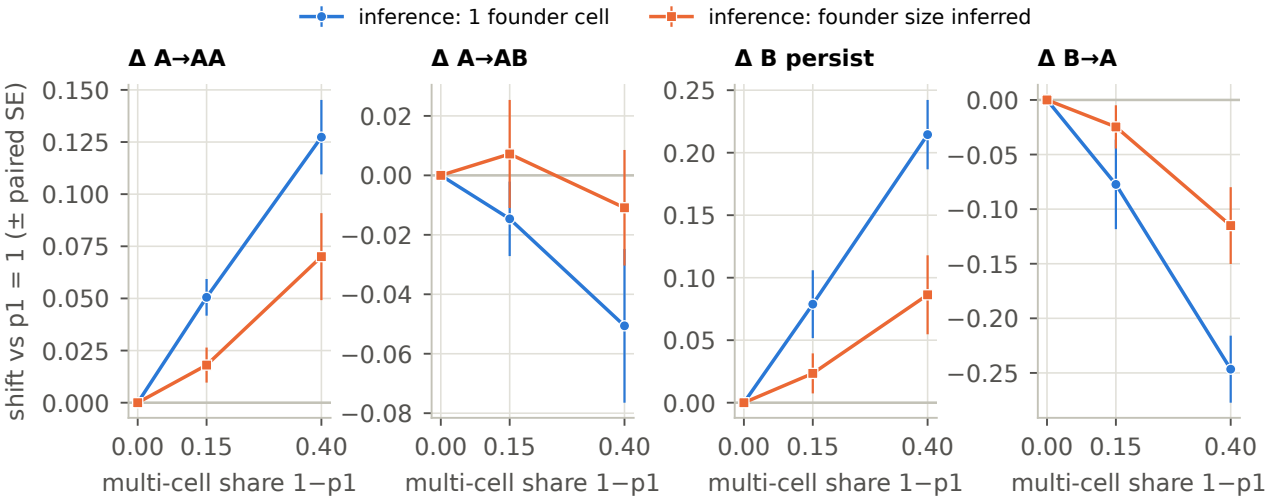


Figure 26: E1a founder number, GW: shift vs the $p_1 = 1$ control of the same arm (± paired SE).

Arm, p_1	A→AA (.55)	B persist (.10)	B→A (.80)	`founder_size.1`	PPC	support (min)
[1], control	0.52 [3/5]	0.13	0.76 [5/5]	fixed	0.004	0.67 (0.44)
[1], 0.85	+0.05 ± .01	+0.08 ± .03	−0.08 [4/5]	fixed	0.042	0.56 (0.21)
[1], 0.6	**+0.13 ± .02** [0/5]	**+0.21 ± .03** [0/5]	**−0.25 ± .03** [0/5]	fixed	**0.117**	**0.16** (0.015)
[1,2,3], control	0.46 [3/5]	0.09	0.83 [5/5]	0.85	0.000	0.55 (0.52)
[1,2,3], 0.85 / 0.6	+0.02 / +0.07 ± .02	+0.02 / +0.09	−0.02 / −0.12 ± .04 [5/5]	0.80 / 0.71	≤ 0.004	0.53 / 0.56

Comment

- **A large, partly detectable error in GW**: 40% multi-cell clones ignored inflate self-renewal (+0.13) and lower B→A (−0.25), all 0/5 covered; the fit is flagged (PPC 0.12, support 0.16). At $p_1 = 0.85$ the shifts are a third of that and the checks borderline.
- **Contrast with BH** (§21): there the same error moved B→A by only −0.07 and was silent — the lifetimes absorb part of the extra growth. A free time scale hides the error.
- **Inferring the founder size helps in GW**: misfit removed (PPC ≤ 0.004), shifts halved and covered, `founder_size.1` tracks p_1 (0.85 / 0.80 / 0.71, contraction ≈ 0.2). Cost on single-cell data: A→AA −0.06 ± 0.01, mean contraction 0.89 → 0.73.
- Consequence: record at seeding whether wells contain multi-cell seeds; if ≳15% of organoids start from several cells, fix the measured distribution or infer it. $p_1 < 0.6$ not tested.

Leads to. Founders are one route by which organoid death enters; per-type cell death also changes which organoids are observed at all (E1b, §23).

23 E1b — Death rates and the observed-organoid fraction (BH and GW)

Why this next. E1a (§20–§22) modelled organoid loss through initial conditions. The real data report only organoids that are still detected, and cell death controls how many go extinct: if the number seeded is known, the observed fraction is an extra datum.

Question

How strongly do A death and B death change the fraction of seeded organoids still detected at the endpoints? Is it an informative extra datum?

Setup

Simulator	BH (simulated, rejection loop) and GW (exact $1 - P$(no captured cell))
Law	`reversible_ab`; A death ∈ {0, .03, .1, .2, .3} × B death ∈ {0, .1, .3, .5}, other entries of each row rescaled in proportion
Data vs inference	no inference used (token budget: 256 simulations, 5 epochs; posteriors meaningless)
Design	real-like: BH 144/240 h, GW generations 6/10; 165/101 recovered organoids
Seeds / runs	1 dataset per cell / 40 (20 BH + 20 GW)
Folder	`results/e1b/bh`, `results/e1b/gw`

Result

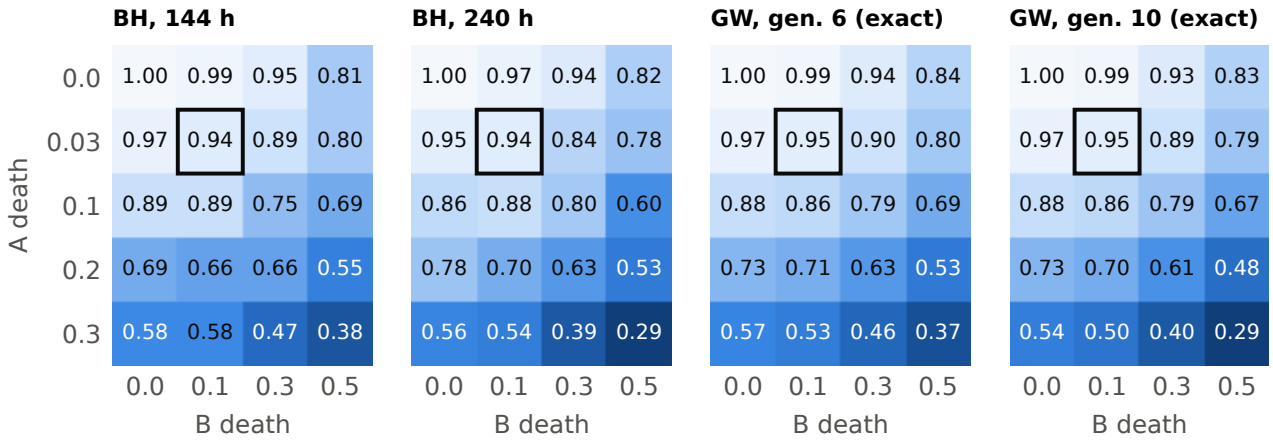


Figure 27: Observed-organoid fraction (detected/seeded) per (A death, B death); box = reference truth. BH: binomial SE ≈ 0.02–0.03 (100–530 seeded clones); GW exact.

(A death, B death)	BH 144 h	BH 240 h	GW gen. 6	GW gen. 10
(0, 0)	1.00	1.00	1.000	1.000
(0.03, 0.1) reference	**0.94**	**0.94**	**0.953**	**0.952**
(0.3, 0)	0.58	0.56	0.565	0.543
(0, 0.5)	0.81	0.82	0.840	0.832
(0.1, 0.3)	0.75	0.80	0.793	0.785
(0.3, 0.5)	0.38	0.29	0.366	0.290

Comment

- A death dominates (0 → 0.3: 1.00 → 0.54–0.58); B death 0 → 0.5 only to ≈ 0.82.
- Losses happen early (day 6 ≈ day 10), except high B death with A death ≥ 0.1.
- Different (A, B) death pairs give the same fraction — confirmed by the exact GW values ((0, 0.5) and (0.1, 0.3): 0.84 vs 0.79; (0.03, 0.5): 0.80): one constraint on a ridge, not two measurements.
- GW exact and BH simulated agree on average to 0.02 (max 0.07, both signs; within ≈ 2 binomial SE of BH) at the same days/generations.
- Consequence: record the number of seeded organoids per well (Q5). A measured fraction well below 0.9 would indicate A death $\gtrsim$ 0.1. Using it as data (`observation=all_clones`) is tested in the follow-up (§24).
- Caveat: death also lowers the growth rate here (rows rescaled).

Leads to. E1b computed the observed fraction without any inference. Does recording the extinct organoids actually improve a fit (§24)?

24 E1b follow-up — all seeded organoids as data, and survivor summaries (BH and GW)

Why this next. E1b (§23): A death controls the fraction of seeded organoids still detected, so recording the number seeded (Q5) looked like an extra datum. Here it enters an actual fit: every seeded organoid is recorded, extinct ones as (0, 0) rows.

Question

E1b showed that death rates change the fraction of seeded organoids still detected. If every seeded organoid is recorded (extinct ones as (0, 0) rows, `observation=all_clones`), are death rates — and the other outcomes — recovered better than from detected organoids only? Does it help to compute the summaries on surviving organoids and add the extinct fraction (`summary=survivor_moments`)?

Setup

Simulator	Bellman–Harris (shape 8) and Galton–Watson
Law (free)	`reversible_ab` (BH 9, GW 7)
Data model	A death $\in \{0.03, 0.1, 0.3\}$ (other A outcomes rescaled as in E1b), B row $\theta^\star$
Arms	*detected*: `observation=perfect` (165/101 detected organoids); *all clones*: `observation=all_clones` (165/101 *seeded* organoids, `moments`); *all clones, survivor*: the same data with `summary=survivor_moments`. Data and inference share the observation model within each arm
Design	real-like: BH 144/240 h, GW generations 6/10
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 90 (3 arms × 3 A death × 2 simulators)
Folder	`results/e1b_all/{detected,all_clones,all_clones_survivor}`

Result

	A death (truth)	MAE all / surv. / det.	90% coverage all / surv. / det.	A death mean (sd) all / surv. / det.	B death contraction all / surv. / det.
GW	0.03	.063 / .057 / **.032**	.89 / .87 / .91	.024 (.013) / .028 (.018) / .029 (.021)	.91 / .87 / .86
GW	0.10	.081 / .068 / **.056**	.84 / .87 / .89	.058 (.027) / .059 (.028) / .048 (.031)	.76 / .72 / .73
GW	0.30	.134 / .119 / **.115**	.60 / .73 / .62	.167 (.069) / .179 (.066) / .141 (.064)	.27 / .30 / .30
BH	0.03	.109 / **.107** / .116	.87 / .80 / .82	.022 (.015) / .034 (.023) / .058 (.045)	.87 / .80 / .63
BH	0.10	.133 / .113 / **.110**	.82 / .87 / .84	.053 (.029) / .057 (.033) / .080 (.057)	.73 / .70 / .56
BH	0.30	.136 / .124 / **.119**	.87 / .78 / .87	.165 (.070) / .153 (.068) / .116 (.074)	.37 / .45 / .53

all = all seeded organoids, `moments`; surv. = all seeded organoids, `survivor_moments`; det. = detected organoids only. Means over 5 seeds; MAE over the outcome probabilities.
Numbers: `results/e1b_all/e1b_all_groups.csv`.

Comment

- **Recording extinct organoids sharpens the death rates**, mostly in BH: A death sd 0.015–0.033 vs 0.045–0.057 (detected only), B death contraction 0.87 vs 0.63 at low death.
- **Overall accuracy does not improve, and in GW it worsens** (MAE 0.063 vs 0.032 at A death 0.03). With the standard summaries the zero rows enter every $\log(1+n)$ statistic together with the survivors and blur the survivor distribution that carries the information on the other outcomes.
- **Survivor summaries recover part of the loss** (GW MAE 0.063 → 0.057, 0.081 → 0.068) and keep the sharper BH death estimates (sd 0.023–0.033), but GW is still worse than detected-only. Part of the gap is design: in the all-clones arms 165/101 are *seeded* organoids, so fewer organoids carry size and composition information.
- High death (0.3) is underestimated in every arm (0.12–0.18): strong death is partly traded against other outcomes.
- Consequence: the number of seeded organoids (Q5) is worth recording mainly for the death rates in BH; enter it as the extinct fraction with survivor-only summaries, not as zero rows in the count statistics.

Leads to. A and B death trade off along one constraint, and the extinct fraction sharpens them only modestly. If death rates must instead be fixed from a guess or an apoptosis marker, do the other parameters compensate (E1c, §25)?

25 E1c — Death rates fixed to wrong values (BH)

Why this next. E1b (§23) showed that different (A, B) death pairs produce the same observed fraction. Fixing deaths is tempting (fewer free parameters), but a wrong value could be compensated elsewhere.

Question

If A and B death are fixed (possibly wrongly) and the rest inferred, do other parameters compensate (flat PPC valley = death not identifiable once fixed)?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	`reversible_ab` with A death, B death fixed (7 free)
Data vs inference	data always $\theta^\star$ (A death 0.03, B death 0.1); inference fixes A death $\in \{0, 0.03, 0.2\}$, B death $\in \{0, 0.1, 0.4\}$
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k (retrained per setting)
Seeds / runs	5 / 45 (3 × 3 grid; the 4 × 4 grid is the GW sweep, §26)
Control	deaths fixed at truth (0.03 / 0.1)
Folder	`results/e1c/bh/`

Result

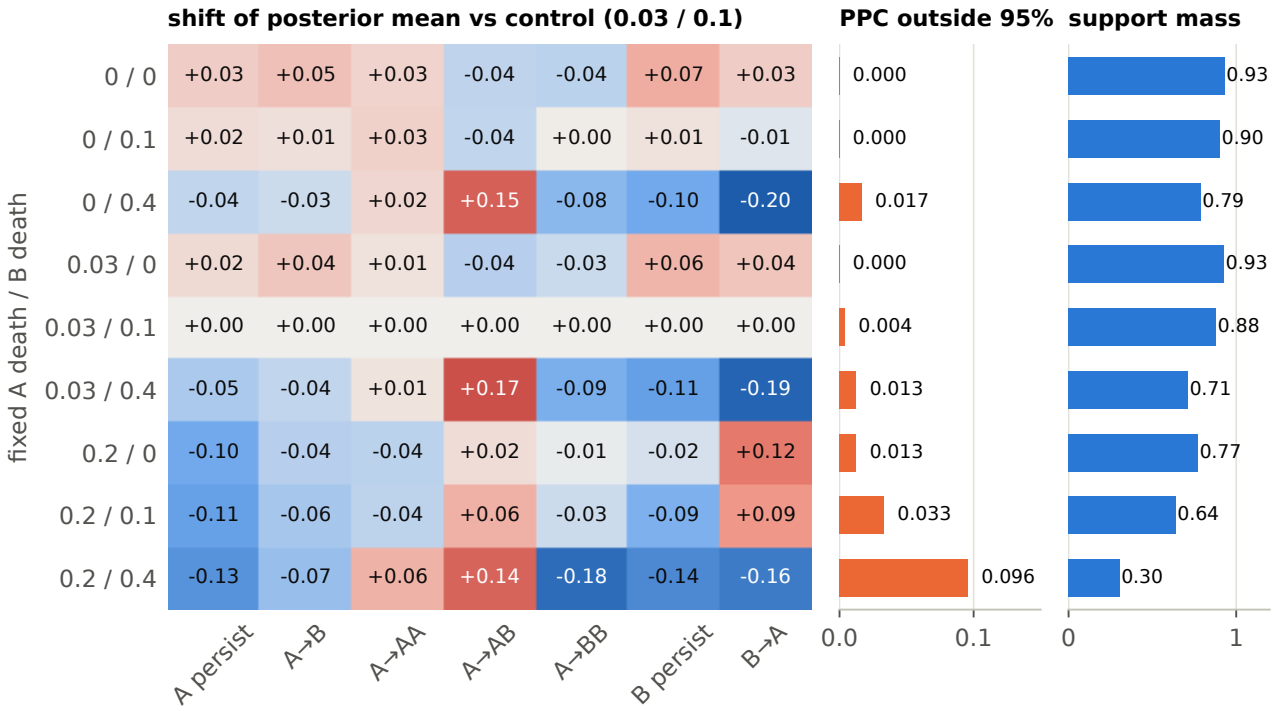


Figure 28: Left: shift of posterior means vs the control (deaths fixed at truth). Right: PPC and posterior support mass (well-specified free-death baseline: support 0.78).

Fixed A / B death	A persist	A→AB	B→A	τ_A	τ_B	PPC	support
0.03 / 0.1 (control)	0.15	0.18	0.67 [5/5]	16.1	18.1	0.004	0.88
0 / 0.4	−0.04	+0.15	−0.20 [0/5]	+0.3	+3.2	0.017	0.79
0.03 / 0.4	−0.05	+0.17	−0.19 [0/5]	−0.3	+2.9	0.013	0.71
0.2 / 0	−0.10	+0.02	+0.12	−2.1	−3.4	0.013	0.77
0.2 / 0.1	−0.11	+0.06	+0.09	−2.7	−2.1	0.033	0.64
0.2 / 0.4	−0.13	+0.14	−0.16 [0/5]	−2.4	+7.9	**0.096**	**0.30**
0 / 0, 0 / 0.1, 0.03 / 0	≤ 0.03	−0.04	≤ 0.04	≤ 0.8	≤ 2.4	0.000	0.90–0.93

Paired SE ≤ 0.03 (probabilities), ≤ 0.35 h (τ_A), ≤ 1.6 h (τ_B).

Comment

- Wrong deaths are absorbed by biologically sensible compensations: B death too high ⇒ less B→A, more A→AB; A death too high ⇒ less quiescence/switching, faster A clock.
- Bias, not information loss: mean contraction *rises* (0.72 → 0.84) while coverage falls (0.76 → 0.38): wrong models are more confident.
- Only the double error is flagged (PPC 0.096, support 0.30; seed 2 support 0.014 ⇒ not an estimate).
- Fixing deaths at the truth gains little vs free deaths (free baseline already constrains them: contraction 0.89 / 0.65).
- Consequence: do not fix death to a guess; keep free or measure (apoptosis marker). C2ST not run; intermediate values (0.1, 0.2) only in the GW sweep (§26).

Leads to. A death too high was absorbed by a faster A clock, again the free time scale (as in §21). The finer GW grid (§26) tests the same errors without it.

26 E1c — Death rates fixed to wrong values (GW, 4 × 4 grid)

Why this next. In BH (§25) only the double error was flagged; A death 0.2 was absorbed silently by a faster clock. Without lifetimes, is a wrong fixed death visible?

Question

As §25, on a finer grid and without the BH ridge: is there a flat PPC valley across fixed death values?

Setup

Simulator	Galton–Watson
Law (free)	`reversible_ab` with A, B death fixed (5 free)
Data vs inference	data always $\theta^\star$; inference fixes A death $\in \{0, .03, .1, .2\}$ × B death $\in \{0, .1, .2, .4\}$
Design	real-like generations 6/10 × 165/101
Inference / budget	NPE amortised, 5k (retrained per setting); bounded SBC (relaunched after the first launch hung; 1 SBC case excluded in 1 run)
Seeds / runs	5 / 80
Control	deaths fixed at truth (0.03 / 0.1)
Folder	`results/e1c/gw/`

Result

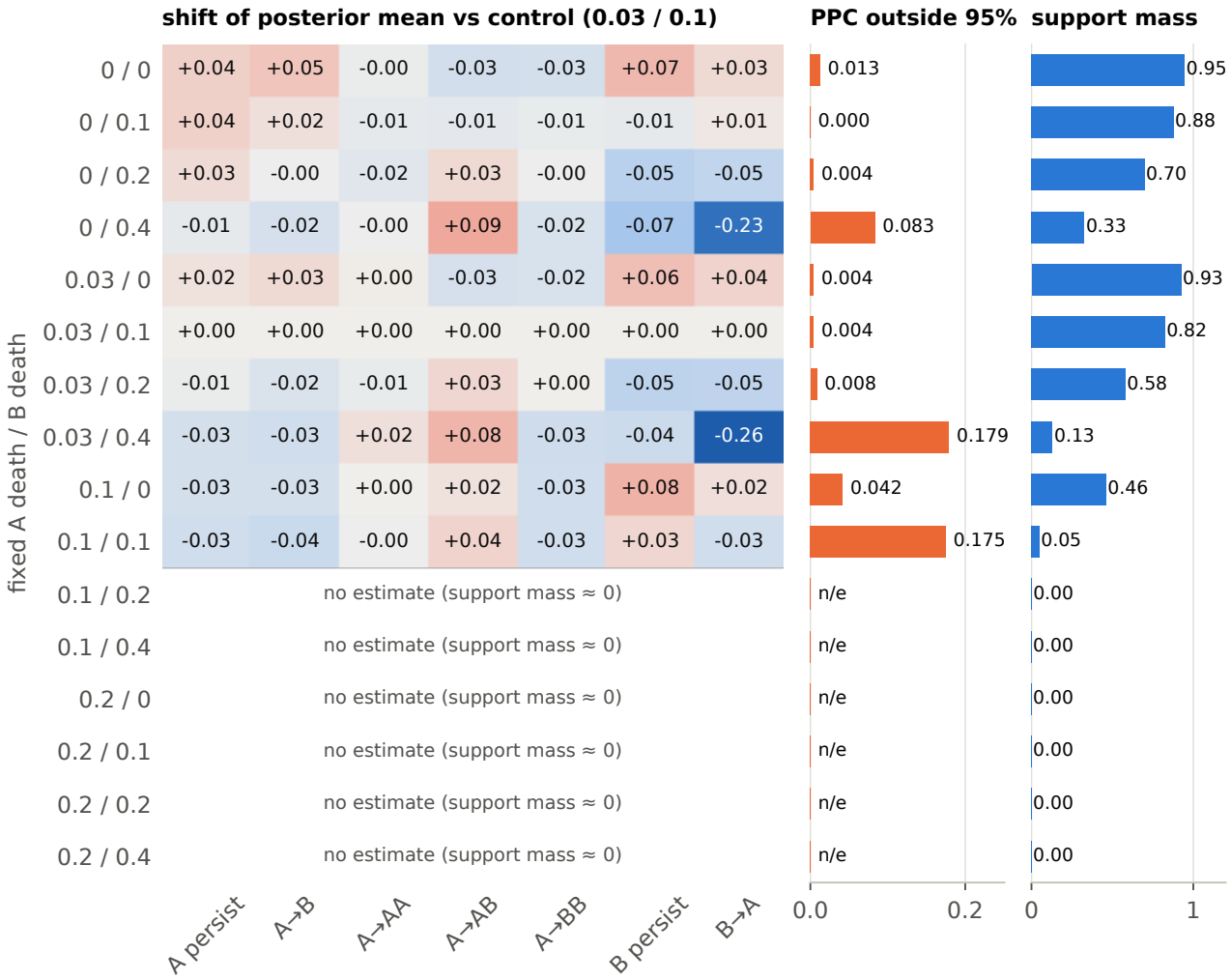


Figure 29: E1c GW. Left: shift of posterior means vs the control (deaths fixed at truth); rows with support mass < 0.02 are not estimates. Right: PPC and posterior support mass.

Fixed A / B death	A persist	A→AB	B persist	B→A	PPC	support
0.03 / 0.1 (control)	0.04	0.10	0.14	0.76 [5/5]	0.004	0.82
0 / 0, 0 / 0.1, 0.03 / 0	$\leq +0.04$	≥ -0.03	$\leq +0.07$	$\leq +0.04$	≤ 0.013	0.88–0.95
0 / 0.2, 0.03 / 0.2	$+0.03$ / -0.01	$+0.03$	-0.05	$-0.05 \pm .02$ [0/5]	≤ 0.008	0.58–0.70
0 / 0.4	-0.01	$+0.09$	-0.07	-0.23 [0/5]	**0.083**	**0.33**
0.03 / 0.4	-0.03 [1/5]	$+0.08$	-0.04	-0.26 [0/5]	**0.179**	**0.13**
0.1 / 0	-0.03 [0/5]	$+0.02$	$+0.08$	$+0.02$	0.042	0.46
0.1 / 0.1	-0.03 [0/5]	$+0.04$	$+0.03$	-0.03	**0.175**	**0.05**
0.1 / 0.2, 0.1 / 0.4	no estimate (support mass ≤ 0.004)				—	**0.00**
0.2 / any	no estimate (17/20 runs without any posterior sample)				—	**0.00**

A→AA never moves ($|\Delta| \leq 0.02$) in an estimable cell. Paired SE ≤ 0.02 for the shifts shown.

Comment

- **The valley is flat downwards, walled upwards for A**: death fixed at 0 fits as well as the truth (shifts ≤ 0.07, PPC ≤ 0.013; small, though up to ≈ 2 posterior sd for GW's narrow posteriors). A death fixed above the truth breaks the fit from 0.1 on (support ≈ 0; at 0.2 no posterior mass in the prior support): GW has one event per generation, so a higher death must be absorbed within A's simplex (A persist and A→B hit 0 first).
- **Contrast with BH** (§25): there A death 0.2 was absorbed silently by a faster A clock (PPC ≤ 0.033, support 0.64–0.77, A persist -0.10, τ_A -2 to -3 h). Whether a wrong fixed death is *detectable* depends on whether the model has a free time scale.
- **B death too high biases B→A in both**: 0.4 $\Rightarrow$ $-0.23/-0.26$ (GW, flagged: support 0.13–0.33) vs $-0.19/-0.20$ (BH, silent). B death 0.2 is silent in GW but already 0/5 covered (B→A -0.05; with B death fixed, B's row has one free coordinate and a narrow posterior).
- Fixing deaths at truth gains little over free deaths (MAE 0.028 vs 0.033). Consequence unchanged: do not fix death rates to guesses. C2ST not run.

Leads to. After seeding and death, the next source of error is the assay: cell types are assigned by clustering, which mislabels cells (E2a/b, §27).

27 E2a / E2b — A↔B mis-typing, known vs ignored (BH)

Why this next. Founders and death (§20–§26) concern the biology. The counts themselves come from marker clustering, with an estimated $\approx 5\%$ double-positive and $\approx 15\%$ double-negative cells assigned to A or B.

Question

Clustering mislabels cells at rate e. If the error is modelled (E2a), do we only lose information? If ignored (E2b), how large is the bias, and is it visible?

Setup

Simulator	Bellman–Harris, shape 8
Law (free)	`reversible_ab` (9)
Data model	symmetric confusion $e \in \{0, .02, .05, .1, .15, .2\}$, capture 1
Inference model	E2a: `observation=noisy` with the same e (network retrained per e); E2b: perfect typing (one network per seed)
Design	real-like 144/240 h × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 60
Control	$e = 0$ (= BH baseline)
Folder	`results/e2/known_bh`, `results/e2/ignored_bh`

Result

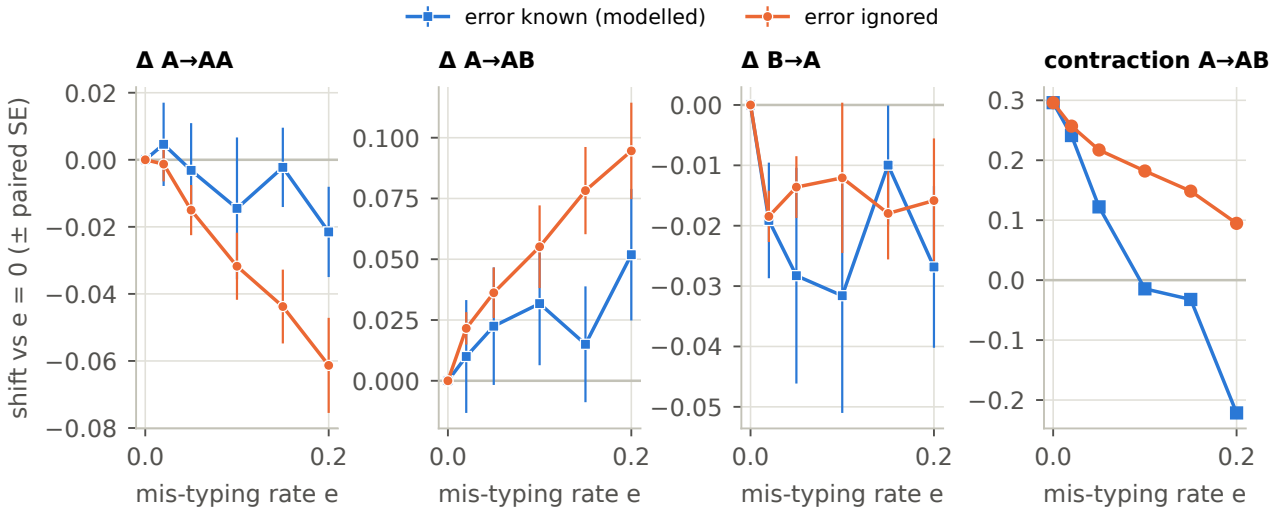


Figure 30: Shift vs $e = 0$ (± paired SE) and contraction of A→AB. Blue: error modelled; orange: error ignored.

e	0.02	0.05	0.1	0.15	0.2	PPC / support
Ignored: A→AB shift	+0.02	+0.04	+0.06	+0.08	+0.09 ± .02	$\leq$ 0.008 / 0.78
Ignored: A→AA shift	−0.00	−0.02	−0.03	−0.04	−0.06± .01	
Ignored: shift / posterior sd (A→AB)	0.2	0.3	0.4	0.6	0.7	
Known: contraction A→AB (0.30 at e=0)	0.24	0.12	−0.01	−0.03	−0.22	$\leq$ 0.008 / n/r
Known: contraction A→AA (0.52)	0.53	0.48	0.43	0.41	0.38	

Comment

- Known error = information loss in the division split; means within 0.05 of control, no trend.
- Ignored error = monotone bias: mislabelling makes clones look mixed, explained by more asymmetric divisions. Coverage of A→AB 3/5 → 0/5 at $e = 0.2$. Invisible to PPC and support mass.
- At the estimated rates (5% ++, 15% −−; Q2) the bias is < 1 sd at this design — tolerable only because posteriors are wide; larger designs (E4) narrow them.
- Consequence: estimate e (2D vs 3D, Q3) and pass it to the model. Asymmetric errors: §29.

Leads to. The bias is tolerable only because BH posteriors are wide; once lifetimes are known (Part III) they narrow to GW-like widths. E2 in GW (§28) shows the tolerance at that precision.

28 E2a / E2b — A↔B mis-typing, known vs ignored (GW)

Why this next. In BH (§27) ignored mis-typing stayed below one posterior sd up to $e = 0.2$, because the posteriors are wide. GW posteriors are $\approx 2.5\times$ narrower: the precision BH would reach with measured division times.

Question

As §27 for Galton–Watson: tolerance to known vs ignored mis-typing when the baseline is unbiased and the posteriors are narrower.

Setup

Simulator	Galton–Watson
Law (free)	`reversible_ab` (7)
Data / inference	symmetric $e \in \{0, .02, .05, .1, .15, .2\}$; modelled (E2a, network retrained per e) or ignored (E2b, one network per seed)
Design	real-like generations 6/10 × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 60
Control	$e = 0$ (= GW baseline)
Folder	`results/e2/known_gw`, `results/e2/ignored_gw`

Result

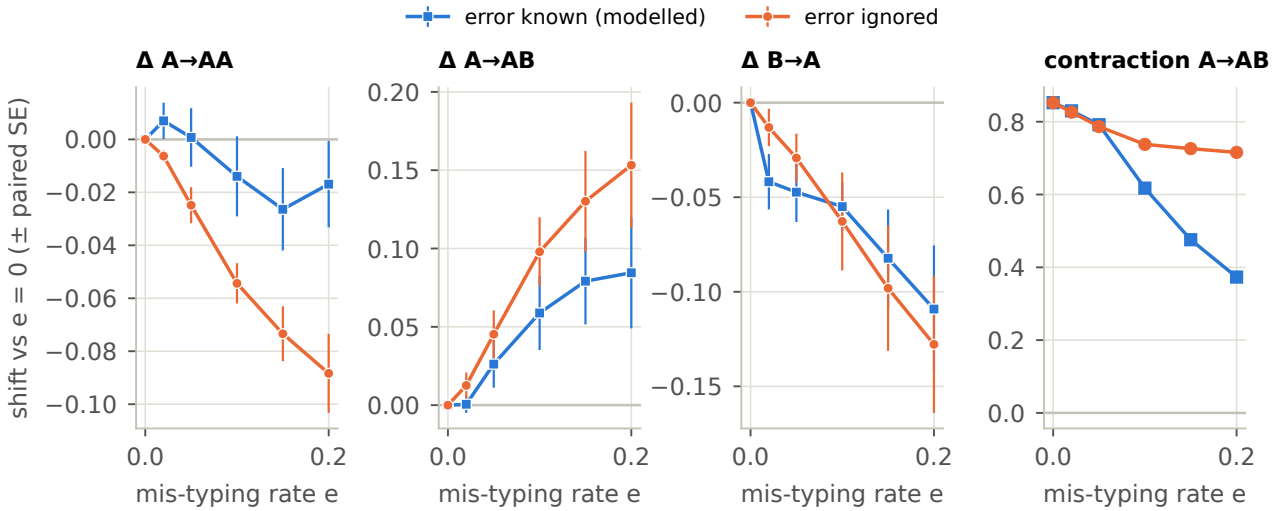


Figure 31: E2 GW: shift vs $e = 0$ (± paired SE) and contraction of A→AB. Blue: error modelled; orange: error ignored.

e	0.02	0.05	0.1	0.15	0.2	BH at 0.2
Ignored: A→AB shift	+0.01	+0.05	+0.10 [0/5]	+0.13	+0.15 ± .04 [0/5]	+0.09
Ignored: A→AA shift	−0.01	−0.02	−0.05 [0/5]	−0.07	−0.09± .01 [0/5]	−0.06
Ignored: B→A shift	−0.01	−0.03	−0.06	−0.10	−0.13± .04 [3/5]	−0.02
Ignored: shift / posterior sd (A→AB)	0.2	0.7	**1.4**	1.8	2.1	0.7
Ignored: PPC / support	.004 / .66	.004 / .64	0 / .59	.012 / .55	.033 / .48	.000 / .78
Known: contraction A→AB (0.85 at e=0)	0.83	0.79	0.62	0.48	0.37	−0.22
Known: B→A shift	−0.04	−0.05	−0.05	−0.08	−0.11± .03 [4/5]	−0.03

Comment

- **Same mechanism as BH**, about 1.5× larger and extending to B's row: mislabelled clones look mixed ⇒ more asymmetric division, less self-renewal, fewer B→A, more B persist (+0.18 at $e = 0.2$).
- **Tolerance is set by the posterior width**: GW posteriors are ≈2.5× narrower (sd of A→AB 0.05 vs 0.12), so the bias exceeds one posterior sd at $e \approx 0.1$ (0/5 covered). Tolerable $e \approx 0.05$ in GW vs ≈ 0.2 in BH.
- **Modelled error = information loss**: A→AB contraction 0.85 → 0.37; means drift towards the prior means (B→A −0.11, A→AB +0.08 at $e = 0.2$) with coverage kept — expected for a wider posterior under a uniform Dirichlet, not a bias.
- **Not detected**: PPC ≤ 0.033; support mass falls 0.67 → 0.48 (min 0.10) — a trend across the sweep, not a single-run warning.
- Consequence: estimate e (Q3) and model it; the BH tolerance will shrink to the GW one once division times are measured. Asymmetric errors: §29.

Leads to. Both E2 sweeps used one symmetric error rate. Real clustering errors need not be symmetric: is one direction worse (§29)?

29 E2 follow-up — asymmetric A↔B mis-typing (GW and BH)

Why this next. E2 (§27, §28) used one symmetric error rate. The clustering error estimates differ by direction (Q2: 5% one way, 15% the other), and A and B play very different roles in the dynamics: A divides and is abundant, B mostly switches back.

Question

Is mis-typing in one direction (A read as B, rate e_A; B read as A, rate e_B) more harmful than the other, when ignored and when modelled? E2 used symmetric errors only.

Setup

Simulator	Galton–Watson and Bellman–Harris (shape 8)
Law (free)	`reversible_ab` (GW 7, BH 9)
Data model	confusion $(e_A, e_B) \in \{(0.10, 0), (0, 0.10), (0.05, 0.15), (0.15, 0.05)\}$, capture 1; dynamics $\theta^\star$
Inference model	ignored: perfect typing; known: `observation=noisy` with the true asymmetric confusion
Design	real-like: GW generations 6/10, BH 144/240 h; 165/101 organoids
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 80 (4 settings × 2 arms × 2 simulators)
Folder	`results/e2_asym/{ignored,known}`

Result

	(e_A, e_B)	MAE ignored → known	coverage ignored → known	support ign.	PPC ign.	B→A (0.80) ign. / known
GW	(0, 0.10)	0.040 → 0.035	0.84 → 0.96	0.70	0.000	0.74 / 0.76
GW	(0.10, 0)	**0.066** → 0.048	0.76 → 0.87	0.50	0.025	0.66 / 0.73
GW	(0.05, 0.15)	0.050 → 0.045	0.82 → 0.91	0.70	0.004	0.72 / 0.72
GW	(0.15, 0.05)	**0.086** → 0.054	**0.64** → 0.91	**0.42**	0.038	0.63 / 0.70
BH	(0, 0.10)	0.121 → 0.110	0.78 → 0.82	0.79	0.000	0.59 / 0.62
BH	(0.10, 0)	0.122 → 0.116	0.78 → 0.87	0.78	0.000	0.62 / 0.62
BH	(0.05, 0.15)	0.125 → 0.113	0.78 → 0.84	0.79	0.000	0.59 / 0.63
BH	(0.15, 0.05)	0.136 → 0.121	0.76 → 0.84	0.78	0.000	0.60 / 0.60

Means over 5 seeds; MAE over the outcome probabilities.
Numbers: `results/e2_asym/e2_asym_groups.csv`.

Comment

- **Stem cells read as differentiated (e_A) is the harmful direction.** In GW, ignoring $e_A = 0.10$–0.15 doubles the error (0.040 → 0.066–0.086) and drops coverage to 0.64; B read as A at the same rate is almost harmless (0.040, 0.84). A is the dividing, abundant type, so its mislabels create spurious B cells and pull B→A down (0.63–0.66).
- **Partly flagged**: in the A→B direction support mass falls to 0.42–0.50 and the PPC rises to 2.5–3.8% (GW); the B→A direction is not flagged.
- **Modelling the known confusion removes the bias** (GW MAE 0.035–0.054, coverage 0.87–0.96).
- **In BH the effect is small** next to the lifetime/self-renewal ridge error (MAE ≈ 0.12 in every setting); modelling still improves coverage (0.76–0.78 → 0.82–0.87). No BH setting is flagged.
- Consequence: an estimate of how often LGR5$^+$ cells are called KRT20$^+$ (Q3) matters more than the reverse; if available, pass it to the model as an asymmetric confusion (`observation=noisy`).

Leads to. E2 treats the double-negative/positive cells as random mislabelling. They may instead be a real third type C with its own dynamics (E2c, §30).

30 E2c — Hidden third type ignored (GW)

Why this next. E2 (§27, §28) treated mislabelled cells as noise. If they are a real population C that divides and differentiates, ignoring it is a model error, not a labelling error.

Question

Data contain a third type C (double negative/positive) read as A with prob. q and as B with $1 - q$. If the model ignores C, are the A and B rows still recovered, and does the PPC flag the two-type model?

Setup

Simulator	Galton–Watson
Data model	`three_type`: A = {death .03, persist .02, A→C, A→AA, A→AC, A→CC}, C = {death .05, persist .10, C→B .40, C→BC .20, C→CC .25}, B = {death .10, persist .90} (B never returns to A). C fraction ≈ 5% (A→AA 0.905) or ≈ 15% (A→AA 0.79); `three_split` with $q \in \{0, 0.5, 1\}$
Inference model	two-type `reversible_ab`, perfect typing (7 free)
Truth for scored names	A death .03, A persist .02, A→AA .905 / .79, A→B = A→AB = A→BB = 0, B death .10, B persist .90, B→A 0
Design	real-like generations 6/10 × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 30
Control	*none* in this sweep (no two-type data with the same A row); arms with C modelled: §31
Folder	`results/e2c/ignore_c/`

Result

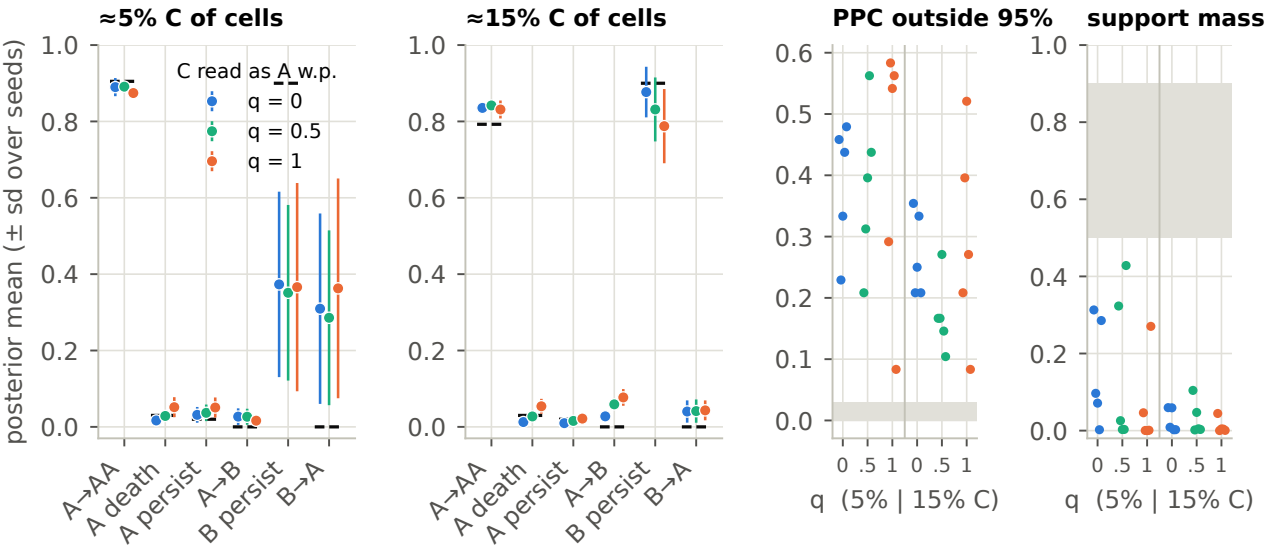


Figure 32: Left: posterior means (± sd over seeds) per q; dashes = truth of the data model. Right: PPC and support mass per run; grey = well-specified ranges.

C share	q	A→AA	A→BB (0)	A→B (0)	B persist (.90)	B→A (0)	PPC	support
≈15%	0	0.84 (.79)	0.10	0.03	0.88	0.04	0.27	0.03
	0.5	0.84	0.05	0.06	0.83	0.04	0.17	0.03
	1	0.83	0.02	0.08	0.79	0.04	0.30	0.01
≈5%	0	0.89 (.905)	0.03	0.03	0.37	0.31	0.39	0.15
	0.5	0.89	0.01	0.03	0.35	0.29	0.38	0.16
	1	0.87	0.01	0.02	0.37	0.36	0.41	0.06

Comment

- **The two-type fit is flagged**: PPC 0.17–0.41 (well specified ≤ 0.03), support mass 0.01–0.16 (well specified 0.5–0.9). Per the reading rules these posteriors are not reliable estimates.
- How C is absorbed: read as B ($q = 0$), A→CC appears as A→BB (0.10 at 15%); read as A ($q = 1$), C→B appears as A→B switching (0.08) and A death rises (0.05). A→AA is overestimated at 15% (+0.05) and slightly under at 5% (−0.02).
- At 5% C, B cells are rare (B only arises via C): B's row stays prior-like (B persist 0.37, contraction 0.25–0.45 for B death).

- **Attribution**: the data truth is far in the prior tail of the inference model (A→AA 0.79–0.905: prior mass above $<0.05\%$; B persist 0.9: 1%), and PPC is *higher* with less C (5% vs 15%). The arms that model C (§31) fail even more although one of them is the data model: the misfit is driven by the prior tail and cannot be attributed to C.
- One run (5%, $q = 1$, seed 2) drew only 74 posterior samples.

Leads to. Ignoring C is flagged, but the misfit cannot yet be attributed to C. Modelling C (§31) should remove it if C is the cause.

31 E2c — Third type modelled: three-arm comparison (GW)

Why this next. The two-type fit to three-type data was flagged (§30), but the PPC rose with *less* C. If the misfit is due to C, modelling it should remove it.

Question

When C is modelled (freely, or with the trajectory constraint C→B only), are A and B recovered, is C's row identifiable without its own marker, and is the misfit of §30 removed?

Setup

Simulator	Galton–Watson
Data model	as §30 (same 30 datasets: $q \in \{0, .5, 1\}$, ≈5% / 15% C)
Inference arms	ignore C: `reversible_ab` (7 free, §30); C free: `three_type` + `three_split`, q known (27 free) = *the data model*; C trajectory: `three_type_traj` (C never produces A; pending Q1) + `three_split` (23 free)
Design	real-like generations 6/10 × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	5 / 30 per arm (90)
Folder	`results/e2c/{ignore_c,model_c_free,model_c_traj}`

Result

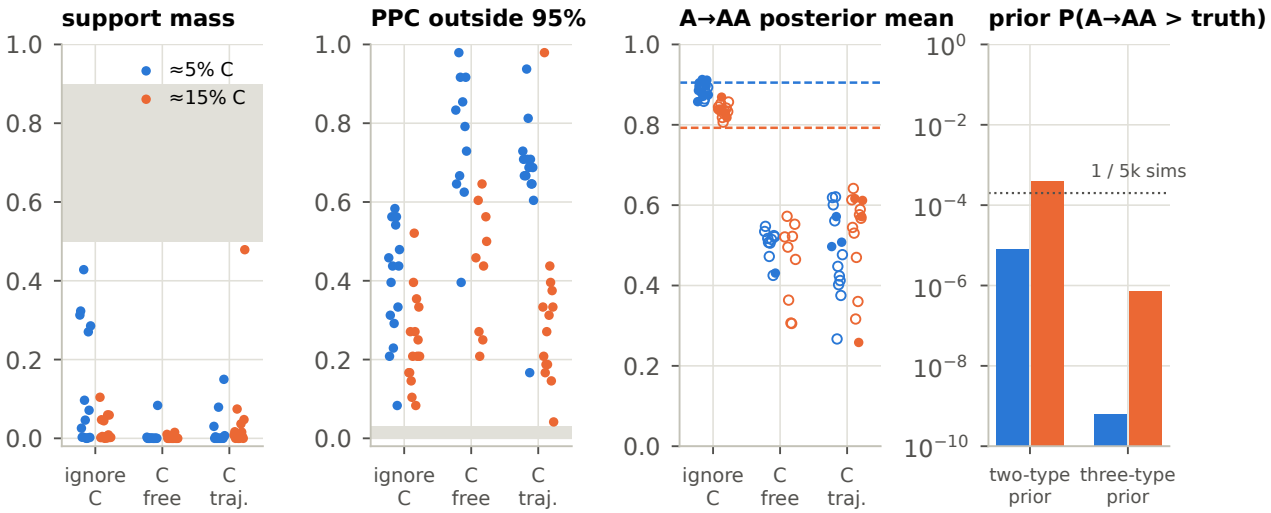


Figure 33: E2c, three inference arms on the same data (one point per run). Grey bands: well-specified ranges. A→AA: open circles = support mass < 0.02 (not estimates); dashed = truth (0.905 at 5% C, 0.79 at 15%). Right: prior probability of A→AA above the truth under each inference model (uniform Dirichlet; dotted = one in 5 000 training simulations).

Arm	C	runs with samples	support: median (max)	PPC (group means)	A→AA (truth)
ignore C	5%	15/15	0.05 (0.43)	0.38–0.41	0.87–0.89 (.905)
	15%	15/15	0.004 (0.10)	0.17–0.30	0.83–0.84 (.79)
C free (well specified)	5%	11/15	0.000 (0.08)	0.63–0.84	*0.48–0.52*
	15%	9/15	0.000 (0.02)	0.37–0.50	*0.33–0.53*
C trajectory	5%	14/15	0.000 (0.15)	0.60–0.75	*0.39–0.60*
	15%	14/15	0.008 (0.48)	0.26–0.38	*0.35–0.59*

Italics: not estimates (support ≈ 0). SBC coverage error 0.024–0.029 in every arm. Prior P(A→AA > truth) at 5% / 15% C: two-type (6 outcomes) $8 \cdot 10^{-6}$ / $4 \cdot 10^{-4}$; three-type (10 outcomes) $6 \cdot 10^{-10}$ / $8 \cdot 10^{-7}$. P(B persist > 0.9): 0.01 (two-type) vs 10^{-9} (three-type).

Comment

- **Modelling C does not remove the misfit — it makes it worse**: support ≈ 0 in almost every run (10/30 free and 2/30 trajectory runs drew no posterior sample at all), PPC 0.26–0.84. No A/B bias and nothing about C (e.g. C→B 0.01–0.12 vs 0.40) can be read from these arms.
- **The failure is a prior-tail failure, not misspecification**: the free arm *is* the data model. Under a 10-outcome uniform Dirichlet the true A→AA (0.79–0.905) and B persist (0.9) have prior probability 10^{-6}–10^{-10}: no training simulation resembles the data, and amortised NPE extrapolates outside the prior support. SBC is fine because it averages over the prior. The two-type model fits *better* than the true model because its prior has more mass near the truth.
- **Hence the ignore-C flag (PPC 0.17–0.41) cannot be attributed to C**; also within ignore-C the PPC is higher with *less* C. Whether the PPC detects a hidden type is still open. The trajectory constraint helps slightly (fewer empty posteriors; one run with support 0.48).
- Follow-up done: the rerun with truths drawn from the three-type prior (§32) confirms that this failure was the tail truth, not the method.

Leads to. The true model fails at a truth with prior probability 10^{-6}–10^{-10}: the same pattern as E0 (§5) and the BH ridge, which X5 (§6, §14) traced to truth position. The rerun with truths drawn from the prior (§32) applies the same remedy.

32 E2c rerun — hidden third type, truths from the prior (GW)

Why this next. The original E2c (§30, §31) could not separate a C effect from a tail-truth failure. X5d (§14) showed that inference is calibrated for truths in the prior bulk; drawing the three-type truths from the prior removes the tail problem and makes both E2c questions answerable.

Question

The original E2c (§30, §31) used a truth deep in the three-type Dirichlet tail, so even the true model failed. With truths in the prior bulk: (1) is ignoring C detected, and (2) does modelling C (free, or trajectory-constrained) recover the A/B rows?

Setup

Simulator	Galton–Watson
Data model	`three_type`, 10 truths *drawn from the three-type prior* (`truth.source=prior`, `data_seed` = seed 0–9); `three_split` with $q \in \{0, 0.5, 1\}$ (C read as A with prob. q). Same 30 datasets in every arm
Prior-drawn truths	generic biology: C is 21–41% of cells at generation 10; B divides (B-division mass 0.37–0.84); C makes A (C→A mass 0.19–0.58)
Inference arms	ignore C: `reversible_ab` (7 free; also forbids B divisions); C free: `three_type` + `three_split`, q known (27 free) = *the data model*; C trajectory: `three_type_traj` (C never makes A; 23 free). Both constrained arms are misspecified for *every* truth, ignore C doubly so
Scored	the 9 A/B parameters present in every arm; "flagged" = support mass < 0.1 or PPC > 0.05; "usable" = at least 100 posterior samples within the 60 s sampling cap; prior-mean baseline = $\lvert\text{truth} - 0.1\rvert$ (10-outcome Dirichlet mean)
Design	real-like generations 6/10 × 165/101
Inference / budget	NPE amortised, 5k
Seeds / runs	10 truths × 3 q × 3 arms = 90
Folder	`results/e2c_prior/{ignore_c,model_c_free,model_c_traj}` (`e2c_prior_runs.csv`, `e2c_prior_groups.csv`)

Result

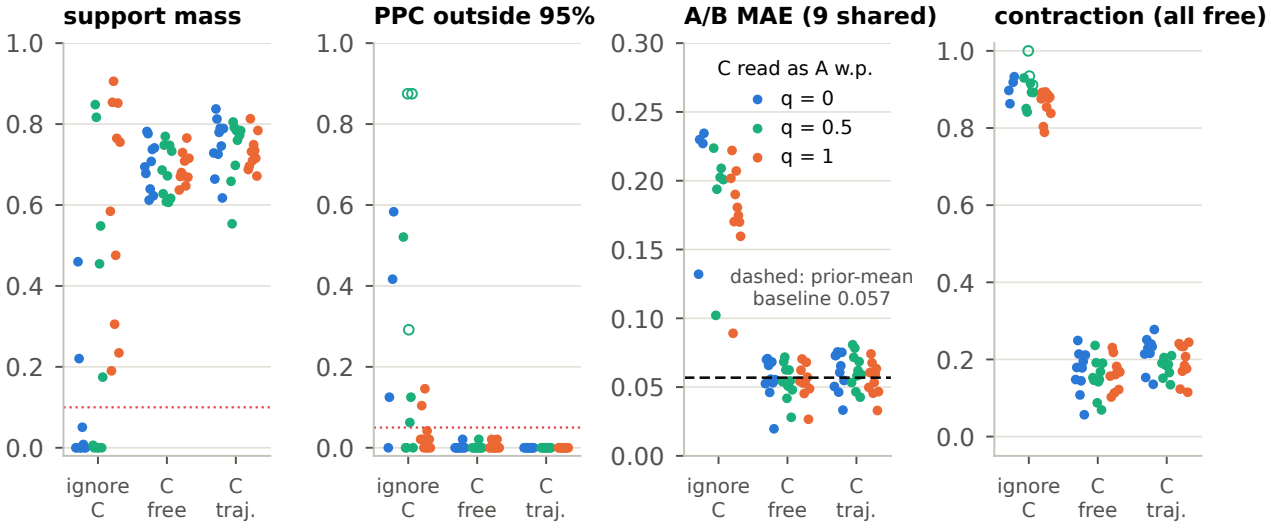


Figure 34: E2c rerun, one point per run (10 prior-drawn truths per q). Dotted red: flag thresholds (support < 0.1, PPC > 0.05). Open circles: fewer than 100 posterior samples (not estimates; A/B MAE not computed). A/B MAE over the 9 shared A/B parameters; dashed = prior-mean baseline. Contraction over all free parameters of each arm.

Arm	q	usable	flagged	support (med.)	PPC	SBC err.	A/B MAE	A/B cov.	contr.
ignore C	0	4/10	**9/10**	0.00	0.28	0.029	0.206	0.11	0.90
	0.5	6/10	**7/10**	0.09	0.31	0.029	0.189	0.20	0.91
	1	10/10	**2/10**	0.67	0.04	0.029	0.177	0.42	0.86
C free (well specified)	0–1	30/30	0	0.68–0.70	0.00	0.026–0.027	0.055	0.93	0.16
C trajectory	0–1	30/30	0	0.72–0.78	0.00	0.024–0.025	0.059	0.92	0.20
Prior-mean baseline (posterior mean replaced by the prior mean 0.1)							0.057		

Model-C rows pool the three q (per q: MAE 0.054–0.056 free, 0.055–0.062 trajectory; contraction 0.15–0.17 and 0.18–0.22). PPC and contraction for ignore C include runs with fewer than 100 samples. Numbers: `tables/e2c_prior.csv`.

Comment

- **Ignoring C is detected when C looks like B, missed when C looks like A.** At $q = 0$ 9/10 runs are flagged (support ≈ 0, PPC up to 0.58; 6/10 runs give no usable posterior). At $q = 1$ only 2/10 are flagged, while the A/B rows are still wrong (MAE 0.18, 90% coverage 0.42) with high contraction (0.86): *confident, silent bias.*
- **Modelling C is calibrated but not informative.** Both model-C arms pass every check (PPC ≈ 0, SBC 0.024–0.027, A/B coverage 0.92–0.93), but contraction is 0.16–0.20 and the A/B error equals the prior-mean baseline (0.055 / 0.059 vs 0.057). With 23–27 free parameters and two endpoints of two labels the data barely move the posterior: correct, but it is the prior speaking.
- **The trajectory constraint is wrong for every truth and nothing detects it** (C→A mass up to 0.58; PPC 0, support 0.72–0.78): the data are too weak to contradict it.
- **Resolves §31**: the "true model fails" result there came from the tail truth, not the method. In the prior bulk the well-specified model is calibrated, but a three-type model is not identifiable from two-label endpoint counts at this design.
- Caveats: 10 truths, NPE 5k, one design. Prior truths have much more C (21–41%) than the biological scenario (5–15%) and B divisions, unlike the original truth; ignore C also forbids B divisions, so its failures are not due to C alone. The A/B "truth" for the two-type arm is the three-type value of the same outcome, which a two-type row cannot match exactly (part of its error is projection).
- Consequence: if a hidden C is suspected, modelling it needs constraints that are actually true (from trajectory data), a C-specific measurement, or much richer designs. Ignoring it is risky mainly when C is read as A (q near 1).

Leads to. This completes the one-at-a-time errors. Section 33 collects which of them the fit itself would reveal; X4 (§34) combines them.

33 Synthesis — which errors are detectable (BH and GW)

Why this next. Every experiment of this part asked whether the PPC or the support mass flags the error, and a pattern recurred: errors that BH absorbs silently through its free time scale are flagged in GW (§22, §26). This section puts the misspecified settings of both simulators on one scale. It contains no new runs.

Question

Across the misspecification experiments — Part IV and, for BH, the timing errors E3a–E3c of Part III — how large is each shift relative to the control's posterior sd, and would the fit itself have revealed it?

Result

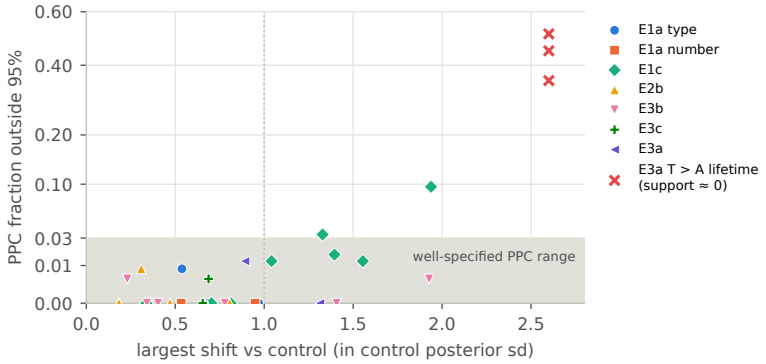


Figure 35: Detectability of BH misspecifications. Each point is one misspecified setting (mean of 5 seeds): largest shift of a posterior mean from its paired well-specified control, in units of the control's posterior sd, against the PPC. Grey band = PPC range of well-specified runs. Crosses: global lifetime $T > \tau_A$ (support mass ≈ 0).

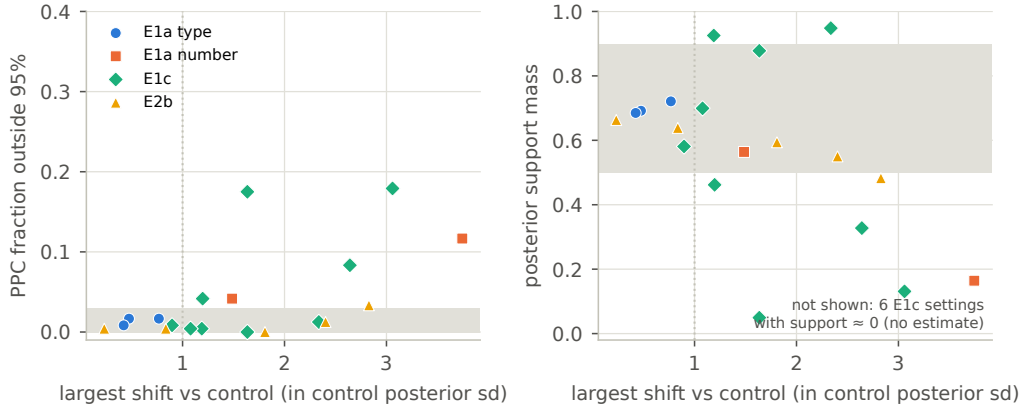


Figure 36: Detectability of GW misspecifications (E1a type and number, E1c, E2b): largest shift from the paired control in control posterior sd, against the PPC (left) and the posterior support mass (right). Grey bands = well-specified ranges. GW flags more errors than BH (no free time scale), but several shifts of 1–3 sd remain inside the well-specified bands.

Comment

- **BH**: the PPC stays ≤ 0.033 for every error except both deaths fixed wrong (0.096, §25) and a global lifetime longer than A's (0.35–0.51, §16).
- **GW**, without lifetimes, flags more: A death fixed too high (support ≈ 0, §26), B death fixed at 4× truth (support 0.13–0.33), 40% multi-cell founders (PPC 0.12, support 0.16, §22).
- **Still silent in GW**: wrong founder type (§20), B death at 2× truth (B→A 0/5 covered), ignored mis-typing up to 2 posterior sd (§28); and a hidden type read as A (§32).
- SBC never detects misspecification. Support mass and PPC vary between well-specified GW runs (support 0.07–0.89, PPC up to 0.10), so single borderline values are not proof.

Leads to. Every error here was tested alone. In the real assay several act together, and some are silent alone: X4 (§34) combines them at the real design and turns the part into an error budget.

34 X4 — Combined realistic errors at the real design (BH)

Why this next. The one-at-a-time errors of this part (and the timing errors of Part III) were mostly small and often silent (§33): founders of the wrong type (§20), mis-typing at a few percent (§27), desynchronised lifetimes (§17), a hidden type (§32). Before the real data are fitted, the question is what they add up to when they act together at the real design and with a well-trained network: the error budget of the baseline model, i.e. how much of an unflagged fit could still be wrong, and whether modelling the known errors removes it.

Question

With the plausible errors of the real assay acting together, how biased is the baseline model (Bellman–Harris, `reversible_ab`, one A founder, perfect typing, lifetime shape 8), is it flagged, and does modelling the known errors help? This is the error budget to quote with real-data fits of the baseline model class.

Setup

Data model	BH, real-like design (144/240 h × 165/101 organoids), $\theta^\star$ (τ_A 20 h, τ_B 30 h)
Arms	*control*: no errors (well specified); *combined*: 10% B founders (E1a), 5% symmetric A↔B mis-typing (E2), lifetime shape 4 = CV 0.5 instead of 8 (E3b); baseline model, errors ignored; *modelled*: same data; founder row inferred, mis-typing 5% known, shape 4; *combined + C*: same errors plus a hidden type C (three-type reference law, C read as A or B with probability 0.5 each, as E2c); baseline model
Inference / budget	NPE amortised, `budget=full`: 50k simulations, SBC 1000, PPC 500
Seeds / runs	5 per arm / 20
Folder	`results/x4/control/`, `combined/`, `modelled/`, `combined_c/` (preset `x4_combined`, arm commands in its header; commit `fa3e96a`, combined + C launched after `bae0b90`); `x4_groups.csv`, `x4_runs.csv` (`scripts/analyse_x4.py`); table `report/tables/x4.csv`

Result

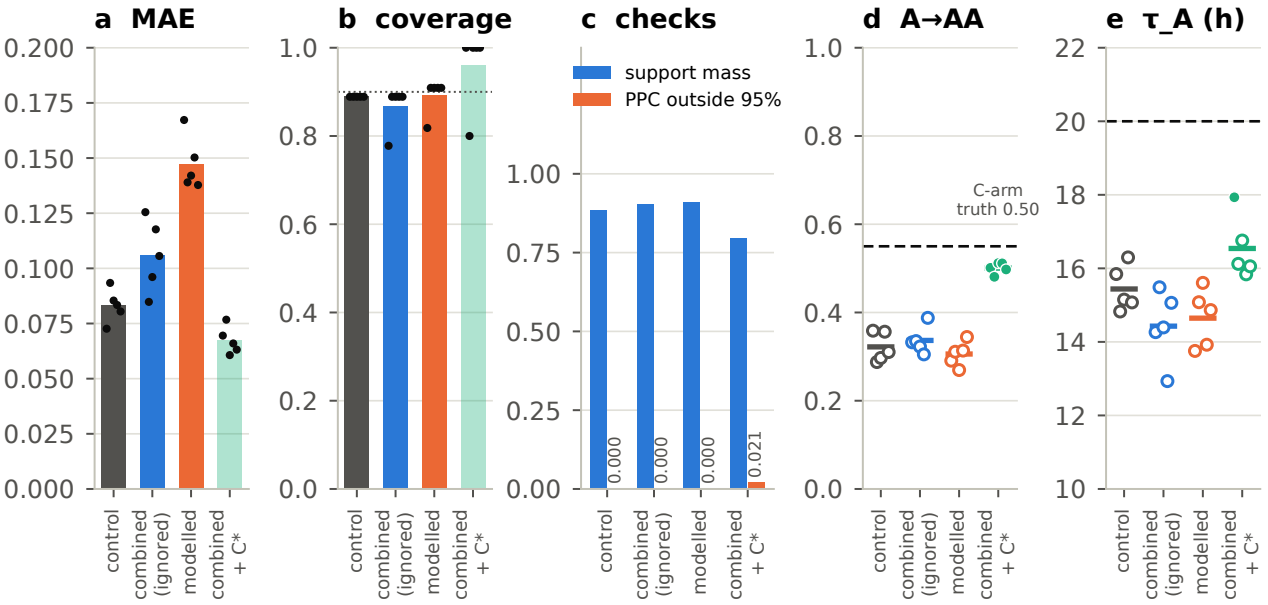


Figure 37: X4: combined realistic errors at the real-like design (BH, 50k simulations). (a) Probability MAE, (b) 90% coverage (bars = mean, dots = seeds), (c) support mass and PPC fraction outside 95% (mean of 5 seeds), (d) A→AA and (e) τ_A per seed (filled = covered, open = not; bar = mean); dashed = truth. *Combined + C: the two-type "truth" is the three-type value of the same outcome (A→AA 0.50, dotted), so its MAE and coverage are not comparable with the other arms.

	control	combined (ignored)	modelled	combined + C
MAE (probabilities)	0.083	0.106	0.147	(0.067)*
90% coverage	0.89	0.87	0.89	(0.96)*
lifetime rel. error	0.33	0.32	0.37	0.46
support mass	0.88	0.90	0.91	0.80
PPC outside 95%	0.000	0.000	0.000	0.021
SBC error	0.016	0.016	0.011	0.016
A→AA (0.55) [covered]	0.32 [0/5]	0.34 [0/5]	0.31 [0/5]	0.50*
τ_A (20 h) [covered]	15.4 [0/5]	14.4 [0/5]	14.6 [0/5]	16.5 [1/5]
τ_B (30 h)	17.0	18.9	15.8	52.5
founder A (0.9)	—	—	0.54 (≈ prior)	—

Means over 5 seeds. *Truth from the three-type reference law (A→AA 0.50, A→AC 0.30, B persist 0.90, . . .): not comparable. Numbers: `results/x4/info.md`, `x4_groups.csv`.

Comment

- **At 50k simulations the lifetime/self-renewal ridge still dominates** (control: A→AA 0.32, τ_A 15.4 h, 0/5 covered, MAE 0.083): the error budget of the baseline is set by the ridge (§13), not by the assay.
- **The realistic errors add a small, silent bias**: MAE +0.023, with PPC 0 and support 0.90 — not flagged. Each error alone was silent or weak (E1a type, E2 at $e = 0.05$, E3b); together they stay silent.
- **"Modelling" the errors makes things worse** (MAE 0.147): inferring the founder row adds a parameter the data cannot determine (founder A 0.54 vs truth 0.9, at its prior; as in E1a, §20), which widens and shifts the other rows. Known mis-typing and shape do not compensate.
- **A hidden C is only weakly flagged** (PPC 2.1%, support 0.80) and is absorbed into B's row: B lifetime 52 h (truth 30 h) and B→A 0.06 — C's slower cycle and the A→C→B flow are read as B.
- Calibration is good in every arm (SBC 0.011–0.016): the network is not the problem.
- **Error budget to quote**: for the baseline model at the real design, ≈ 0.08 MAE from the ridge plus ≈ 0.02 from the assay errors, none of it flagged; lifetimes ≈ 25–35% short. It applies to the baseline model class, which the CRC0322 data reject (Part V); the final real-data model has its own synthetic recovery check (§37).
- Caveats: one truth, 5 seeds per arm; one combination of error sizes; the C arm's two-type parameters have no exact truth.

Leads to. The baseline model at the real design carries ≈ 0.1 MAE of error that no check flags, most of it the ridge, and freeing parameters the data cannot determine does not help. A clean synthetic fit is therefore not enough: the real-data analysis must first ask whether the baseline model class describes the organoids at all, and move beyond it if the checks say no (Part V).

Part V
Real data (CRC0322)

Motivation. Parts I–IV used synthetic data only, all simulated from shared parameters. They established what endpoint counts can identify (B's row and the founder mix cannot be learned at realistic sizes: E0 and the exact likelihood, E1a), how the checks behave when the model is wrong (support mass ≈ 0 and PPC well above 0.03 flag large errors; a clean fit is not identifiability) and which time model to use (BH with lifetime information; GW only with a measured generation length, X3). This part fits the measured CRC0322 counts with those lessons: the data (§35), a ladder of twelve models whose failures are read with the checks of Parts I–IV (§36), and the final model with the identifiability of each estimate (§37). Write-up: `results/real/info.md`.

35 R — CRC0322 data and fitting procedure

Why this next. The end goal of every experiment so far. Before fitting: what do the organoids look like in the terms of the model, and which features could the synthetic baseline (one A founder, shared parameters, constant lifetimes) fail to produce?

Question

What do the measured CRC0322 organoids look like in the terms of the model, and how are they fitted?

Setup

Organoids	CRC0322, day 6 (165 organoids) and day 10 (101); per organoid, counts of A = LGR5$^+$ and B = KRT20$^+$ cells. Every cell is called A or B (no unlabelled cells: double-positive/negative cells are already forced into one of the two)
Data variants	*3D mean* (primary): rounded mean of the three 3D whole-organoid estimates; *3D sample 1–3*: each estimate alone (organoids with a failed 0-cell estimate dropped: 1/4/4 at day 6, 1/1/2 at day 10); *2D*: the section counts
Relation 3D–2D	3D ≈ 4–5× 2D (median ratio 4.0 at day 6, 4.8 at day 10) with the same A fraction per organoid (correlation 0.996–0.998): the 3D estimates are the 2D section scaled to the whole organoid, not independent measurements
Models	12 candidate models (Table 3), each trained once at the real-like design (BH 144/240 h; 165/101 organoids) with `budget=full`: 50k simulations, SBC on 1000 prior draws
Fit	each trained network conditioned on each data variant (`bio-sbi-infer ... --ppc 500`): posterior support mass and PPC on the 48 summary features (500 predictive datasets)
Criterion	a model *fits* if its support mass is in the synthetic well-specified range (0.5–0.9) and few features lie outside their 95% predictive interval (well-specified synthetic runs: 0–0.10)
Folder	`data/real/` (`scripts/prepare_real_data.py`, `data/real/README.md`); `results/real/<model>/fit_<variant>/`; table `results/real/real_fits.csv`

Result

Feature (across organoids)	3D mean		2D sections	
	day 6	day 10	day 6	day 10
Mean cells per organoid	52	157	13	31
Growth day 6 → 10	×3.0 (after ≈5.7 doublings in 6 days)		×2.4	
sd of log total cells	0.55	0.73	0.33	0.53
A fraction per organoid, mean (sd)	0.68 (0.28)	0.63 (0.30)	0.68 (0.28)	0.63 (0.30)
corr(A, B) across organoids, counts	−0.13	0.00	−0.45	−0.19
corr(A, B) across organoids, $\log(1+n)$	−0.28	−0.08	−0.53	−0.29
Organoids larger than 2^{day} cells	43 / 165	0 / 101	0	0

Comment

- **Growth decelerates**: ≈25 h per doubling up to day 6, ≈60 h from day 6 to day 10 (3D).
- **A-rich and B-rich organoids coexist**: the A fraction varies widely between organoids (sd ≈ 0.3) and A and B counts are uncorrelated or *negatively* correlated across organoids, although larger organoids have more cells of both types under any shared-parameter branching model.
- GW at one generation per day cannot reach the day-6 3D sizes (43 organoids exceed 2^6 cells); all real fits except one 2D control use BH.
- 3D and 2D differ in scale by a near-constant factor; parameters tied to the size scale (lifetimes, K) are expected to differ between them, composition parameters not.

Leads to. Two features — decelerating growth and coexisting A-rich and B-rich organoids — are outside what the synthetic baseline was built to produce. The ladder (§36) tests the baseline first and adds one ingredient at a time.

36 R — Model ladder: what the organoids require

Why this next. The baseline of all synthetic work is `reversible_ab` with one A founder (E0, §8; E1a, §20). If the data reject it, Part IV says how it would show: support mass ≈ 0 and a PPC far above the well-specified 0–0.03 (E1a/E1c GW, §33); and posterior means from such fits are not estimates (E2c, §31).

Question

Which model ingredients do the organoids require? Starting from the synthetic baseline (`reversible_ab`, one A founder, constant lifetimes), each step adds one ingredient suggested by the previous failure.

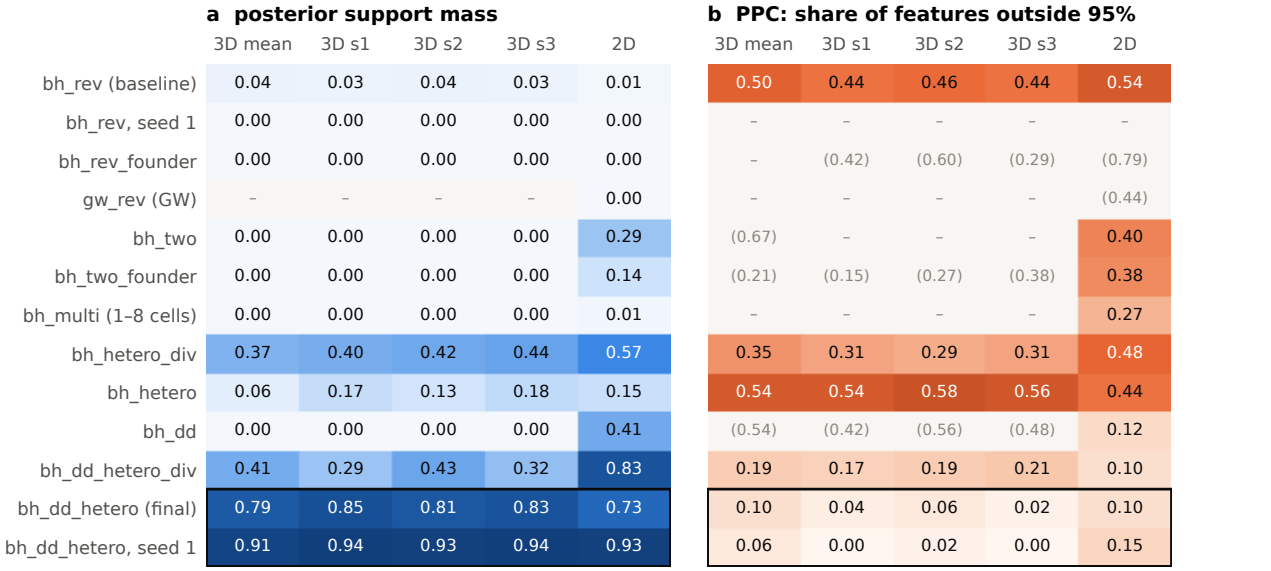
Setup

Simulator	Bellman–Harris, shape 8 (`gw_rev`: GW, one generation per day, 2D only); `bellman_harris_dd` for density dependence
Laws	`reversible_ab` (B does not divide), `two_type` (all 12 outcomes free), `multi_founder` (1, 2, 4, 8 founder cells), `two_type_hetero` (organoid heterogeneity of whole rows, or of the divisions only: `_div`); founder type fixed to A or inferred
Budget	`budget=full` (50k simulations, SBC 1000); one training seed per model, two for `bh_rev` and the final model
Variants	3D mean, 3D samples 1–3, 2D (`gw_rev`: 2D only)
Folder	`results/real/<model>/`

Result

Table 3: Real-data model ladder. Support mass: 3D mean / range over the three 3D samples / 2D (well-specified synthetic fits: 0.5–0.9). PPC: share of the 48 features outside the 95% predictive interval (3D mean / 2D; in brackets: from fewer than 100 posterior samples, unreliable). corr(A,B) z: (observed − predictive mean)/predictive sd of the log-count A–B correlation, day 6 / day 10, 3D mean (`gw_rev`: 2D). Support near 0 means the network extrapolates: such posterior means are not estimates. Full table: `results/real/real_fits.csv`.

Model	B div.	founder	heterog.	dens. dep.	support 3D mean / samples / 2D	PPC 3D / 2D	corr z d6 / d10
`bh_rev` (synthetic baseline)	no	A	–	–	0.04 / 0.03–0.04 / 0.01	0.50 / 0.54	−21 / −60
`bh_rev_s1` (2nd seed)	no	A	–	–	0.00 everywhere	–	–
`bh_rev_founder`	no	inferred	–	–	0.00 everywhere	–	–
`gw_rev` (GW, 2D only)	no	A	–	–	– / – / 0.00	– / (0.44)	−28 / −22
`bh_two`	yes	A	–	–	0.00 / 0.00 / 0.29	(0.67) / 0.40	−16 / −48
`bh_two_founder`	yes	inferred	–	–	0.00 / 0.00 / 0.14	(0.21) / 0.38	−2.6 / −3.8
`bh_multi` (1–8 cells)	yes	inferred	–	–	0.00 / 0.00 / 0.01	– / 0.27	–
`bh_hetero_div`	yes	inferred	divisions	–	0.37 / 0.40–0.44 / 0.57	0.35 / 0.48	−2.0 / −2.2
`bh_hetero`	yes	inferred	all rows	–	0.06 / 0.13–0.18 / 0.15	0.54 / 0.44	+1.3 / +0.4
`bh_dd`	yes	inferred	–	yes	0.00 / 0.00 / 0.41	(0.54) / 0.12	−2.2 / −1.6
`bh_dd_hetero_div`	yes	inferred	divisions	yes	0.41 / 0.29–0.43 / 0.83	0.19 / 0.10	−0.2 / +2.5
`bh_dd_hetero`	yes	inferred	all rows	yes	**0.79 / 0.81–0.85 / 0.73**	**0.10 / 0.10**	−0.7 / +0.8
second seed (`_s1`)					0.91 / 0.93–0.94 / 0.93	0.06 / 0.15	−1.2 / +0.04



–: not fitted (gw_rev: 2D only) or no posterior sample inside the prior. Grey (0.xx): PPC from < 100 posterior samples (support ≈ 0), not reliable.

Figure 38: Real-data model ladder: posterior support mass (a) and PPC share of features outside the 95% predictive interval (b) for every model × data variant. Box: final model, both training seeds. Grey values in brackets: PPC from fewer than 100 posterior samples inside the prior (support ≈ 0), unreliable. Plotted numbers: `tables/real_ladder.csv`.

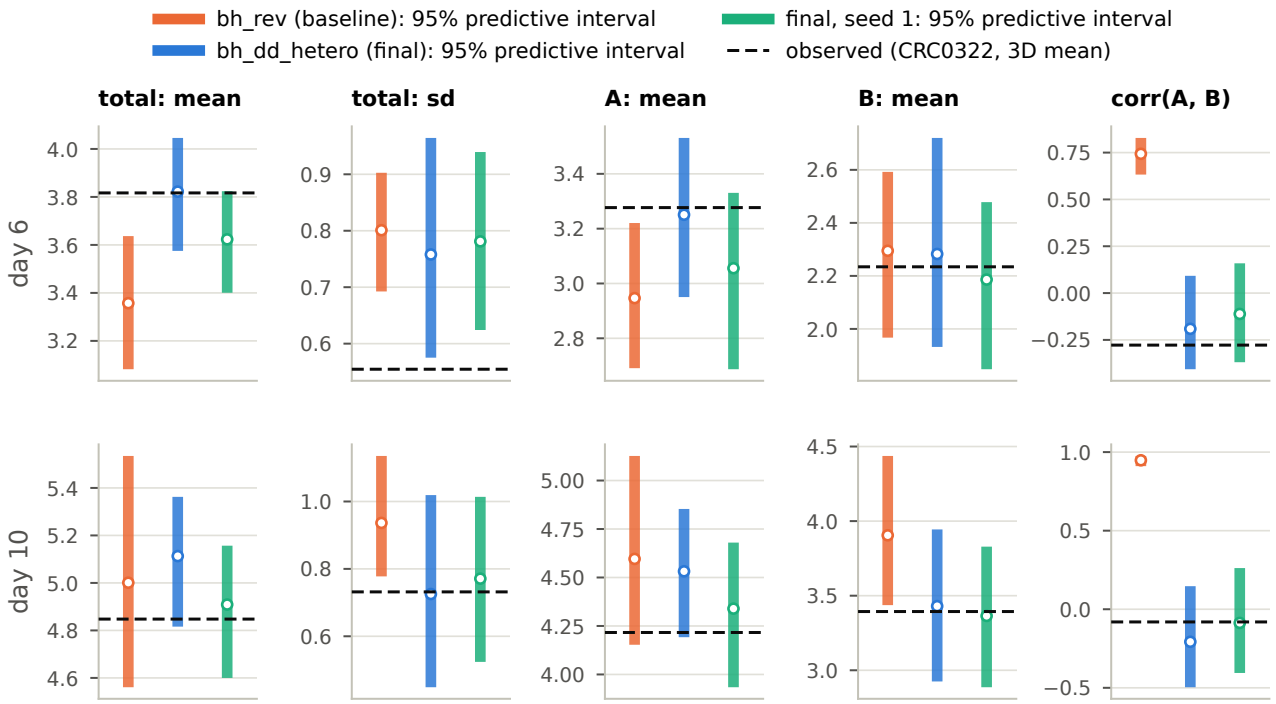


Figure 39: Posterior predictive check on the 3D-mean data, key features of $\log(1+n)$ per organoid (mean of total, A and B counts; sd of the total; correlation of log A and log B across organoids), day 6 (top) and day 10 (bottom). Bars: 95% predictive interval (circle: predictive mean) of the synthetic baseline `bh_rev` (orange) and the final model `bh_dd_hetero` (blue: seed 0; green: seed 1). Dashed: observed. The baseline predicts a strongly positive A–B correlation and too wide, too small day-6 organoids. Plotted numbers: `tables/real_ppc.csv`.

Comment

- **Single-founder branching with shared parameters cannot produce the organoids** (`bh_rev`, `bh_two`, `bh_rev_founder`): it predicts a strongly positive A–B correlation across organoids (larger organoids have more of both), where the data are negative, with A-rich and B-rich organoids side by side. B never dividing (the synthetic baseline) is the worst; letting B divide and inferring the founder type removes most of the correlation misfit (z $-21 \to -2.6$) but not the misfit of sizes (3D support 0.00). The baseline's second training seed does not even put posterior mass inside the prior.
- **Organoids differ in fate**: letting each organoid draw its own outcome probabilities around the shared ones is the first change that brings 3D support into the synthetic range (0.37–0.44 with `bh_hetero_div`).
- **Growth decelerates**: without density dependence day-6 organoids are too small and day-10 organoids too large. Density dependence alone fails (`bh_dd`: 3D support 0.00); with fate-only heterogeneity it fits day 6 but makes day-10 sizes too uniform (z of the day-10 total sd +2.8); with whole-row heterogeneity (organoids differ also in death and persistence) both days fit (`bh_dd_hetero`).
- **Multi-cell seeding does not help** (`bh_multi`, 1–8 founder cells: support ≤ 0.01); its 2D fit, unreliable as it is (support 0.01), puts 76% on single-cell founders.
- **The failures are the ones the synthetic experiments warned about.** Support mass and PPC flag the misfit as in E1a/E1c (GW) and the E2c rerun; the posterior means of the failing models are extrapolations, not estimates (support ≈ 0, as in E2c, §31). GW at 24 h per generation fails as τ longer than A's cycle did in X3 (§19). What the synthetic experiments could not anticipate is that the organoids are not exchangeable copies of one process: every synthetic data set was simulated from shared parameters.

Leads to. Only `bh_dd_hetero` fits both data types, robustly across the 3D samples. Which of its parameters are estimates is a separate question, answered by a synthetic check of the same network (§37).

37 R — Final model: fate heterogeneity and density-dependent growth

Why this next. A model that fits is not necessarily identifiable: E0 at 4096 organoids (§8) showed a clean PPC next to a prior-driven B row, and E1a (§20) that the founder type is not learnable in BH. The final model frees both B's divisions and the founder type, so its trained network is checked on synthetic data before its estimates are read.

Question

What does the model that fits say about the organoids, and which of its estimates can be trusted?

Setup

Model	`bh_dd_hetero`: Bellman–Harris, lifetime shape 8; all 12 outcomes free (`two_type_hetero`); founder type inferred; organoid-level heterogeneity of every outcome row (κ); density-dependent lifetimes $\tau_k(1 + N/K)$ (§2.5). 15 free parameters
Priors	uniform Dirichlet per row and for the founder type; lifetimes log-uniform on [10, 80] h; K log-uniform on $[10, 10^5]$; κ log-uniform on [1, 1000]
Training	real-like design, `budget=full` (50k simulations, SBC 1000); two training seeds (`bh_dd_hetero`, `bh_dd_hetero_s1`)
Synthetic check	the trained network on one synthetic dataset from the reference law with $K = 300$, $\kappa = 10$ (`recovery.csv` of the training run)
Folder	`results/real/bh_dd_hetero/`, `results/real/bh_dd_hetero_s1/`

Result

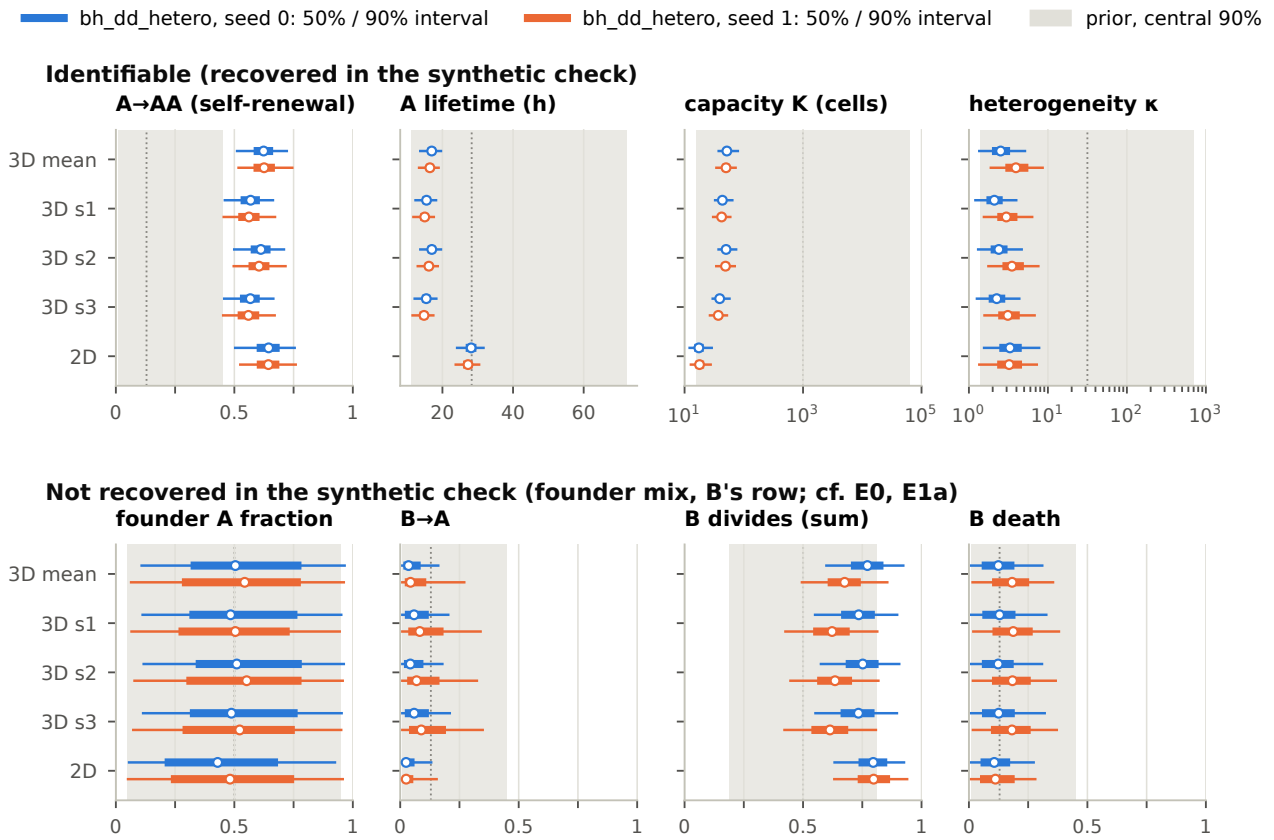


Figure 40: Final model, posterior marginals for every data variant and both training seeds (thick: 50% interval, thin: 90%, circle: median); grey band: central 90% of the prior (dotted: prior median). Top: parameters recovered in the synthetic check. Bottom: founder mix and B's row, not recovered in the synthetic check (B divides = B→AA + AB + BB). Size-scale parameters (A lifetime, K) differ between 3D and 2D by the 3D/2D scale factor. Plotted numbers: `tables/real_posterior.csv`.

Parameter	3D mean	2D	Identifiable? (synthetic check)
A→AA (self-renewal)	**0.62 ± 0.06**	0.64 ± 0.08	yes: 0.51 vs truth 0.55, contraction 0.76
A death / persist / →B	0.09 / 0.06 / 0.09	0.06 / 0.03 / 0.09	moderately (covered; contraction 0.55–0.88)
A→AB / A→BB	0.03 / 0.11	0.08 / 0.10	moderately (A→BB 0.07 vs 0.29, not covered)
A mean lifetime (small organoid)	**16.9 ± 1.7 h**	28.0 ± 2.3 h	yes: 17.3 vs 20 h, contraction 0.99; scale differs 3D vs 2D
Carrying capacity K	**54 ± 13 cells**	18 ± 5	yes: median 216 vs 300, contraction 0.98; scale differs
Heterogeneity κ	**2.8 ± 1.2**	3.8 ± 2.3	yes: median 16 vs 10, contraction 1.00
B mean lifetime	17.2 ± 2.3 h	27.2 h	weakly: 36 vs 30 h, contraction 0.60
B row: →A / →AA / →AB / →BB / death	0.06 / 0.25 / 0.09 / 0.42 / 0.13		**no**: B→A 0.18 vs 0.80, contraction 0.05 (cf. E0)
Founder A fraction	0.54 ± 0.28	0.46	**no**: contraction 0.10, stays at the prior (cf. E1a)

Posterior mean ± sd, seed 0. Synthetic check of the trained network: SBC coverage error 0.012, support 0.89, 90% coverage 0.85. Robustness: over the three 3D samples support 0.81–0.85, PPC 2–6%; a second training seed gives support 0.91–0.94 (3D) and 0.93 (2D), PPC 0–6% (3D) and 15% (2D), and the same estimates: A→AA 0.56–0.63 vs 0.56–0.62, A lifetime 14.7–16.4 h vs 15.4–16.9 h, K 51 vs 54, founder A 0.50–0.54; κ 3.4–4.5 vs 2.3–2.8 (strong heterogeneity either way; the exact value is less stable).

Comment

- **The organoids need fate heterogeneity and slowing growth.** $\kappa \approx 3$ means the per-organoid outcome probabilities vary widely around the population mean (for A→AA ≈ 0.6: sd ≈ 0.25 between organoids); $K \approx 50$ means the division time doubles by ≈50 cells in the 3D counts. Neither alone fits; together they reproduce sizes, size spread, composition spread and the A–B anticorrelation on both days (Fig. 39; the day-6 total sd is still at the lower edge of the predictive interval, z −2.1).
- **Identifiable**: stem-like cells self-renew in ≈62% of their fates and divide every ≈17 h early on. In the synthetic check A→AA came out at 0.51 (truth 0.55), not on the BH ridge value 0.32 of the constant-lifetime model (§13) — possibly because the deceleration carries timing information; one synthetic dataset only.
- **Not identifiable**: how KRT20$^+$ cells contribute (dividing vs reverting to LGR5$^+$) and how many organoids start from each type. The reported B row is shaped by the prior and the model structure, as E0 (§8), the exact likelihood (practically non-identifiable at 266 organoids, §9) and E1a (§20) predicted; it is not evidence that KRT20$^+$ cells divide.
- **2D and 3D agree on composition** (A→AA, A's other outcomes, κ) and differ on the size scale (lifetime, K), as expected if 2D sections sample ≈1/4 of the cells.
- Caveats: one trained network per model except the final one (two seeds); support values near 0 depend on extrapolation. The heterogeneity and density-dependence forms are simple choices (Dirichlet around the mean rows; $\tau(1 + N/K)$ with N at birth); alternatives may fit equally well. Mis-typing is not modelled (cf. §29: A read as B is the harmful direction), and the 3D counts are derived from the 2D sections (Q3). The synthetic conclusions of Parts I–IV (design, error sizes) were obtained under the homogeneous model and should be re-checked under this model class.

Leads to. The real-data estimates rest on one model class and on unmodelled mis-typing; Part VI lists the follow-ups and the measurements that would settle B's row.

Part VI

Planned follow-ups

Motivation. Every experiment of the original plan is finished: E0–E4, the E2c rerun, X1–X5, P0, the follow-ups of E1b and E2 and the real-data fits (§3, §4, Parts I–V). The items below are follow-ups that the results suggest; nothing is running; the exact BH posteriors exist only as a preliminary, not converged result (§13.1). Each is listed with the earlier result that motivates it.

38 Follow-ups and measurements

Why this next. The final real-data model (§37) ignores mis-typing, whose harmful direction is now known (§29), and uses one simple form of heterogeneity and density dependence. X4 (§34) showed that the baseline carries an unflagged error budget and that freeing undeterminable parameters makes estimates worse. Measurements: the ridge at realistic sizes is a property of the data (§12.1), B's row is practically unidentifiable at feasible designs (§8, §9) and stays so on the real data (§37); X1 (§4) showed that no inference method changes this.

Question

- **Mis-typing in the real-data model**: add an asymmetric confusion matrix to the final model (A read as B is the harmful direction, §29), ideally with the rate measured (Q3) rather than inferred.
- **Alternative forms of heterogeneity and density dependence**: per-organoid lifetimes, a mixture of organoid classes, other capacity laws; do they fit the CRC0322 data equally well, and do the identifiable estimates (A→AA, τ_A, K, κ) change?
- **Synthetic checks under the final model class**: more truths and designs (which conclusions of Parts I–IV carry over), and an X4-type error budget for the final model.
- **Survivor summaries in real fits**: the extinct fraction with survivor summaries (§24) once the number of seeded organoids is recorded (Q5).
- **Biology questions Q1–Q5**: the trajectory and proportion of a possible third type C (Q1, Q2), 2D vs 3D mislabelling (Q3), the meaning of inactivation (Q4), whether the number of seeded organoids is known (Q5). Each decides whether a model extension is needed and how it can be constrained.
- **Converged exact BH posteriors** (sweep stopped; preliminary results in §13.1): real-like design with B divisions fixed and free, and the ridge at $\{96,144,240\}$ h $\times$ 512, on the NPE datasets. The preliminary runs suggest that part of NPE's ridge bias (A→AA 0.32, τ_A 16 h) is the approximation's; confirming it needs $\approx 10^5$ evaluations per run (2–3 days for the 15 runs on 16 threads) or a faster sampler (HMC with the population covariance as metric, a reparametrisation along the ridge).
- **NPE vs exact on the early designs and at large sizes**: NPE on $\{3\} \times 266$ and $\{2,3\} \times 133$ (command in the `exact_gw_early` preset header); NPE variants at 4096 per endpoint (sequential or truncated NPE, early-endpoint summaries, larger budget) to see whether an approximation can match the exact posterior (§9).
- **Measurements, then a refit** (implied by §8, §12.1, §34 and Part V): A's division time, whether B cells divide, the LGR5→KRT20 mis-typing rate and the number of seeded organoids. Once measured, refit with them as informative priors carrying their measurement uncertainty (as in X5b), with $\geq$ 20k simulations, amortised NPE checked against NRE (§4).

Setup

Blocking	Q1 (C trajectory), Q2 (C proportions), Q3 (2D vs 3D mislabelling), Q4 (meaning of inactivation), Q5 (seeded organoids known?)

A Device and batch-size benchmark

Why this next. Engineering, not part of the argument. E0's 50k-simulation and SNPE runs (§5) were the slowest of the project, and wall times recorded in run metrics were measured under heavy contention. Before the large E1–E4 sweeps: does a GPU or a larger batch help?

Question

Do GPU training and/or a larger batch (512 vs 128) make runs faster without degrading the posterior?

Setup

Configuration	GW, `two_type`, generations 4/8/12 × 32, NPE 20k, seed 0, SBC 200, PPC 200, 4 threads
Hardware	20-core CPU; NVIDIA RTX A1000 (8 GB, also driving the display)
Varied	device (CPU/GPU) × batch (128/512); SNPE 4 rounds on CPU and GPU
Protocol	one job at a time, contention checked every 15 s (all 6 jobs clean); single measurements
Folder	`results/gpu_bench/` (branch `gpu-bench`, `811d8ca`)

Result

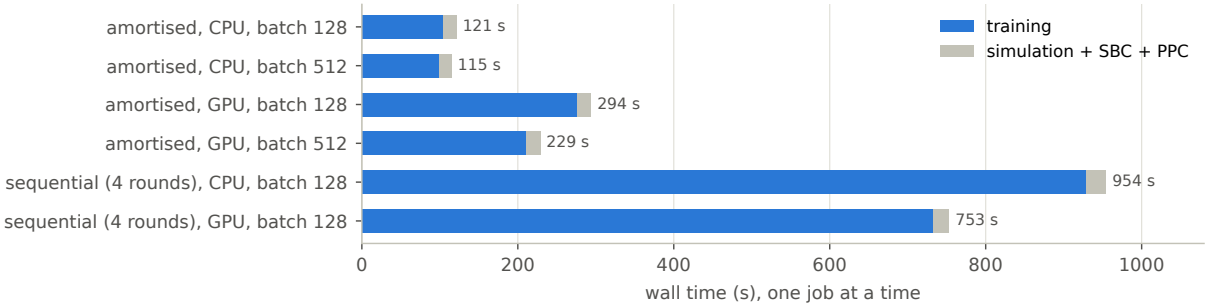


Figure 41: Wall time per job (blue = network training).

Job	wall (s)	s/epoch	epochs	MAE	SBC error
amortised, CPU, 128	121	1.47	71	0.140	0.032
amortised, CPU, 512	115	0.65	151	0.140	0.028
amortised, GPU, 128	294	4.07	68	0.137	0.033
amortised, GPU, 512	229	1.27	166	0.138	0.031
SNPE, CPU, 128	954	4.3	215	0.135	—
SNPE, GPU, 128	753	3.2	226	0.129	—

Comment

- Amortised NPE: GPU 2–2.4× slower (small flows, launch overhead). Batch 512: epochs 2.3× faster but 2× more epochs: −5%.
- SNPE: GPU 21% faster. Quality identical across devices.
- Keep CPU + batch 128; GPU only for SNPE or as extra capacity. Together with P0 (§A.1) these are the only valid timings in this report.

A.1 P0 — Bellman–Harris timing on an idle machine

> **Why this next.** Reading rule (5): the run times recorded in the sweep metrics were measured under heavy contention, and BH was assumed to be much slower than GW. The device benchmark above used GW only.

Question

What does one BH run cost at the real-like design, measured on an idle machine? (Run times in the sweep metrics were recorded under heavy contention.)

Setup

Configuration	`law=reversible_ab`, real-like design (BH 144/240 h, GW generations 6/10; 165/101 clones), NPE, `workers=4 torch_threads=4`
Protocol	one job at a time, nothing else of ours running (load 0.1 at start); single measurement each
Folder	`results/p0_bh_timing/` (`smoke/`, `dev/`, `gw_dev/`)

Result

Run	Budget	Total (s)	Simulation (s)	Training (s)	Epochs
BH	smoke (256 simulations)	3	0.7	0.5	6
BH	dev (5k simulations, SBC 200, PPC 200)	**42**	4	35	102
GW (reference)	dev	45	3	38	80

Comment

- At the real-like design a BH run costs the same as a GW run (≈ 45 s at `budget=dev`); simulation is $< 10\%$ of the time, training $\approx 85\%$.
- The minutes per run recorded in the sweeps reflect machine contention, not BH cost. Larger designs scale simulation roughly linearly in clones (E4: 4096 clones per endpoint take several minutes).

B Speed: exact likelihood vs NPE

> **Why this next.** Engineering, as the device benchmark and P0. The exact likelihood (§9, §13.1) is the reference for what the data contain; whether it can replace or only check NPE depends on its cost per posterior and per study.

Question

What does each method cost per likelihood evaluation, per posterior for one dataset, and for 100 datasets (e.g. an SBC or a design study)?

Setup

Machine	20 cores (Intel Core Ultra 7 265), one job at a time, contention sampled every 15 s (`load.log`)
Exact kernels	GW: numpy reference (`gwexact.py`, fixed grids L) and the production kernel (`bio_sbi.inference.gw_exact`, Numba, data-sized grids, 1 thread); BH: `bhexact.py` (Numba DOP853), grids 512–1024 $\times$ 512, dt 1 and 2 h, shape 8 and 4, 1/4/20 threads
NPE	full pipeline at the real-like design, `budget=dev` (5k simulations; 4 and 20 threads) and `full` (50k; 20 threads); sampling timed on the run's own (well-specified) reference dataset
Cost model	MCMC per posterior: 8 000 MH evaluations or 40 000 NUTS gradient evaluations (generic); compared with the measured `exact_gw` runs; for `exact_bh` the measured SMC run (40k evaluations in 91 min, 16 threads) scaled to the $\approx 10^5$ evaluations of a converged posterior
Folder	`results/exact_benchmark/` (`info.md`, `summary.md`, `timings.csv`; `scripts/benchmark_exact.py`, `scripts/run_exact_benchmark.sh`; commit `523708e`); table `report/tables/exact_speed.csv`

Result

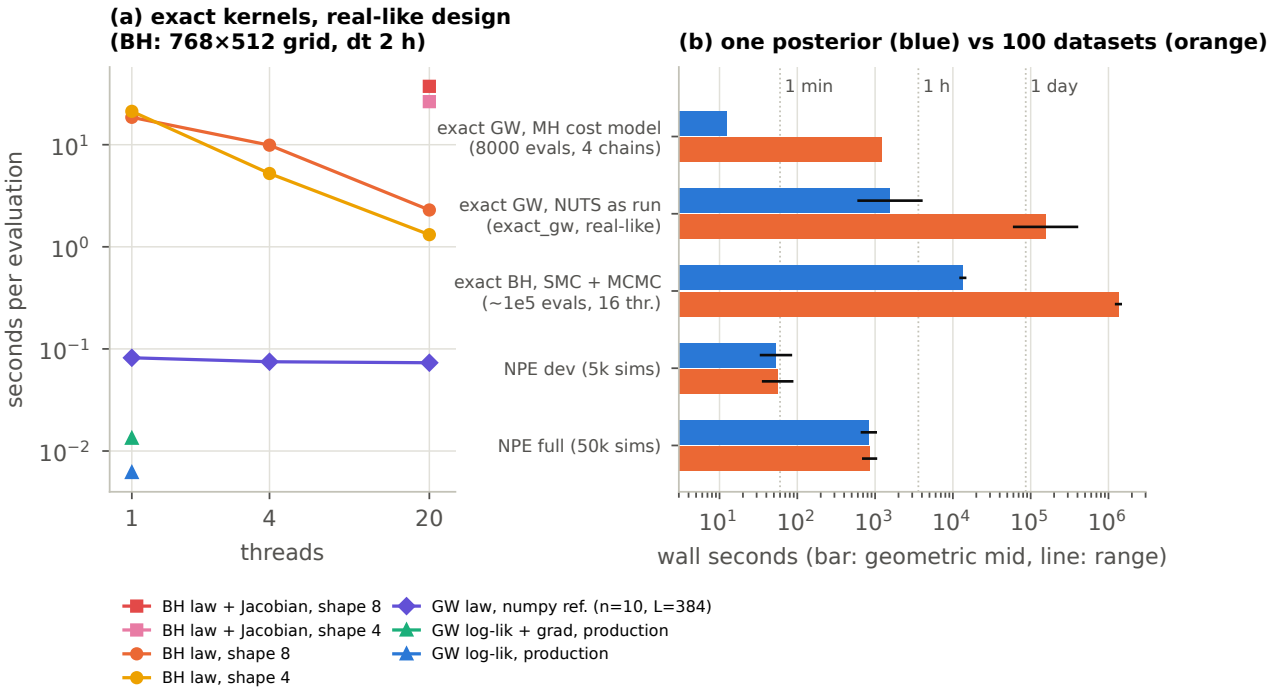


Figure 42: (a) Seconds per evaluation against threads (log scales): BH law and law + Jacobian at 768 × 512, dt = 2 h, shapes 8 and 4; GW numpy reference law (generation 10, $L = 384$) and the production GW kernel (real-like data, 1 thread). (b) Wall seconds for one posterior (blue) and for 100 datasets (orange), real-like design; bar = geometric mid, line = range. Exact GW: MH cost model and the `exact_gw` NUTS runs as measured; exact BH: SMC + adaptive MCMC, ≈ 10^5 evaluations at the wall time per evaluation of the real-like SMC run (`results/exact_bh/smc_check_rev0_c`: 16 threads, dt 3 h; range 3.5–4 h); NPE: training + 1 or 100 × sampling. Plotted numbers: `tables/exact_speed.csv`.

Method (real-like design)	s per evaluation	one posterior	100 datasets
Exact GW (production kernel, 1 thread)	0.006 log-lik; 0.013 with gradient	12 s (MH model) / 2 min (NUTS model); **as run: 10–15 min** (fixed) and **25–66 min** (free)	20 min–3.7 h (models); 17–110 h as run
Exact BH (768 × 512, dt 2 h, 20 threads)	2.3 law / 2.65 log-lik (shape 8); 1.3 / 1.1 (shape 4); Jacobian 37 (shape 8), 26 (shape 4)	**SMC + MCMC ≈ 3.5–4 h** (≈ 10^5 evaluations, 16 threads; 7–8 h on 8); AMIS 1–2 h but unreliable; MH model 6 h; NUTS model 17 days	SMC 15–17 days; MH 25 days
NPE `dev` (5k simulations)	sampling 0.02–0.03 per dataset	training 33–81 s	≈ 1–1.5 min
NPE `full` (50k simulations)	sampling 0.06–0.28 per dataset	training 11–17 min	11–17 min

Other data, exact GW production kernel: E0 at 4096 clones 0.011 s (0.024 s with gradient; runs as measured: 14–22 min of MCMC); real CRC0322 counts 0.003 s (0.006 s). BH law per evaluation at 20 threads: 512 / 768 / 1024 × 512 grid 1.5 / 2.3 / 2.9 s (shape 8, dt 2 h); dt 1 h doubles it. GW kernels do not speed up with threads; BH scales 8–20× from 1 to 20 threads.

Comment

- **Exact GW is cheap per evaluation; exact BH is not.** The production GW kernel takes 3–24 ms per log-likelihood (7–12× faster than the numpy reference, because it sizes the grid from the data); one BH law at the real-like design takes 1.3–2.3 s on 20 threads and 26–75 s with the Jacobian, 200–400× the GW likelihood and 2 000–6 000× the GW gradient.
- **Per dataset, exact GW is comparable to NPE; per 100 datasets NPE wins by orders of magnitude.** An exact GW posterior costs 12 s (MH model) to 2 min (NUTS model); the actual NUTS runs took 10–66 min because wide or multimodal posteriors needed 45k–340k gradient evaluations at 20–40 ms each on the shared machine. NPE costs 0.5–1.5 min (`dev`) or 11–17 min (`full`) for the first dataset and ≈ 0.03 s for every further one, so an SBC on 100 datasets costs about the training time.
- **Exact BH is a validation tool, not a workhorse**: a reliable posterior needs ≈ 10^5 evaluations, ≈ 3.5–4 h on 16 threads at the real-like design (AMIS with 1 000–1 300 evaluations, ≈ 1–2 h, gave unreliable importance weights), about two weeks for 100 datasets; gradient-based sampling (NUTS) is out of reach. Use it on a few datasets to check NPE (§9), not for SBC.

- Caveats. (i) 118 of 126 jobs are flagged "not clean", but the foreign CPU was the benchmark's own timed subprocess (a runner bug: its grandchild `_exec` process was not counted as its own) plus another user's notebook (≈ 30% of one core; residual median 21% of one core on 20 cores): the timings are effectively uncontended. (ii) The "60 s per dataset" NPE sampling in `summary.md` is the 60-s rejection-sampling cap, hit because the new dataset was the real CRC0322 data, on which the baseline model has support ≈ 0 (misspecified); well-specified sampling costs 0.02–0.09 s (also P0: 0.02 s). (iii) The MCMC cost model (8 000 MH or 40 000 NUTS evaluations) is generic; the real `exact_gw` runs took 16–69 min each on average including SBC/PPC, and the `exact_bh` estimate scales the one SMC run (40k evaluations in 91 min on 16 threads, *dt* 3 h, data-sized grid, chains evaluated in parallel: 0.14 s wall per evaluation) to 10^5 evaluations; AMIS (≈ 1 000–1 300 evaluations) was cheaper but unreliable. Single measurements; single-thread BH timings are noisy.

C Reproducibility

Code versions. `manifest.json` in each run records `git_commit` (HEAD at run time; since `6b45d23` also the commit of the loaded code). Nearly all runs have `git_dirty = true`. Several sweeps span several HEAD values because commits were made while they ran; the loaded code is that of the launch commit (verified for GW E1a: identical networks across settings per seed).

Experiment	Launch commit	Command (preset; `hydra.sweep.dir=results/...`)
X2 (§3)	`503ec72`	`+experiment=x2_seeds [simulator=bellman_harris]`; training arm `seed=0,...,9`; data arm `data_seed=0,...,9 seed=0` (folders `results/x2/<sim>_training`, `<sim>_data`)
X1 (§4)	`fa3e96a`; NPE/NRE E0-free reruns `a64d17e`	`scripts/run_x1.sh <npe|snpe|nle|nre>` (one lane per method; settings in sequence; `SETTINGS=...` with `scripts/run_x1_settings.sh`); per setting: `simulator=galton_watson law=two_type|reversible_ab design=default budget=dev` (E0 free / fixed), `+experiment=x5b_lifetime_prior` (BH ridge; loud: wrong bounds as X5b), `+experiment=e3a_lifetime_fixed lifetime_setting={B:30,T:18.75}` (silent); methods: `inference=npe` [`inference.method.rounds=4 budget.sbc_cases=0`] | `inference=nle|nre budget.sbc_cases=50`; `seed=0,...,4` (`results/x1/<setting>/<method>`); analysis `scripts/analyse_x1.py`
E0 (§5)	`194e31f`, `b3cbae0`	`+experiment=b_row_identifiability law=two_type|reversible_ab`; `inference.method.rounds=4`; `+experiment=b_row_flat_prior`
X5a (§6)	`d403cac`; bio `7f4a11f`	`+experiment=x5a_truth_position law=two_type|reversible_ab '++law.concentration={},<Jeffreys>,<flat B.to_A>'` biological priors: same with `'++law.concentration=<bio-B>,<bio>'` (folders `results/x5/a_*_bio`) (exact lists in the preset header and `results/x5/a_*/info.md`)
X5c (§7)	`d403cac`	`+experiment=x5c_data_vs_prior`
E0 at 4096 (§8)	`503ec72`	`+experiment=e4_design law=two_type design.clones=4096 'design.horizons=[4,6,10]' seed=0,1,2,3,4` (folder `results/e4/gw_xxl_two_type_5k`); BH: add `simulator=bellman_harris` with `'design.horizons=[96,144,240]'` (`bh_xxl_two_type_5k`); 20k: add `budget.simulations=20000 budget.max_epochs=500` (`*_two_type_20k`)
Design comparison (§10)	`b3cbae0`	`+experiment=design_comparison` (laws two_type, reversible_ab)
E4 GW (§11)	`19a10ea`	`+experiment=e4_design`
	`5cb6018` (1024, 2048)	`+experiment=e4_design design.clones=1024,2048` (folder `results/e4/gw_large`)
	`07fb7fe` (4096)	`+experiment=e4_design design.clones=4096` (folder `results/e4/gw_xxl`)
	`07fb7fe` (20k; docs `503ec72`)	`+experiment=e4_design design.clones=4096 'design.horizons=[4,6,10]' budget.simulations=20000 seed=0,1,2,3,4` (folder `results/e4/gw_xxl_20k`)
E4 BH (§12)	`43e9e44`	`+experiment=e4_design simulator=bellman_harris design.horizons=[240],[144,240],[96,144,240],[144,240,336]`
	`5cb6018` (1024, 2048)	same + `design.clones=1024,2048` (folder `results/e4/bh_large`)
	`07fb7fe` (4096)	same + `design.clones=4096` (folder `results/e4/bh_xxl`)
	`07fb7fe` (20k; docs `503ec72`)	`simulator=bellman_harris design.clones=4096 'design.horizons=[96,144,240]' budget.simulations=20000 seed=0,1,2,3,4` (folder `results/e4/bh_xxl_20k`)
	`503ec72` (20k, real-like, 512)	`+experiment=x5b_lifetime_prior budget.simulations=20000 budget.max_epochs=500 seed=0,1,2,3,4` (folder `results/e4/bh_real_20k`) `+experiment=e4_design simulator=bellman_harris design.clones=512 'design.horizons=[96,144,240]' budget.simulations=20000 budget.max_epochs=500 seed=0,1,2,3,4` (folder `results/e4/bh_512_20k`)

Experiment	Launch commit	Command (continued)
BH ridge (§13)	19a10ea	+experiment=bh_ridge (+ lifetimes fixed / A concentration)
X5d (§14)	d403cac	+experiment=x5d_prior_average simulator=galton_watson|bellman_harris
X5b (§15)	d403cac	vague: +experiment=x5b_lifetime_prior right / wrong: add '~simulator.model.lifetime_mean_bounds' and '+simulator.model.lifetime_mean_bounds={A:[16,25],B:[24,37.5]}' resp. {A:[24,37.5],B:[36,56.25]}
E3a / E3b / E3c (§16–18)	27b1d61–c919397	e3a_lifetime_free, e3a_lifetime_fixed, e3b_desync, e3c_persist
X3 (§19)	d59df24; BH control a64d17e	+experiment=x3_gw_fit_bh (folder results/x3/gw; τ = hours_per_generation=16,20,24,30) control: simulator=bellman_harris law=reversible_ab design=real_like budget=dev seed=0,1,2,3,4 (results/x3/bh_control); analysis scripts/analyse_x3.py

Experiment	Launch commit	Command (continued)
E1a type (§20)	BH 27b1d61; GW 19a10ea/d46b4c2	+experiment=e1a_founder_type simulator=... (fixed; inferred with '~law.fixed.founder')
E1a number BH (§21)	27b1d61	+experiment=e1a_founder_number simulator=bellman_harris
E1a number GW (§22)	b0789d3	+experiment=e1a_founder_number simulator=galton_watson
E1b (§23)	BH 43e9e44; GW d46b4c2	+experiment=e1b_death_detection simulator=...
E1b follow-up (§24)	a64d17e; survivor a63b3a8	+experiment=e1b_all_clones (results/e1b_all/all_clones); add observation=perfect (detected); add summary=survivor_moments (all_clones_survivor)
E1c BH (§25)	43e9e44/c919397	+experiment=e1c_fixed_death simulator=bellman_harris law.fixed.A.death=0,0.03,0.2 law.fixed.B.death=0,0.1,0.4
E1c GW (§26)	d46b4c2	+experiment=e1c_fixed_death (4 × 4 grid; relaunched with bounded SBC)
E2a/E2b BH (§27)	27b1d61–c919397	+experiment=e2a_typing_known, +experiment=e2b_typing_ignored
E2a/E2b GW (§28)	12c0827	+experiment=e2a_typing_known|e2b_typing_ignored simulator=galton_watson
E2 asymmetric (§29)	ignored: BH fa3e96a, GW a64d17e; known a64d17e	ignored: +experiment=e2_asymmetric (results/e2_asym/ignored) known: add observation=noisy observation.capture=1.0 'observation.confusion=${data.observation.confusion}' (results/e2_asym/known)
E2c ignore C (§30)	19a10ea	+experiment=e2c_third_type
E2c C modelled (§31)	free d46b4c2; traj. 12c0827	+experiment=e2c_third_type law=three_type|three_type_traj observation=three_split
E2c rerun (prior truths) (§32)	36d7b07	ignore C: +experiment=e2c_prior_truth C free: +experiment=e2c_prior_truth law=three_type observation=three_split C traj.: +experiment=e2c_prior_truth law=three_type_traj observation=three_split (folders results/e2c_prior/<arm>)
X4 (§34)	fa3e96a; combined + C after bae0b90	+experiment=x4_combined (control); combined: add '++data.law.fixed.founder={A:0.9,B:0.1}' observation@data.observation=noisy data.observation.capture=1.0 data.observation.confusion=0.05 data.simulator.model.lifetime_shape=4; modelled: add '~law.fixed.founder' observation=noisy observation.capture=1.0 observation.confusion=0.05 simulator.model.lifetime_shape=4; combined + C: law@data.law=three_type with C read 50/50 (exact commands in the preset header); results/x4/<arm>; analysis scripts/analyse_x4.py

Experiment	Launch commit	Command (continued)
R real data (Part V): data	341e149	uv run python scripts/prepare_real_data.py (writes data/real/); table uv run python scripts/analyse_real.py; runner scripts/run_real.sh
R: training	bh_rev, bh_rev_founder: 7d16a44; bh_two, bh_two_founder, bh_multi, gw_rev: 08b62fc; bh_rev_s1, bh_hetero, bh_hetero_div: de468b4; bh_dd, bh_dd_hetero, bh_dd_hetero_div: bae0b90; bh_dd_hetero_s1: a63b3a8	single runs: uv run bio-sbi <model> design=real_like budget=full hydra.run.dir=results/real/<name>, <model>: bh_rev: simulator=bellman_harris law=reversible_ab (_s1: seed=1; _founder: '~law.fixed.founder') gw_rev: simulator=galton_watson law=reversible_ab bh_two: law=two_type (_founder as above) bh_multi: law=multi_founder 'law.founder_sizes=[1,2,4,8]' '++law.concentration={}' bh_hetero: law=two_type_hetero '~law.fixed.founder'; _div: add '+law.clone_heterogeneity.outcomes=[div_AA,div_AB,div_BB]' bh_dd*: the same with simulator=bellman_harris_dd
R: fits	as training	uv run bio-sbi-infer results/real/<name> data/real/crc0322_<variant>_bh.csv --out results/real/<name>/fit_<variant> --ppc 500 (_gw files for gw_rev)
Benchmark (App. A)	811d8ca (gpu-bench)	scripts/run_device_benchmark.sh
P0 timing (App. A.1)	f0924e6	single runs (no -m): simulator=bellman_harris law=reversible_ab design=real_like budget=smoke|dev; GW reference simulator=galton_watson ... budget=dev; hydra.run.dir=results/p0_bh_timing/{smoke,dev,gw_dev}

Experiment	Launch commit	Command (continued: exact likelihood)
Exact GW, XG (§9)	41d1ae4 (exact-gw; merged 6e7d13a)	+experiment=exact_gw_e0_4096 (results/exact_gw/e0_4096) +experiment=exact_gw_e0 (results/exact_gw/e0; two_type rerun with law=two_type inference.sbc_cases=0, split by seed into two_type_a|b|c) +experiment=exact_gw_real_like (results/exact_gw/real_like) +experiment=exact_gw_early (results/exact_gw/early); analysis scripts/analyse_exact_gw.py
Exact BH, XB (§13.1; preliminary)	523708e (AMIS); SMC run db3ee16 (merged c2b639c)	+experiment=exact_bh_real_like (results/exact_bh/real_like_*); +experiment=exact_bh_ridge 'design.horizons=[96,144,240]' design.clones=512 (results/exact_bh/ridge_512_*) (AMIS, stopped); SMC: single run +experiment=exact_bh_real_like law=reversible_ab seed=0 with local_k = 60 (results/exact_bh/smc_check_rev0_c); sampler study scripts/exact_bh_sampler/; BH Fisher information: scripts/b_row_theory/bhfisher.py, bhexact_validate.py
Speed (App. B)	523708e	scripts/run_exact_benchmark.sh results/exact_benchmark full (one job at a time); derived tables uv run python scripts/benchmark_exact.py derive

Rows follow the order of this report (exact-likelihood rows in the last table). Every command has the form

```
uv run bio-sbi -m <overrides> torch_threads=4 workers=4 hydra.sweep.dir=<folder>
```

The exact command, grid and fixed settings are in each folder's `info.md`; each run's `.hydra/config.yaml` reproduces it. Packages: `bio-sbi` 0.2.0, `sbi` 0.27.0, `torch` 2.14.1+cu126, `numpy` 2.5.4, `numba` 0.68.0, `hydra-core` 1.3.8, Python 3.12.13.

Rebuilding this report (from the repository root). This report reuses the figures of the ID-ordered report (`report/figures/`, via `\graphicspath`); it has no figure script of its own:

```
uv run python report/make_figures.py      # report/figures/*.pdf, report/tables/*.csv
cd report_sequential && /home/matteo/miniconda3/envs/latex/bin/tectonic report.tex
```

`make_figures.py` only reads `results/`; incomplete sweeps would be skipped and shown as pending boxes. Numbers in tables come from the `info.md` analyses in `results/` (overview: `results/overview.md`) or from `scripts/summarize.py` functions (`mae_probabilities`, per-parameter rows); plotted numbers are in `report/tables/`.

D Glossary

A, B (C)	LGR5$^+$ stem-like, KRT20$^+$ differentiated (hidden third type)
Amortised NPE	one network trained on prior simulations, usable for any dataset
BH / GW	Bellman–Harris (continuous time, lifetimes) / Galton–Watson (generations)
C2ST	classifier two-sample test; accuracy 0.5 = indistinguishable
Clone / organoid	all descendants of one seeding event; recovered if ≥ 1 cell detected
Contraction	$1 - \mathrm{Var}_{\mathrm{post}}/\mathrm{Var}_{\mathrm{prior}}$
Cramér–Rao (CR) sd	lower bound on the sd of an unbiased estimator, from the inverse Fisher information of the exact likelihood (local, large-sample)
Exact likelihood	$p(\text{counts} \mid \theta)$ computed without simulation (GW: pgf + FFT; BH: method of stages + FFT); `exact_gw` (NUTS), `exact_bh` (SMC + adaptive MCMC; AMIS first look)
Control (paired)	well-specified run of the same sweep with the same seeds and datasets
Coverage (90%)	seeds whose 90% credible interval contains the truth
Design	endpoints (horizons) × organoids per endpoint
Endpoint / horizon	destructive measurement time (generations for GW, hours for BH)
Founder	seeded cell; `founder` = type mix, `founder_size` = cells per clone
Heterogeneity κ	concentration of the per-organoid outcome rows (§2.5); small = organoids differ strongly
Carrying capacity K	organoid size at which density-dependent lifetimes have doubled, $\tau_k(1 + N/K)$
Data variant	one version of the CRC0322 counts: 3D mean, 3D sample 1–3, 2D sections (§35)
Lifetime τ_A, τ_B	mean time from birth to the cell's event (BH), Gamma with shape 8
MAE	mean absolute error of posterior means over the outcome probabilities
Observed fraction	detected / seeded organoids
Paired SE	SE over seeds of the difference misspecified − control
PPC	posterior predictive check; fraction of summaries outside the 95% predictive interval
Practical / structural non-identifiability	parameter determined in principle (nonsingular Fisher information) but not at feasible data sizes / not determined by any amount of data
Ridge	direction in parameter space along which the data are (nearly) constant
SBC	simulation-based calibration over prior draws
SNPE	sequential NPE (rounds focus on the observed data)
Support mass	share of the NPE posterior inside the prior support
$\theta^\star$	reference truth (Table 2)
Well specified	data model = inference model